HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

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1 Basics

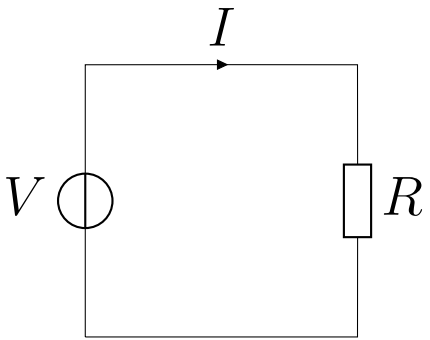
1.1 Quantities, Units and Ohm's Law

Quantities and units

quantity	symbol	unit	definition
voltage	V	volt, V	energy per charge, J/C
current	I	ampere, A	charge per time, C/s
resistance	R	ohm, Ω	$R = V/I$
power	P	watt, W	$P = V \cdot I$
capacitance	C	farad, F	$C = Q/V$
time, frequency	t, f	s, Hz	$f = 1/T$

- prefixes: p (10^{-12}), n (10^{-9}), µ (10^{-6}), m (10^{-3}), k (10^{3}), M (10^{6})
- useful combination: V / kΩ = mA, mA · kΩ = V

Ohm's law



Ohm's law

$$V = R \cdot I \qquad I = \frac{V}{R} \qquad R = \frac{V}{I}$$

- a resistor is **linear**: double voltage $\Rightarrow$ double current
- the conductance $G = 1/R$ is measured in siemens, S

Power and power rating

Power in a resistor

$$P = V \cdot I = I^2 R = \frac{V^2}{R}$$

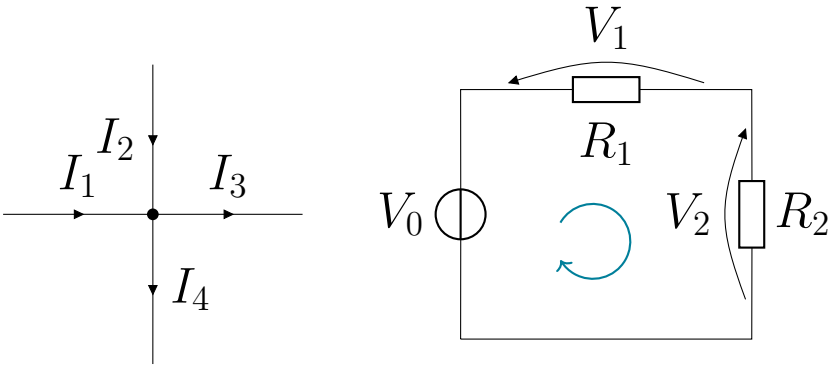
- the power is turned into heat; every resistor has a **power rating**
  - SMD 0603: 0.1 W, SMD 1206: 0.25 W, through-hole (0207): 0.25 W … 0.6 W
- design rule: stay below about 50 % of the rating
- small signal circuits (kΩ, mA) dissipate only mW
- example: $100\,\Omega$ at 5 V: $P = 25\,\text{V}^2/100\,\Omega = 250$ mW, too much for a 0.25 W resistor

Standard values: the E-series

- resistors are not available in every value, but in logarithmic series
- E12 ($10\,\%$ tolerance): 1.0 1.2 1.5 1.8 2.2 2.7 3.3 3.9 4.7 5.6 6.8 8.2
- E24 ($5\,\%$) adds: 1.1 1.3 1.6 2.0 2.4 3.0 3.6 4.3 5.1 6.2 7.5 9.1
- E96 ($1\,\%$): 96 values per decade, standard for metal film resistors today
- each value exists in every decade: $4.7\,\Omega$, $47\,\Omega$, ..., $4.7\,\mathrm{M}\Omega$

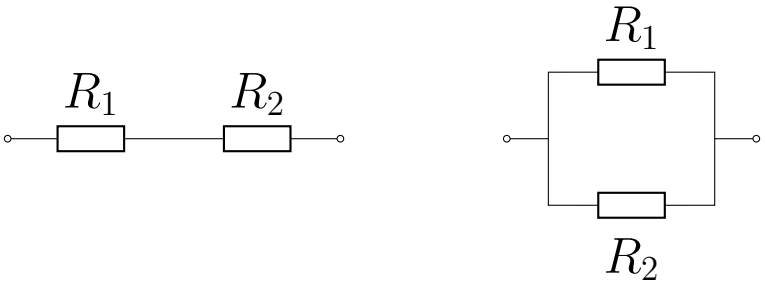
1.2 Kirchhoff's Laws and Networks

Kirchhoff's current law (KCL) and voltage law (KVL)



- KCL: at every node the sum of incoming currents equals the sum of outgoing currents: $I_1 + I_2 = I_3 + I_4$
- KVL: around every closed loop the sum of voltages is zero: $V_0 = V_1 + V_2$

Series and parallel connection

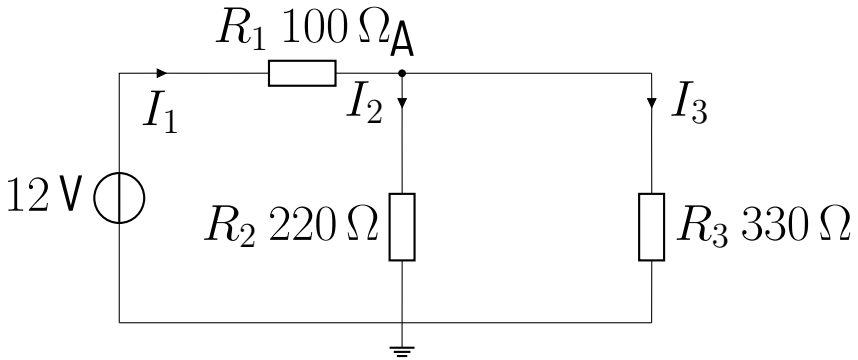


Series and parallel

series: $R = R_1 + R_2$ parallel: $R = R_1 \parallel R_2 = \dfrac{R_1 R_2}{R_1 + R_2}$

- series: same current, the voltages add up
- parallel: same voltage, the currents add up
- parallel resistance is always smaller than the smallest resistor; $R \parallel R = R/2$

Example: series-parallel network



- $R_2 \parallel R_3 = 132\,\Omega$, total $R = 232\,\Omega$, $I_1 = 12\,\text{V}/232\,\Omega = 51.7\,\text{mA}$
- $V_A = I_1 \cdot 132\,\Omega = 6.83\,\text{V}$; $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ (KCL: $I_1 = I_2 + I_3$)

▶ Falstad: `examples/basics/network.txt`

Exercise 1.1: Power in a series-parallel network

Use the network above ($12\,\mathrm{V}$, $R_1 = 100\,\Omega$, $R_2 = 220\,\Omega$, $R_3 = 330\,\Omega$). All resistors are $0.25\,\mathrm{W}$ types.

a) Calculate the power dissipated in each resistor and the power delivered by the source.
b) Which resistor is overloaded? Which rating would you choose instead?
c) Check the currents and powers in Falstad.

Solution 1.1

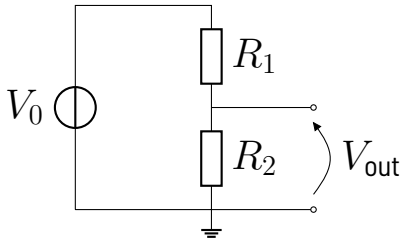
a) $P_1 = I_1^2 R_1 = (51.7\,\text{mA})^2 \cdot 100\,\Omega = 268\,\text{mW}$, $P_2 = V_A^2/R_2 = (6.83\,\text{V})^2/220\,\Omega = 212\,\text{mW}$, $P_3 = (6.83\,\text{V})^2/330\,\Omega = 141\,\text{mW}$. Source: $P = 12\,\text{V} \cdot 51.7\,\text{mA} = 621\,\text{mW} = P_1 + P_2 + P_3$.

b) R_1 exceeds $0.25\,\text{W}$; R_2 is at $85\,\%$ of its rating, which is too hot as well. With the $50\,\%$ rule R_1 needs $0.6\,\text{W}$ and R_2 $0.5\,\text{W}$ (better: $1\,\text{W}$ for R_1).

c) ▶ Falstad: `examples/basics/network.txt`: `VA` $= 6.83\,\text{V}$; hovering over the resistors shows $I_1 = 51.7\,\text{mA}$, $I_2 = 31.0\,\text{mA}$, $I_3 = 20.7\,\text{mA}$ and the power of each resistor.

1.3 The Voltage Divider

Unloaded divider



- the same current flows through R_1 and R_2: $I = V_0/(R_1 + R_2)$
- the voltages divide in the ratio of the resistances

Voltage divider

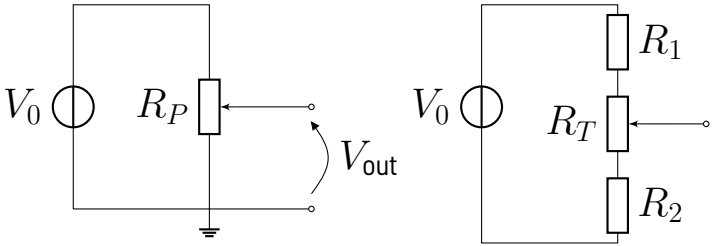
$$V_{\text{out}} = V_0 \frac{R_2}{R_1 + R_2}$$

Loaded divider

- a load R_L appears in parallel to R_2: $V_{\text{out}} = V_0 \dfrac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L}$
- seen from the load the divider is a real source with $V_{\text{th}} = V_0\, R_2/(R_1 + R_2)$ and $R_{\text{th}} = R_1 \parallel R_2$
- rule of thumb: $R_L \geq 10\,(R_1 \parallel R_2)$ keeps the error below $10\,\%$
- for an error below $1\,\%$: $R_L \geq 100\,(R_1 \parallel R_2)$, or buffer the tap with an op-amp

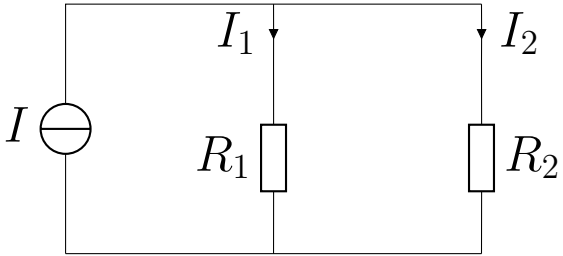
▶ Falstad: `examples/basics/divider_loaded.txt`

Potentiometer: adjustable divider



- the wiper divides the track: $V_{\text{out}} = V_0 \cdot x, x = 0 \ldots 1$ (linear taper)
- trimmer between two fixed resistors: fine adjustment over a small range
- the wiper must not carry a large current; load it lightly

Current divider



Current divider

$$I_1 = I \frac{R_2}{R_1 + R_2} \qquad I_2 = I \frac{R_1}{R_1 + R_2}$$

- the **smaller** resistor takes the **larger** current
- example: 10 mA into $1\,\text{k}\Omega \,||\, 4.7\,\text{k}\Omega$: $I_1 = 8.25$ mA, $I_2 = 1.75$ mA

▶ Falstad: `examples/basics/current_divider.txt`

Exercise 1.2: Loaded voltage divider

A divider with $R_1 = R_2 = 10\,\text{k}\Omega$ is supplied with $V_0 = 10\,\text{V}$.

a) Calculate V_{out} without load.
b) Calculate V_{out} with a load $R_L = 10\,\text{k}\Omega$.
c) Check both results in Falstad.

Solution 1.2

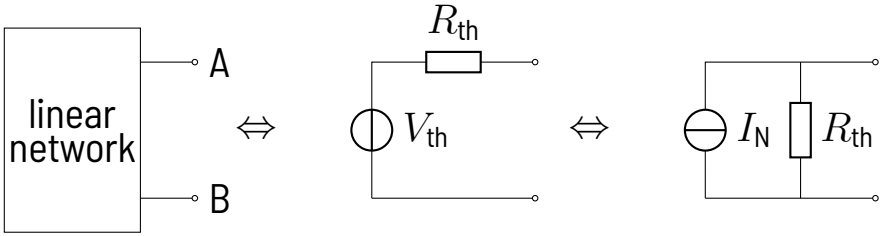
a) $V_{\text{out}} = 10\,\text{V} \cdot \frac{10}{10+10} = 5\,\text{V}$.

b) $R_2 \parallel R_L = 5\,\text{k}\Omega$, so $V_{\text{out}} = 10\,\text{V} \cdot \frac{5}{10+5} = 3.33\,\text{V}$. The load costs one third of the output voltage, because R_L is only twice the source resistance $R_{\text{th}} = 5\,\text{k}\Omega$.

c) ▶ Falstad: `examples/basics/divider_loaded.txt`
the labels show `Vout_open` $= 5.00\,\text{V}$ and `Vout_load` $= 3.33\,\text{V}$ (move the mouse over the label).

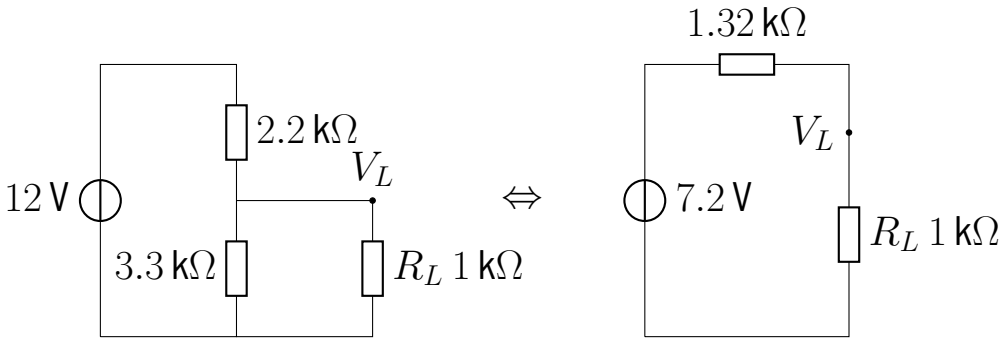
1.4 Equivalent Sources and Superposition

Thevenin and Norton equivalent



- open-circuit voltage: $V_{\text{th}} = V_{\text{oc}}$ (nothing connected to A–B)
- short-circuit current: $I_{\text{N}} = I_{\text{sc}}$ (A and B shorted)
- internal resistance $R_{\text{th}} = V_{\text{oc}}/I_{\text{sc}}$, or: set all sources to zero (voltage source $\rightarrow$ wire, current source $\rightarrow$ open) and calculate the resistance between A and B

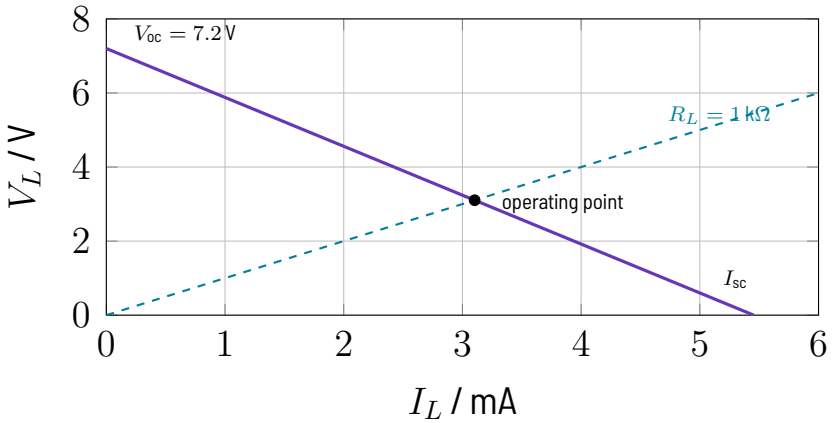
Example: Thevenin equivalent of a divider



- $V_{\text{oc}} = 12\,\text{V} \cdot 3.3/5.5 = 7.2\,\text{V}, \quad I_{\text{sc}} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$
- $R_{\text{th}} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega = 2.2\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega$
- with R_L: $V_L = 7.2\,\text{V} \cdot 1/(1 + 1.32) = 3.10\,\text{V}$ in both circuits

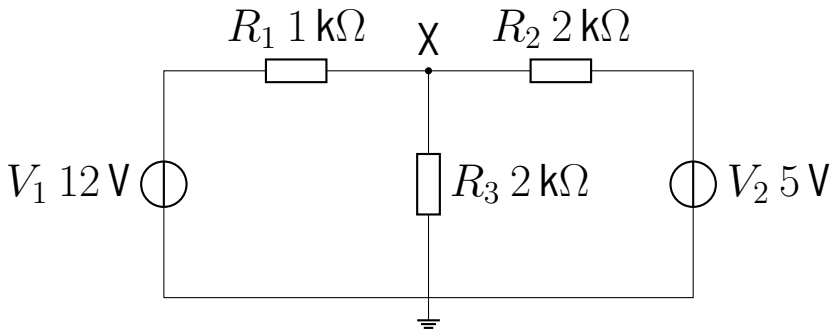
▶ Falstad: `examples/basics/thevenin.txt`

Load line of a real source



- the source can deliver every point on the line $V_L = V_{th} - R_{th}\, I_L$
- the load forces $V_L = R_L\, I_L$; the intersection is the operating point

Superposition



- in a linear network the effect of all sources is the sum of the effects of each source alone
- the other sources are set to zero: voltage source $\rightarrow$ wire, current source $\rightarrow$ open
- V_1 alone: $V_{X1} = 12\,\text{V} \cdot \frac{R_2 \parallel R_3}{R_1 + R_2 \parallel R_3} = 6\,\text{V}$; V_2 alone: $V_{X2} = 5\,\text{V} \cdot \frac{R_1 \parallel R_3}{R_2 + R_1 \parallel R_3} = 1.25\,\text{V}$
- both: $V_X = 7.25\,\text{V}$ ▶ Falstad: `examples/basics/superposition.txt`

Exercise 1.3: Thevenin equivalent from open-circuit voltage and short-circuit current

The divider $12\,\mathrm{V}$, $R_1 = 2.2\,\mathrm{k\Omega}$ (top), $R_2 = 3.3\,\mathrm{k\Omega}$ (bottom) feeds a load $R_L = 1\,\mathrm{k\Omega}$.

a) Determine V_{oc} and I_{sc} at the tap and from them R_{th}.
b) Calculate V_L with the equivalent circuit.
c) Which load would reduce the open-circuit voltage by only $5\,\%$?
d) Compare the original circuit and the equivalent in Falstad.

Solution 1.3

a) $V_{\text{oc}} = 12\,\text{V} \cdot 3.3/(2.2+3.3) = 7.2\,\text{V}$. With the tap shorted no current flows in R_2: $I_{\text{sc}} = 12\,\text{V}/2.2\,\text{k}\Omega = 5.45\,\text{mA}$. $R_{\text{th}} = 7.2\,\text{V}/5.45\,\text{mA} = 1.32\,\text{k}\Omega$.

b) $V_L = 7.2\,\text{V} \cdot \frac{1}{1+1.32} = 3.10\,\text{V}$.

c) $V_L/V_{\text{oc}} = R_L/(R_L + R_{\text{th}}) = 0.95 \Rightarrow R_L = 19\,R_{\text{th}} = 25.1\,\text{k}\Omega$ (E12: $27\,\text{k}\Omega$).

d) ▶ Falstad: `examples/basics/thevenin.txt`: `VL_orig` = `VL_equiv` = $3.10\,\text{V}$. Replace R_L by a wire and hover over it: the short-circuit current is $5.45\,\text{mA}$ in both circuits.

Exercise 1.4: Superposition

Use the circuit of the superposition slide.

a) Calculate V_X with superposition.
b) Check the result with KCL at node X.
c) How large is the current through R_2, and in which direction does it flow?
d) Check the three cases in Falstad.

Solution 1.4

a) V_1 alone: $R_2 \parallel R_3 = 1\,\text{k}\Omega$, $V_{X1} = 12\,\text{V} \cdot 1/2 = 6\,\text{V}$. V_2 alone: $R_1 \parallel R_3 = 0.667\,\text{k}\Omega$, $V_{X2} = 5\,\text{V} \cdot 0.667/2.667 = 1.25\,\text{V}$. $V_X = 7.25\,\text{V}$.

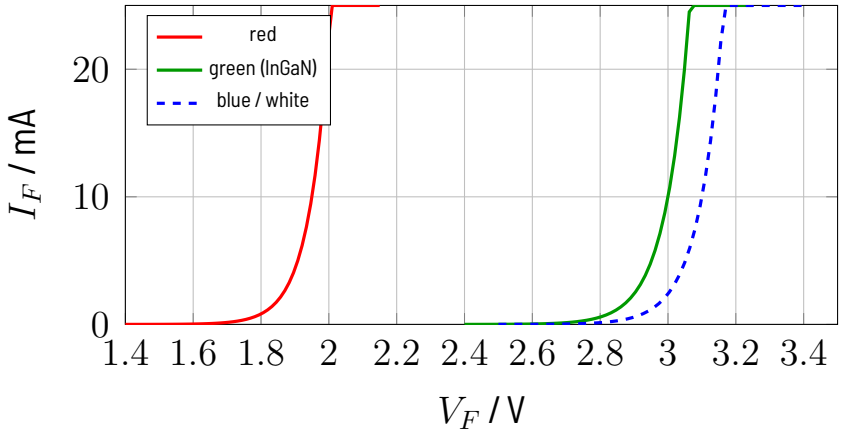
b) $\frac{12-7.25}{1} = 4.75\,\text{mA}$ flows in through R_1; $\frac{7.25}{2} = 3.625\,\text{mA}$ flows out through R_3 and $\frac{7.25-5}{2} = 1.125\,\text{mA}$ through R_2: $4.75 = 3.625 + 1.125$ ✓

c) $1.125\,\text{mA}$ from X into the $5\,\text{V}$ source: V_2 is charged (absorbs power).

d) ▶ Falstad: `examples/basics/superposition.txt`: `VX1` $= 6.00\,\text{V}$, `VX2` $= 1.25\,\text{V}$, `VX` $= 7.25\,\text{V}$.

1.5 The LED with Series Resistor

LED characteristic



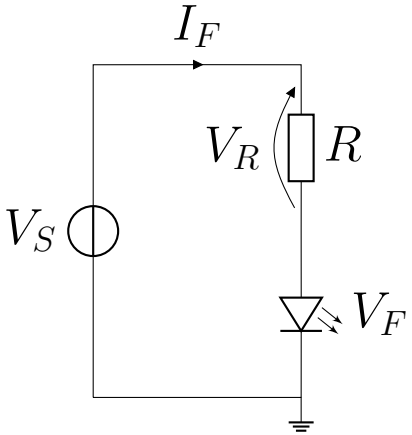
- below a threshold almost no current, above it the current rises steeply
- simple model: constant forward voltage V_F at the operating current

Forward voltage and current

colour	material	V_F at $10\,\text{mA} \ldots 20\,\text{mA}$
infrared	GaAs	$1.2\,\text{V} \ldots 1.5\,\text{V}$
red	AlGaInP	$1.8\,\text{V} \ldots 2.1\,\text{V}$
yellow, orange	AlGaInP	$2.0\,\text{V} \ldots 2.2\,\text{V}$
green (classic, yellow-green)	GaP	$2.1\,\text{V} \ldots 2.4\,\text{V}$
green (true green), blue, white	InGaN	$2.8\,\text{V} \ldots 3.4\,\text{V}$

- standard $3\,\text{mm}$/$5\,\text{mm}$ LEDs: $I_F \leq 20\,\text{mA}$; modern LEDs are clearly visible at $2\,\text{mA} \ldots 5\,\text{mA}$
- reverse voltage only about $5\,\text{V}$: an LED is not a rectifier

Dimensioning the series resistor



LED series resistor

$$R = \frac{V_S - V_F}{I_F} \qquad P_R = (V_S - V_F) \cdot I_F$$

- example: $V_S = 5\,\text{V}$, red LED $V_F = 2.0\,\text{V}$, $I_F = 10\,\text{mA}$: $R = 300\,\Omega \Rightarrow$ E12: $330\,\Omega$, $I_F = 9.1\,\text{mA}$, $P_R = 27\,\text{mW}$

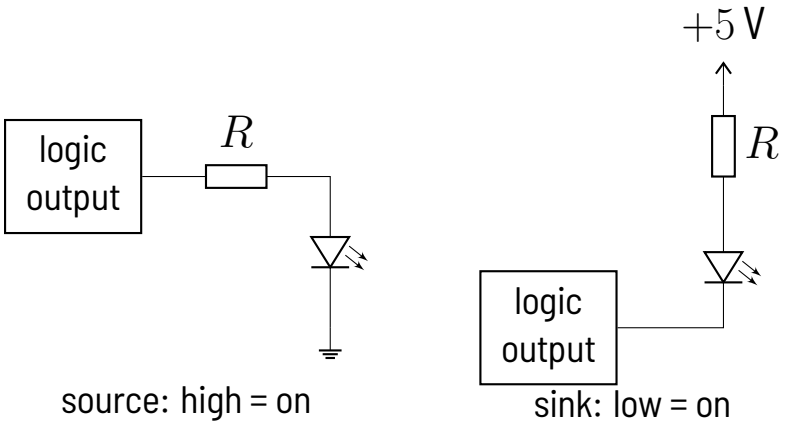
Design recipe

Design card: LED with series resistor

1. take V_F from the datasheet at the desired current (or the table: red 2.0 V, blue/white 3.0 V)
2. choose I_F: 2 mA … 10 mA for an indicator, check the maximum current of the driver
3. $R = (V_S - V_F)/I_F$, round **up** to the next E12/E24 value
4. recalculate I_F and check $P_R = I_F^2 R$ against the rating

- rounding up keeps the current below the target, which protects LED and driver
- the larger $V_S - V_F$, the less I_F depends on the spread of V_F

Driving an LED from a logic output



- a logic output is a switch to V_S or to ground with an output resistance of some $10\,\Omega$
- 74HC at 5 V: ± 4 mA with guaranteed levels, absolute maximum 25 mA per pin
- microcontrollers: some mA per pin and a limit for the whole port
- $330\,\Omega$ at 5 V gives about 9 mA, 1 kΩ about 3 mA

▶ Falstad: `examples/basics/led_resistor.txt`

Exercise 1.5: LED on a $5\,\mathrm{V}$ logic output

A red LED ($V_F = 2.0\,\mathrm{V}$ at $10\,\mathrm{mA}$) is to be driven by a $5\,\mathrm{V}$ logic output.

a) Calculate the series resistor for $10\,\mathrm{mA}$ and choose an E12 value.
b) Calculate the actual current and the power in the resistor.
c) The same LED is to be driven from a $3.3\,\mathrm{V}$ microcontroller pin with at most $4\,\mathrm{mA}$. Choose the resistor.
d) Falstad uses an LED model with $V_F \approx 1.8\,\mathrm{V}$. Which current does the simulation show with $330\,\Omega$?

Solution 1.5

a) $R = (5\,\mathrm{V} - 2.0\,\mathrm{V})/10\,\mathrm{mA} = 300\,\Omega$; rounded up to E12: $330\,\Omega$.

b) $I_F = 3.0\,\mathrm{V}/330\,\Omega = 9.1\,\mathrm{mA}$, $P_R = 3.0\,\mathrm{V} \cdot 9.1\,\mathrm{mA} = 27\,\mathrm{mW}$: any resistor will do, even SMD 0402.

c) $R \geq (3.3\,\mathrm{V} - 2.0\,\mathrm{V})/4\,\mathrm{mA} = 325\,\Omega$, again $330\,\Omega$ ($I_F = 3.9\,\mathrm{mA}$; V_F is a bit lower at this current, so the real current is slightly higher).

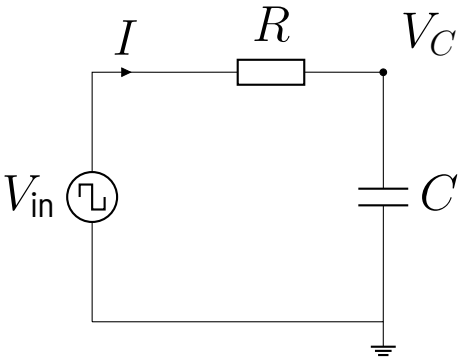
d) ▶ Falstad: `examples/basics/led_resistor.txt`: $\mathtt{VLED} = 1.78\,\mathrm{V}$, so $I_F = (5 - 1.78)\,\mathrm{V}/330\,\Omega = 9.76\,\mathrm{mA}$ (hover over the resistor). Use the slider to see how the current changes with R.

1.6 Capacitor and RC Circuits

The capacitor

- charge proportional to voltage: $Q = C \cdot V$
- current = change of charge: $I = C \dfrac{dV}{dt}$
  - the voltage across a capacitor cannot jump (that would need an infinite current)
  - DC: once charged, no current flows (open circuit)
- energy: $E = \frac{1}{2} C V^2$
- types: ceramic (pF to 100 µF; COG stable, X7R voltage dependent), film (precise, for filters and hold capacitors), electrolytic (large, polarized)

Charging through a resistor

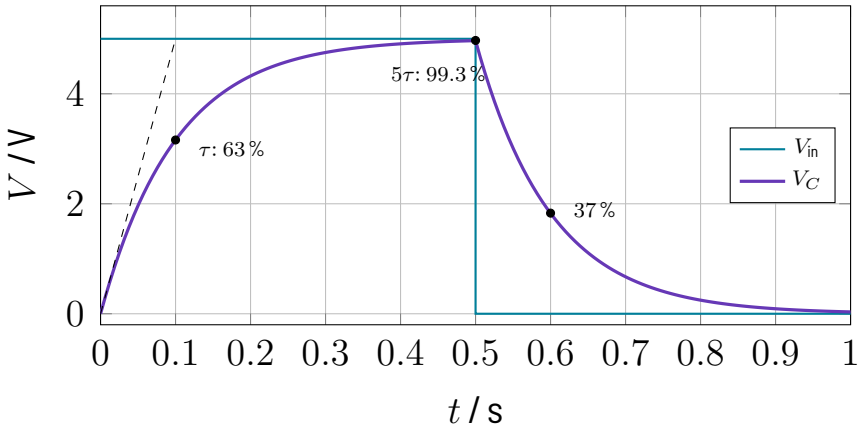


RC charging and discharging

$$V_C(t) = V_0\left(1 - e^{-t/\tau}\right) \qquad V_C(t) = V_0\, e^{-t/\tau} \qquad \tau = R\,C$$

- step of V_0 at the input; the current $I = (V_{\text{in}} - V_C)/R$ decreases as V_C rises
- time to reach a voltage V_1: $t = \tau \ \ln \frac{V_0}{V_0 - V_1}$

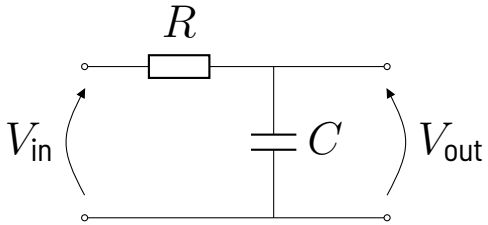
The time constant

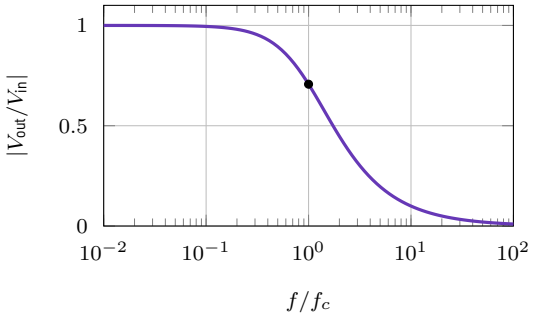


- after τ: $63\,\%$, after 3τ: $95\,\%$, after 5τ: $99.3\,\%$ (“settled”)
- the tangent at $t = 0$ reaches the final value after exactly τ
- example $R = 10\,\text{k}\Omega$, $C = 10\,\mu\text{F}$: $\tau = 0.1\,\text{s}$

▶ Falstad: `examples/basics/rc_charging.txt`

Preview: the RC low-pass





Cut-off frequency

$$f_c = \frac{1}{2\pi RC} \qquad \left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{1}{\sqrt{1+(f/f_c)^2}}$$

- at f_c the amplitude drops to $1/\sqrt{2} = 0.707$ (-3 dB), phase $-45°$
- example: $1.6\,\text{k}\Omega$, $100\,\text{nF}$: $f_c = 995\,\text{Hz}$ ▶ Falstad: `examples/basics/rc_lowpass.txt`

Exercise 1.6: RC charging

A capacitor $C = 10\,\mu\text{F}$ is charged through $R = 10\,\text{k}\Omega$ from a square wave that switches between $0\,\text{V}$ and $5\,\text{V}$ at $1\,\text{Hz}$.

a) Calculate τ and V_C after τ and after $0.5\,\text{s}$.
b) After which time does V_C reach $4\,\text{V}$?
c) To which voltage does C discharge $0.1\,\text{s}$ after the falling edge?
d) Read the values from the Falstad scope.

Solution 1.6

a) $\tau = 10\,\text{k}\Omega \cdot 10\,\mu\text{F} = 0.1\,\text{s}$. $V_C(\tau) = 5\,\text{V}\,(1 - e^{-1}) = 3.16\,\text{V}$; $V_C(0.5\,\text{s}) = V_C(5\tau) = 5\,\text{V}\,(1 - e^{-5}) = 4.97\,\text{V}$.

b) $t = \tau \ln \frac{5}{5-4} = 0.1\,\text{s} \cdot \ln 5 = 0.161\,\text{s}$.

c) The discharge starts at $4.97\,\text{V}$: $V_C = 4.97\,\text{V} \cdot e^{-1} = 1.83\,\text{V}$.

d) ▶ Falstad: `examples/basics/rc_charging.txt`: the scope shows $3.16\,\text{V}$ at $0.1\,\text{s}$, $4\,\text{V}$ at $0.161\,\text{s}$, $4.97\,\text{V}$ at the falling edge and $1.83\,\text{V}$ $0.1\,\text{s}$ later.

1.7 Working with Falstad

Opening and building circuits

- `https://www.falstad.com/circuit/circuitjs.html`
- open an example: click its Falstad box in the script, or **File** → **Import From Text** and paste the content of `examples/basics/*.txt`
- new circuit: **Circuits** → **Blank Circuit**
- place elements: **Draw** menu or right click; frequent ones have shortcuts (`r` resistor, `c` capacitor, `w` wire, `g` ground)
- drag to place, double click to edit values (`4.7k`, `100n`, `10u` are accepted)
- save your work: **File** → **Export As Text** or **Export As Link**

Pitfalls when drawing

- wires connect **only at their end points**: a wire passing over a terminal is not connected; split the wire there
- unconnected terminals are shown as red dots
- every circuit needs a ground (or a source that defines the potential)
- **labeled nodes** (**Draw** $\rightarrow$ **Wiring** $\rightarrow$ **Labeled Node**) with the same name are connected without a wire, and their name can be used for scopes
- "singular matrix": usually a floating node or two ideal voltage sources in parallel

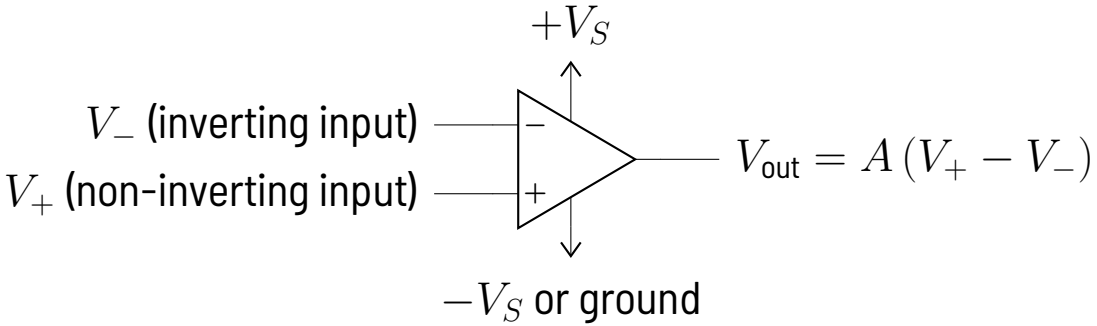
Measuring

- hover the mouse over an element: the bottom right corner shows voltage, current and power
- hover over a labeled node: its voltage to ground
- voltage colours: green positive, red negative, grey zero; yellow dots: current
- scope: right click on an element → **View in New Scope**; right click on a scope for settings (scale, several traces, **Max Scale**)
- sliders: **Edit** dialog of an element, check **Show slider**; the examples have sliders for the values worth changing
- simulation speed: slider on the right; the time step is in **Options → Other Options**

2 Op-Amp Basics

2.1 The Op-Amp in a Nutshell

Symbol and model



- an amplifier for the **difference** of two voltages, with a huge gain $A = 10^5 \ldots 10^7$
- inputs draw (almost) no current; the output can deliver some 10 mA
- the output stays inside the supply range; supply pins are often not drawn

Negative feedback

- a part of V_{out} is fed back to the **inverting** input
- V_{out} rises $\Rightarrow V_-$ rises $\Rightarrow V_+ - V_-$ shrinks $\Rightarrow V_{\text{out}}$ is pulled back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$ ($50\,\mu\text{V}$ for $V_{\text{out}} = 5\,\text{V}$, $A = 10^5$)

Rule 2.1 ***Golden rules*** *for an op-amp with negative feedback:*

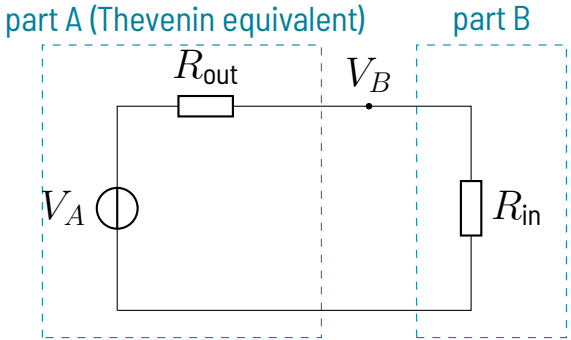
1. *The output does whatever is necessary to make the voltage difference between the inputs zero:* $V_+ = V_-$.
2. *The inputs draw no current:* $I_+ = I_- = 0$.

Real op-amps on a single supply

- most circuits in the lab run from a single supply (5 V or 12 V) and ground
- **input common-mode range**: the input voltages must stay inside it (often the supply minus 1.5 V)
- **output swing**: a rail-to-rail output reaches the rails within some 10 mV, older types (LM358) only within about 1.5 V of the positive rail
- **output current**: typically 10 mA … 40 mA, short-circuit protected
- examples: MCP6002, TLV9002 (CMOS, rail-to-rail, 5 V), LM358 (bipolar, up to 32 V)

2.2 The Unity-Gain Buffer

Every connection is a voltage divider

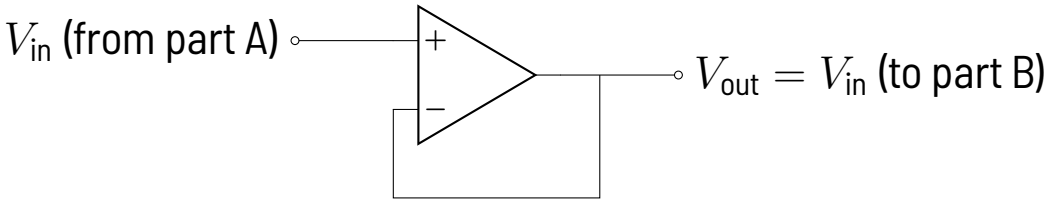


Loading

$$V_B = V_A \frac{R_{\text{in}}}{R_{\text{in}} + R_{\text{out}}} \qquad \text{error} \approx \frac{R_{\text{out}}}{R_{\text{in}}} \text{ for } R_{\text{in}} \gg R_{\text{out}}$$

- rule of thumb: $R_{\text{in}} \geq 100\, R_{\text{out}}$ for an error below $1\,\%$

The voltage follower



- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$ (gain 1)
- rule 2: input current ≈ 0, input resistance GΩ ...TΩ: part A is **not loaded**
- output resistance with feedback $r_o/(1+A)$: $50\,\Omega/10^5 = 0.5$ mΩ: part B sees an **ideal source**

Unity-gain buffer

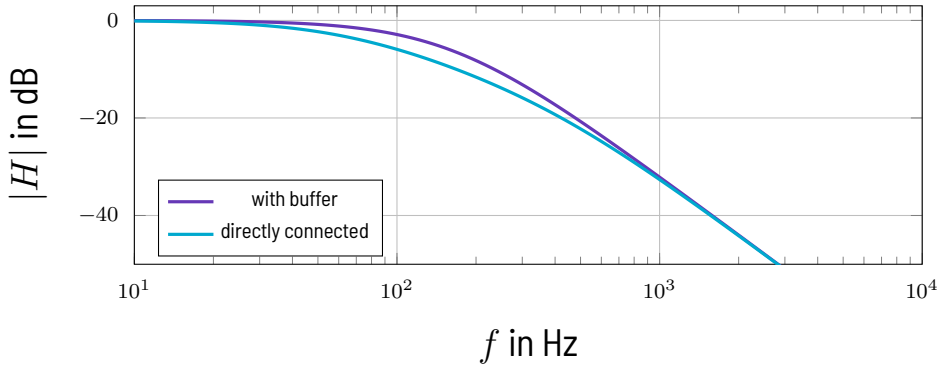
$V_{\text{out}} = V_{\text{in}} \qquad R_{\text{in}} \to \infty \qquad R_{\text{out}} \to 0$

Example: potentiometer as a set-point

- potentiometer $10\,\text{k}\Omega$ at $5\,\text{V}$, wiper in the middle: $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$
- load $1\,\text{k}\Omega$ directly at the wiper: $V_B = 2.5\,\text{V} \cdot \frac{1}{1+2.5} = 0.714\,\text{V}$
- load behind a follower: $2.50\,\text{V}$; the op-amp delivers $2.5\,\text{mA}$
- the wiper itself still has $2.50\,\text{V}$: the follower does not load it

▶ Falstad: `examples/opampbasics/buffer_pot.txt`

Example: two RC low-passes in cascade



- each stage $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 159\,\text{Hz}$
- with buffer: $H = H_1 \cdot H_2 = \frac{1}{(1+j\omega RC)^2}$, at f_c: $|H| = 1/2$ (-6 dB)
- directly: $H = \frac{1}{1+3j\omega RC-(\omega RC)^2}$, at f_c: $|H| = 1/3$ (-9.5 dB)

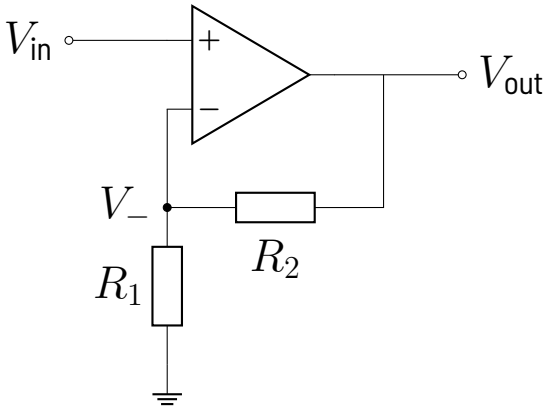
▶ Falstad: `examples/opampbasics/buffer_rc_cascade.txt`

Where buffers are used – and their limits

- in this lecture: behind a voltage reference (chapter 4), at bridge taps (chapter 5), between filter stages (chapter 7), behind the hold capacitor (chapter 8)
- limits
  - output current (some 10 mA) and output swing
  - input common-mode range: e.g. LM358 inputs only up to $V_S - 1.5$ V
  - capacitive loads (long cables): the follower may oscillate; add $50\,\Omega$ … $100\,\Omega$ in series with the output
  - offset voltage: $V_{\text{out}} = V_{\text{in}} + V_{\text{OS}}$ (µV …mV)

2.3 The Non-Inverting Amplifier

Non-inverting amplifier



- R_1, R_2 divide the output: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$
- $R_2 = 0$ (or $R_1 \to \infty$): the follower with $G = 1$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$, output resistance ≈ 0

Exercise 2.1: Buffering a potentiometer

A $10\,\mathrm{k}\Omega$ potentiometer between $5\,\mathrm{V}$ and ground sets a voltage. The wiper is in the middle position. The voltage is used by a following stage with an input resistance of $R_L = 1\,\mathrm{k}\Omega$.

a) Draw the Thevenin equivalent of the wiper. In which wiper position is R_{out} largest?

b) Calculate the voltage at R_L without a buffer and the error in percent.

c) How large must R_L be for an error below $1\,\%$ without a buffer?

d) Now a voltage follower (op-amp at $5\,\mathrm{V}$, rail-to-rail) is placed between wiper and R_L. Calculate the voltage at R_L, the current from the wiper and the output current of the op-amp.

e) The wiper is turned to the upper end ($5\,\mathrm{V}$). What does an LM358 at $5\,\mathrm{V}$ supply deliver, and why?

f) Check (b) and (d) with Falstad.

Solution 2.1

a) $V_A = 2.5\,\text{V}$, $R_{\text{out}} = 5\,\text{k}\Omega \parallel 5\,\text{k}\Omega = 2.5\,\text{k}\Omega$. For a wiper position x (0 …1): $R_{\text{out}} = x(1-x)\,10\,\text{k}\Omega$, largest in the middle ($R/4$).

b) $V_B = 2.5\,\text{V} \cdot 1\,\text{k}\Omega/(1\,\text{k}\Omega + 2.5\,\text{k}\Omega) = 0.714\,\text{V}$, error $-71\,\%$.

c) $R_L \geq 100 \cdot 2.5\,\text{k}\Omega = 250\,\text{k}\Omega$.

d) $V_B = 2.50\,\text{V}$; wiper current ≈ 0 (picoamperes for a CMOS op-amp); op-amp output current $2.5\,\text{V}/1\,\text{k}\Omega = 2.5\,\text{mA}$, which comes from the $5\,\text{V}$ supply.

e) The input common-mode range of the LM358 ends at about $V_S - 1.5\,\text{V} = 3.5\,\text{V}$, and its output does not reach the positive rail either: the output stays at about $3.5\,\text{V}$. For signals up to the supply a rail-to-rail op-amp is needed.

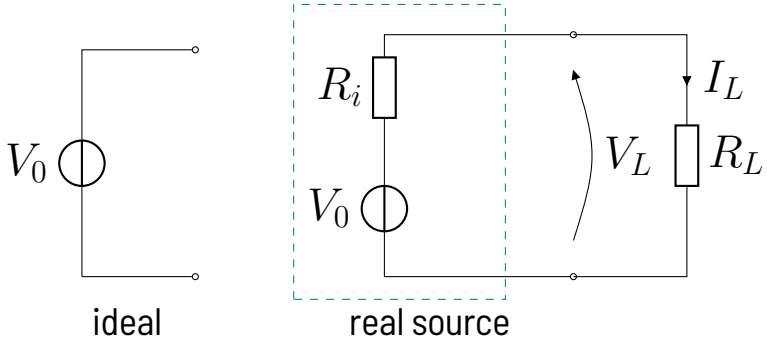
f) ▶ Falstad: `examples/opampbasics/buffer_pot.txt`

`V_direct` $= 0.714\,\text{V}$, `V_wiper` $=$ `V_buffered` $= 2.50\,\text{V}$.

3 Voltage and Current Sources

3.1 Ideal and Real Voltage Sources

Ideal and real voltage source



- ideal: the voltage is V_0 for every current
- real: the terminal voltage drops with the load current

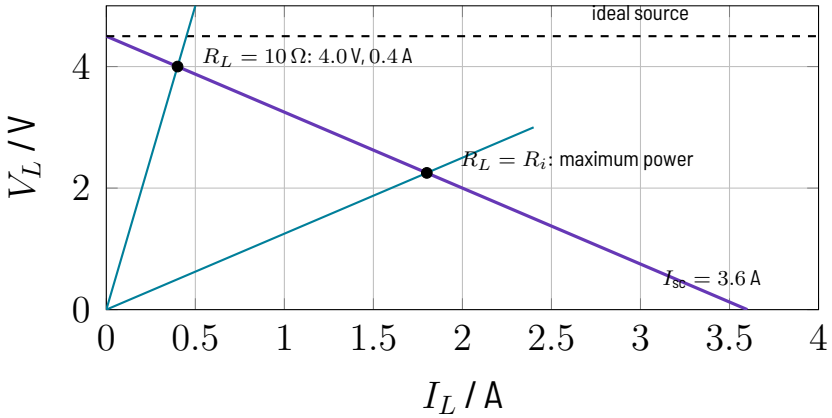
Terminal voltage of a real source

$$V_L = V_0 - R_i\, I_L = V_0\, \frac{R_L}{R_i + R_L}$$

Examples of internal resistances

- AA alkaline cell: $V_0 = 1.5\,\text{V}$, $R_i \approx 0.1\,\Omega \ldots 0.3\,\Omega$ (rises as the cell discharges)
- $9\,\text{V}$ block battery: $R_i \approx 1\,\Omega \ldots 2\,\Omega$
- laboratory power supply in voltage mode: some $\text{m}\Omega$, plus the leads
- voltage divider: $R_i = R_1 \parallel R_2$ (chapter Basics)
- logic output: $20\,\Omega \ldots 50\,\Omega$
- op-amp output with feedback: well below $1\,\Omega$ at low frequencies

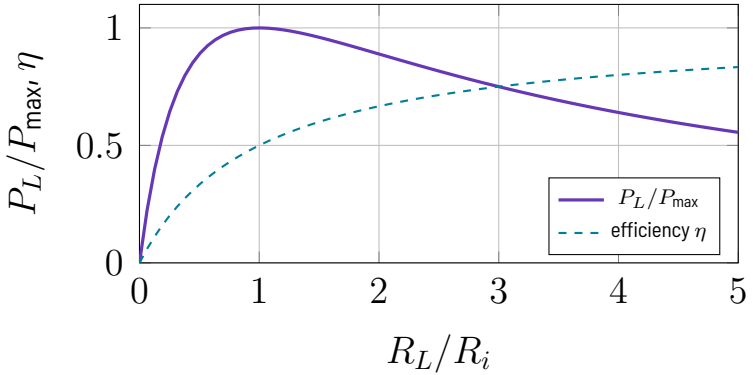
Load line and operating point



- example: $V_0 = 4.5\,\text{V}$, $R_i = 1.25\,\Omega$; load lines of $R_L = 10\,\Omega$ and $R_L = R_i$
- the operating point is the intersection of source line and load line

▶ Falstad: `examples/sources/real_source.txt`

Power matching



- load power $P_L = V_0^2 \, R_L/(R_i + R_L)^2$ is maximum for $R_L = R_i$: $P_{\max} = V_0^2/(4R_i)$
- then half of the power is lost in the source: efficiency $\eta = 50\,\%$
- power supplies and signal sources for measurement: $R_L \gg R_i$ (voltage matching)

Exercise 3.1: Measuring a real voltage source

A battery pack shows 4.5 V without load and 4.0 V with a load of $10\,\Omega$.

a) Calculate the load current and the internal resistance.
b) Calculate the short-circuit current.
c) Which load takes the maximum power from the battery? How large is this power and the efficiency?
d) Check (a) in Falstad and use the slider to find the maximum load power.

Solution 3.1

a) $I_L = 4.0\,\text{V}/10\,\Omega = 0.4\,\text{A}$, $R_i = (4.5\,\text{V} - 4.0\,\text{V})/0.4\,\text{A} = 1.25\,\Omega$.

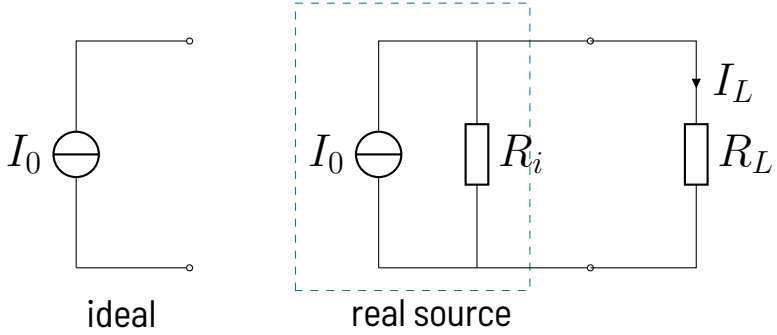
b) $I_{\text{sc}} = V_0/R_i = 4.5\,\text{V}/1.25\,\Omega = 3.6\,\text{A}$.

c) $R_L = R_i = 1.25\,\Omega$: $P_{\text{max}} = (4.5\,\text{V})^2/(4 \cdot 1.25\,\Omega) = 4.05\,\text{W}$, $V_L = 2.25\,\text{V}$, $\eta = 50\,\%$; the battery heats up with the same $4.05\,\text{W}$.

d) ▶ Falstad: `examples/sources/real_source.txt`: `V_ideal` $= 4.50\,\text{V}$, `V_term` $= 4.00\,\text{V}$, $I_L = 0.400\,\text{A}$ (hover over R_L). With the slider at $R_L = 1.25\,\Omega$ the resistor shows $4.05\,\text{W}$.

3.2 Current Sources and Source Transformation

Ideal and real current source

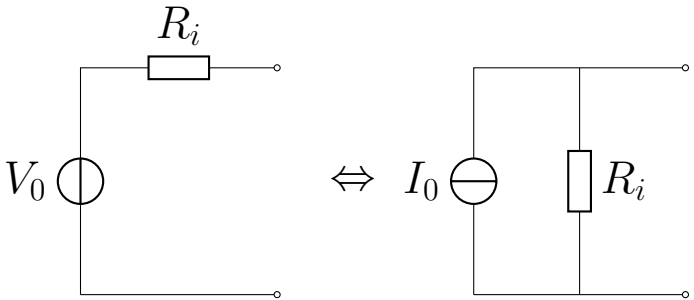


Load current of a real current source

$$I_L = I_0 - \frac{V_L}{R_i} = I_0 \frac{R_i}{R_i + R_L}$$

- a good current source has a **large** internal resistance ($R_i \gg R_L$)
- an ideal current source must never be left open: its voltage would rise without limit
- real current sources work only in a limited voltage range: the **compliance range**

Source transformation



- both have the same open-circuit voltage V_0 and the same short-circuit current $I_0 = V_0/R_i$
- therefore they behave identically for **every** load (Thevenin $\leftrightarrow$ Norton)
- inside, they are different: the power lost in R_i is not the same
- example: $10\,\text{V}$ with $10\,\text{k}\Omega \Leftrightarrow 1\,\text{mA}$ with $10\,\text{k}\Omega$; with $R_L = 4.7\,\text{k}\Omega$ both give $3.20\,\text{V}$

▶ Falstad: `examples/sources/source_transform.txt`

Exercise 3.2: Source transformation

A source consists of $10\,\text{V}$ in series with $10\,\text{k}\Omega$. It feeds $R_L = 4.7\,\text{k}\Omega$.

a) Transform it into a current source with a parallel resistor.
b) Calculate the load voltage with both models.
c) Calculate the power dissipated in the internal resistance for both models. What do you conclude?
d) Check (b) in Falstad.

Solution 3.2

a) $I_0 = 10\,\text{V}/10\,\text{k}\Omega = 1\,\text{mA}$ parallel to $10\,\text{k}\Omega$.

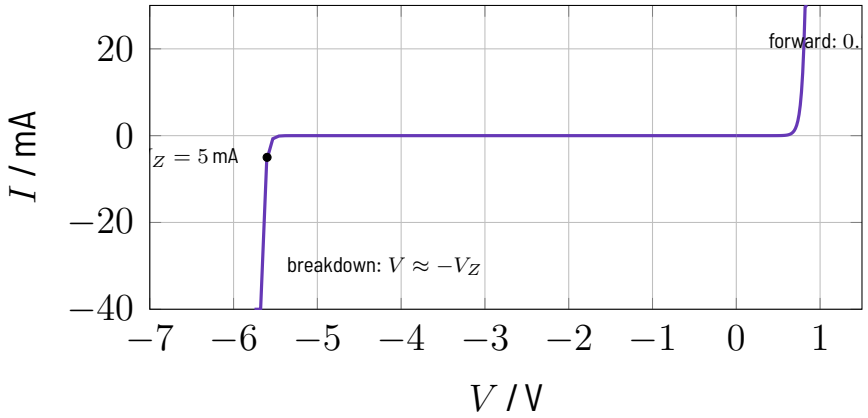
b) Voltage source: $V_L = 10\,\text{V} \cdot 4.7/14.7 = 3.20\,\text{V}$. Current source: $V_L = 1\,\text{mA} \cdot (10\,\text{k}\Omega \parallel 4.7\,\text{k}\Omega) = 1\,\text{mA} \cdot 3.20\,\text{k}\Omega = 3.20\,\text{V}$.

c) Voltage source: $I = 0.680\,\text{mA}$ flows through R_i, $P = 4.63\,\text{mW}$. Current source: R_i sees $3.20\,\text{V}$, $P = 1.02\,\text{mW}$. The models are equivalent only at the terminals, not inside.

d) ▶ Falstad: `examples/sources/source_transform.txt`: `VL_thevenin` = `VL_norton` = $3.20\,\text{V}$.

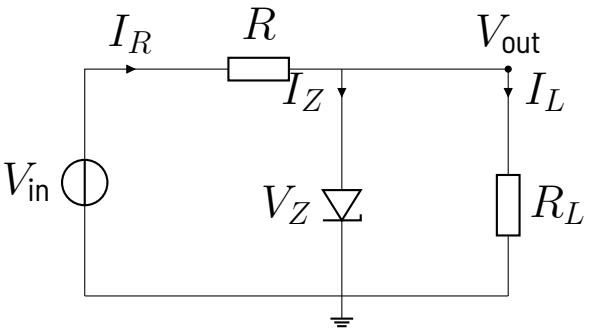
3.3 The Zener Diode Shunt Regulator

Zener diode characteristic



- breakdown voltages 2.4 V … 75 V in E24 steps (BZX55, BZX79: 0.5 W; 1N47xx: 1.3 W)
- datasheet: V_Z at a test current (often 5 mA), dynamic resistance $r_Z = \Delta V / \Delta I$

Shunt regulator



- the resistor takes the voltage difference: $I_R = (V_{\text{in}} - V_Z)/R$
- the Zener takes what the load does not need: $I_Z = I_R - I_L$
- the Zener must keep a minimum current $I_{Z,\text{min}}$ (about 5 mA, or $10\,\%$ of $I_{Z,\text{max}}$)

Designing a shunt regulator

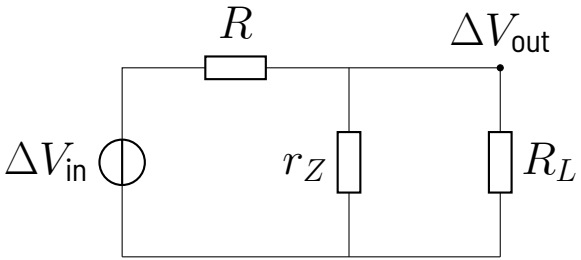
Design card: Zener shunt regulator

1. worst case for regulation: lowest V_{in}, largest load current $\Rightarrow \quad R \leq \dfrac{V_{\text{in,min}} - V_Z}{I_{L,\text{max}} + I_{Z,\text{min}}}$, choose the next **smaller** E-value
2. worst case for the Zener: highest V_{in}, no load $\Rightarrow \quad I_{Z,\text{max}} = \dfrac{V_{\text{in,max}} - V_Z}{R}$, $P_Z = V_Z \, I_{Z,\text{max}}$
3. power of the resistor: $P_R = (V_{\text{in,max}} - V_Z)^2 / R$

- example: $12\,\text{V}$, $5.6\,\text{V}$, $I_L = 0\,\text{mA} \ldots 20\,\text{mA}$: $R \leq 6.4\,\text{V}/25\,\text{mA} = 256\,\Omega \Rightarrow 220\,\Omega$
- the efficiency is poor: the full current flows all the time, even without load

▶ Falstad: `examples/sources/zener_shunt.txt`

Regulation and ripple



- for small changes the Zener is a resistor r_Z: small-signal equivalent circuit
- line regulation: $\dfrac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \dfrac{r_Z \parallel R_L}{R + r_Z \parallel R_L} \approx \dfrac{r_Z}{R}$
- load regulation: output resistance $R_{\text{out}} = R \parallel r_Z \approx r_Z$
- example $r_Z = 20\,\Omega$, $R = 220\,\Omega$: 1 V ripple at the input $\rightarrow 83$ mV at the output

Exercise 3.3: Design of a Zener shunt regulator

A load needs 5.6 V and draws 0 mA … 20 mA. The input voltage is 12 V. Use a 5.6 V Zener diode with $I_{Z,\text{min}} = 5$ mA.

a) Calculate the series resistor and choose an E12 value.
b) Calculate the maximum Zener current and the power of Zener diode and resistor. Which ratings do you choose?
c) Does the circuit still regulate at full load when the input drops to 11 V?
d) Read the output voltage without load and with $R_L = 270\,\Omega$ in Falstad and calculate I_Z.

Solution 3.3

a) $R \leq (12\,\text{V} - 5.6\,\text{V})/(20\,\text{mA} + 5\,\text{mA}) = 256\,\Omega \Rightarrow 220\,\Omega$.

b) Without load: $I_{Z,\text{max}} = 6.4\,\text{V}/220\,\Omega = 29.1\,\text{mA}$, $P_Z = 5.6\,\text{V} \cdot 29.1\,\text{mA} = 163\,\text{mW}$ $\Rightarrow 0.5\,\text{W}$ Zener (BZX79C5V6). $P_R = (6.4\,\text{V})^2/220\,\Omega = 186\,\text{mW} \Rightarrow 0.5\,\text{W}$ resistor.

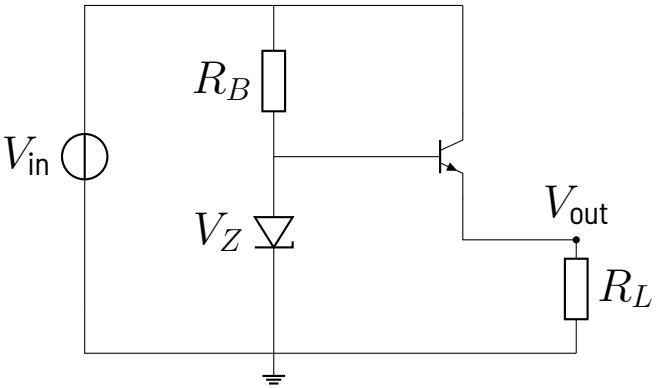
c) $I_R = (11\,\text{V} - 5.6\,\text{V})/220\,\Omega = 24.5\,\text{mA}$, $I_Z = 4.5\,\text{mA}$: just below $I_{Z,\text{min}}$; the voltage drops slightly, but the circuit still works. For a safe margin choose $200\,\Omega$ (E24).

d) ▶ Falstad: `examples/sources/zener_shunt.txt`

`Vout_open` $= 5.65\,\text{V}$ ($I_Z = I_R = 28.9\,\text{mA}$), `Vout_load` $= 5.61\,\text{V}$, $I_L = 5.61\,\text{V}/270\,\Omega = 20.8\,\text{mA}$, $I_R = (12 - 5.61)\,\text{V}/220\,\Omega = 29.0\,\text{mA}$, $I_Z = 8.3\,\text{mA}$.

3.4 Series Regulators

Emitter follower with Zener diode



Emitter follower regulator

$$V_{\text{out}} = V_Z - V_{BE} \approx V_Z - 0.65\,\text{V} \qquad I_B = \frac{I_L}{\beta + 1}$$

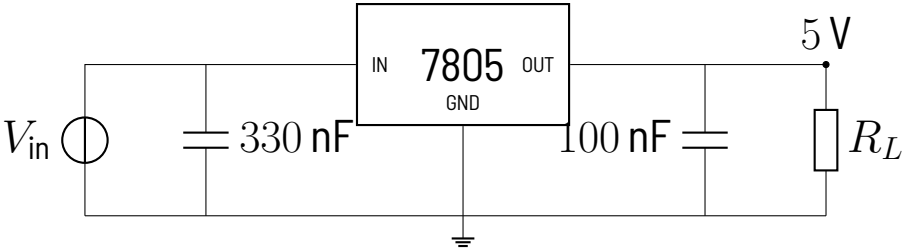
- the Zener sees only I_B: load changes hardly affect V_Z

▶ Falstad: `examples/sources/series_regulator.txt`

Properties of the emitter follower regulator

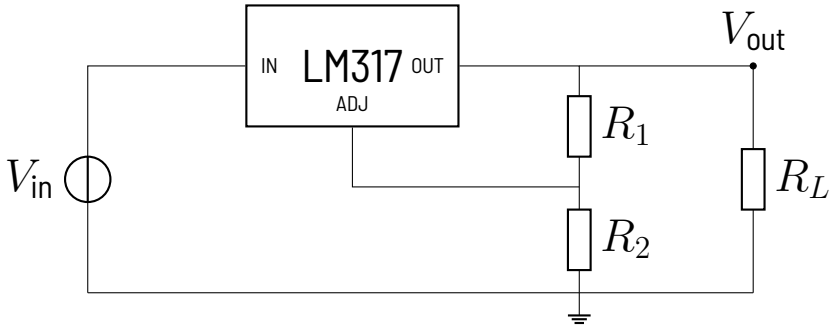
- output resistance $\approx r_e + (R_B \parallel r_Z)/(\beta + 1)$ with $r_e = V_T/I_E$ ($0.25\,\Omega$ at 100 mA): much smaller than with the Zener alone
- V_{BE} depends on current and temperature (-2 mV/K): V_{out} is only accurate to about 0.1 V
- transistor power: $P_Q = (V_{\text{in}} - V_{\text{out}}) \cdot I_L$ (heat sink!)
- design of R_B: $\dfrac{V_{\text{in,min}} - V_Z}{R_B} \geq I_{B,\text{max}} + I_{Z,\text{min}}$
- no short-circuit protection: a short at the output destroys the transistor

Integrated fixed regulator 78xx



- 7805, 7812, ...: fixed output voltage, 1 A (TO-220), 0.1 A (78L05); 79xx for negative voltages
- dropout about 2 V: $V_{\text{in}} \geq V_{\text{out}} + 2$ V; LDO regulators: 0.1 V ... 0.5 V
- internal current limit and thermal shutdown; capacitors close to the pins for stability

Adjustable regulator LM317



LM317

$$V_{\text{out}} = 1.25\,\text{V}\left(1 + \frac{R_2}{R_1}\right) + I_{\text{ADJ}}\,R_2 \qquad I_{\text{ADJ}} \approx 50\,\mu\text{A}$$

- the regulator keeps $1.25\,\text{V}$ between OUT and ADJ; $R_1 = 240\,\Omega$ sets $5.2\,\text{mA}$, enough for the minimum load

Exercise 3.4: Emitter follower regulator

A Zener diode $5.6\,\text{V}$ with $R_B = 1\,\text{k}\Omega$ drives the base of an npn transistor ($\beta = 100$) from $V_{\text{in}} = 12\,\text{V}$. The load is $R_L = 47\,\Omega$.

a) Estimate V_{out} and the load current with $V_{BE} = 0.7\,\text{V}$ (high current).

b) Calculate base current and Zener current.

c) Calculate the power of the transistor. Is a TO-92 transistor ($0.6\,\text{W}$) sufficient?

d) Which series resistor would a Zener shunt regulator need for the same load? Compare the idle current.

e) Compare with Falstad for $R_L = 47\,\Omega$ and $470\,\Omega$.

Solution 3.4

a) $V_{\text{out}} \approx 5.6\,\text{V} - 0.7\,\text{V} = 4.9\,\text{V}$, $I_L = 4.9\,\text{V}/47\,\Omega = 104\,\text{mA}$.

b) $I_B = 104\,\text{mA}/101 = 1.03\,\text{mA}$; $I_{R_B} = 6.4\,\text{V}/1\,\text{k}\Omega = 6.4\,\text{mA}$, $I_Z = 5.4\,\text{mA}$.

c) $P_Q = (12\,\text{V} - 4.9\,\text{V}) \cdot 104\,\text{mA} = 0.74\,\text{W}$: too much for TO-92; use a BD139 (TO-126) with a small heat sink.

d) $R \leq 6.4\,\text{V}/(104\,\text{mA} + 5\,\text{mA}) = 59\,\Omega \Rightarrow 56\,\Omega$. Without load the Zener takes $114\,\text{mA}$ ($0.64\,\text{W}$); the series regulator only draws $6.4\,\text{mA}$.

e) ▶ Falstad: `examples/sources/series_regulator.txt`: $R_L = 47\,\Omega$: $V_{\text{out}} = 4.89\,\text{V}$, $I_L = 104\,\text{mA}$, $V_Z = 5.60\,\text{V}$; $R_L = 470\,\Omega$: $V_{\text{out}} = 4.95\,\text{V}$, $V_Z = 5.61\,\text{V}$.

Exercise 3.5: Adjustable regulator LM317

a) Design an LM317 regulator for $5\,\text{V}$ with $R_1 = 240\,\Omega$. Choose R_2 from E24 and calculate the actual output voltage (with $I_{\text{ADJ}} = 50\,\mu\text{A}$).

b) Which input voltage is needed at least (dropout $2\,\text{V}$)? What is the power of the LM317 at $12\,\text{V}$ input and $300\,\text{mA}$?

c) The LM317 is used as a current source for an LED string with $20\,\text{mA}$. Calculate the resistor.

Solution 3.5

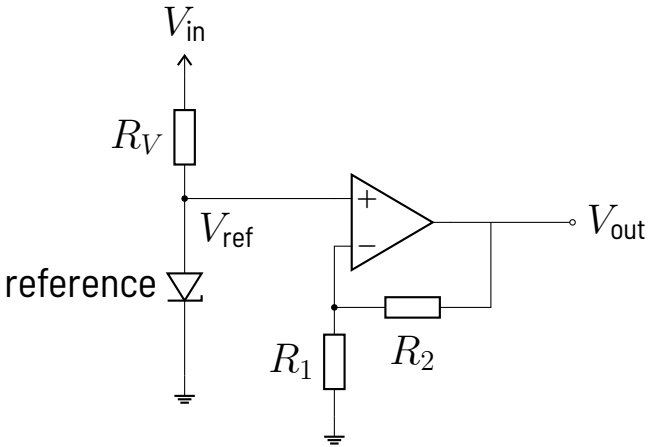
a) $R_2 = R_1\,(5/1.25 - 1) = 3\,R_1 = 720\,\Omega \Rightarrow$ E24: $750\,\Omega$. $V_{\text{out}} = 1.25\,\text{V}\,(1 + 750/240) + 50\,\mu\text{A} \cdot 750\,\Omega = 5.16\,\text{V} + 0.04\,\text{V} = 5.20\,\text{V}$. (With E96: $R_2 = 715\,\Omega$ gives $4.97\,\text{V}$.)

b) $V_{\text{in}} \geq 7.2\,\text{V}$. $P = (12\,\text{V} - 5.2\,\text{V}) \cdot 0.3\,\text{A} = 2.0\,\text{W}$: TO-220 with heat sink.

c) $R = 1.25\,\text{V}/20\,\text{mA} = 62.5\,\Omega \Rightarrow$ E24: $62\,\Omega$ ($20.2\,\text{mA}$); the LM317 then needs $V_{\text{in}} \geq V_{\text{LEDs}} + 1.25\,\text{V} + 2\,\text{V}$.

3.5 Op-Amp Voltage Sources

Reference and non-inverting amplifier



Op-amp voltage source

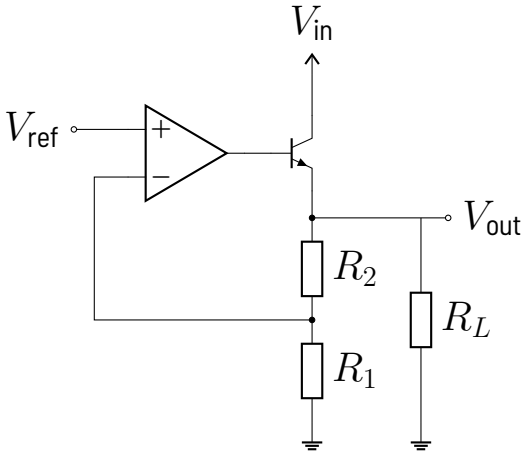
$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_2}{R_1}\right) \qquad R_{\text{out}} \approx \frac{r_o}{1 + A\beta}, \; \beta = \frac{R_1}{R_1 + R_2}$$

- the reference is not loaded; V_{out} is as accurate as V_{ref} and the ratio R_2/R_1

Output resistance and output current

- example: $r_o = 50\,\Omega$, $A = 10^5$, $R_1 = R_2$: $R_{\text{out}} = 50\,\Omega/(1+0.5\cdot 10^5) = 1\,\text{m}\Omega$
  - 10 mA more load current changes V_{out} by only 10 µV
- but: the op-amp itself delivers only some 10 mA
- and: V_{out} must stay inside the output swing of the op-amp

More current: transistor inside the loop



- npn emitter follower as **booster**: the op-amp delivers only $I_B = I_L/(\beta + 1)$
- V_{BE} lies inside the loop: the op-amp output moves to $V_{\text{out}} + V_{BE}$, V_{out} stays exact
- transistor power $(V_{\text{in}} - V_{\text{out}}) \cdot I_L$; the op-amp output must reach $V_{\text{out}} + V_{BE}$

Example: $5.00\,\text{V}$ from $12\,\text{V}$, up to $100\,\text{mA}$

- reference LM4040-2.5 ($2.500\,\text{V}$, $0.1\,\%$) with $R_V = 4.7\,\text{k}\Omega$: $2.0\,\text{mA}$
- $G = 2$: $R_1 = R_2 = 10\,\text{k}\Omega$ ($0.1\,\%$) $\Rightarrow V_{\text{out}} = 5.00\,\text{V}$
- booster BD139 ($\beta \approx 100$): op-amp output $5.7\,\text{V}$ and $1\,\text{mA}$ at $100\,\text{mA}$ load

	$R_L = 470\,\Omega$	$R_L = 47\,\Omega$
emitter follower with Zener (section 3.4)	$4.95\,\text{V}$	$4.89\,\text{V}$
op-amp and booster	$5.000\,\text{V}$	$5.000\,\text{V}$

▶ Falstad: `examples/sources/opamp_voltage_source.txt`

Exercise 3.6: Op-amp voltage source for $3.3\,\mathrm{V}$

A $3.3\,\mathrm{V}$ supply for a sensor board is built from a $2.5\,\mathrm{V}$ reference, an op-amp and an npn booster ($\beta = 100$, $V_{BE} = 0.75\,\mathrm{V}$). The input voltage is $12\,\mathrm{V}$, the load current up to $200\,\mathrm{mA}$.

a) Choose R_2 for $R_1 = 10\,\mathrm{k\Omega}$ from E24 and from E96. Calculate the output voltages.

b) Which voltage and current must the op-amp output deliver at full load?

c) Calculate the power of the transistor. Compare with the power delivered to the load.

d) In the Falstad example, turn the R_L slider from $1\,\mathrm{k\Omega}$ to $20\,\Omega$. What happens to the output voltage, and what would limit the output current in a real circuit?

Solution 3.6

a) $R_2 = R_1\,(3.3/2.5 - 1) = 3.2\,\text{k}\Omega$. E24: $3.3\,\text{k}\Omega$, $V_\text{out} = 2.5\,\text{V} \cdot 1.33 = 3.325\,\text{V}$ ($0.8\,\%$); E96: $3.24\,\text{k}\Omega$, $V_\text{out} = 3.310\,\text{V}$ ($0.3\,\%$).

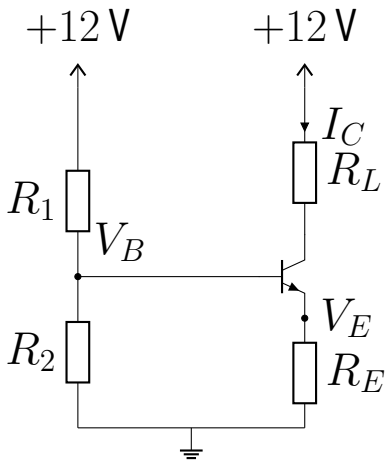
b) $V_\text{out} + V_{BE} \approx 4.06\,\text{V}$; $I_B = 200\,\text{mA}/101 = 1.98\,\text{mA}$ (the divider takes only $0.25\,\text{mA}$ more from the emitter).

c) $P_Q = (12\,\text{V} - 3.31\,\text{V}) \cdot 0.2\,\text{A} = 1.74\,\text{W}$, the load gets only $0.66\,\text{W}$: efficiency $28\,\%$. A heat sink is needed; a switching regulator would be the better choice for this current.

d) ▶ Falstad: `examples/sources/opamp_voltage_source.txt`: `Vout_100mA` stays at $5.000\,\text{V}$ over the whole slider range ($250\,\text{mA}$ at $20\,\Omega$). The simulation has no current limit; in reality the transistor power ($1.75\,\text{W}$), its safe operating area and the op-amp output current (I_L/β) limit the current.

3.6 Transistor Current Sources

BJT current source



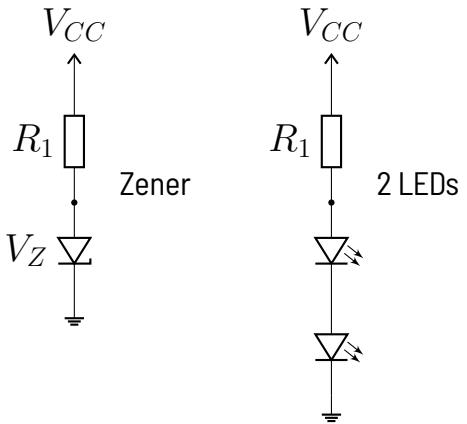
BJT current source

$$I_C \approx I_E = \frac{V_B - V_{BE}}{R_E} \approx \frac{V_B - 0.65\,\text{V}}{R_E}$$

- the emitter follows the base: $V_E = V_B - V_{BE}$ is fixed, so the emitter current is fixed
- the load sits in the collector and may vary within the compliance range

Compliance range and base voltage

- the transistor must stay active: $V_{CE} \geq$ about $0.3\,\mathrm{V}$, i.e. $V_C \geq V_E + 0.3\,\mathrm{V}$
- compliance: $R_{L,\mathrm{max}} = \dfrac{V_{CC} - V_E - 0.3\,\mathrm{V}}{I_C}$; a larger load saturates the transistor and the current drops
- stiff base voltage: divider current $\geq 10\,I_B$; base current error $I_B \cdot (R_1 \parallel R_2)$
- temperature: V_{BE} drops by $2\,\mathrm{mV/K}$; a large V_E ($1\,\mathrm{V}$ and more) makes the current stable

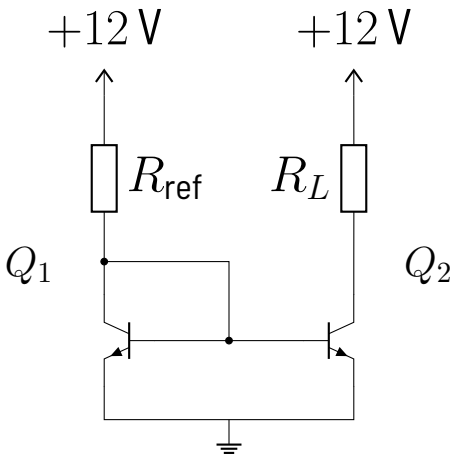


Example: 1 mA current source

- $V_{CC} = 12\,\text{V}$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$: $V_B = 2.16\,\text{V}$ (unloaded)
- $R_E = 1.5\,\text{k}\Omega$: $I_E = (2.16\,\text{V} - 0.65\,\text{V})/1.5\,\text{k}\Omega = 1.0\,\text{mA}$
- loads $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$: the same current; $15\,\text{k}\Omega$ is outside the compliance range

▶ Falstad: `examples/sources/bjt_current_source.txt`

Current mirror



- Q_1 is connected as a diode: it sets V_{BE} for the reference current
- Q_2 has the same V_{BE} and therefore the same collector current: $I_{\text{out}} = I_{\text{ref}}\,\beta/(\beta+2)$
- $I_{\text{ref}} = (V_{CC} - V_{BE})/R_{\text{ref}}$; with $11\,\text{k}\Omega$: $1.04\,\text{mA} \rightarrow 1.02\,\text{mA}$

▶ Falstad: `examples/sources/current_mirror.txt`

Exercise 3.7: BJT current source for a resistive sensor

A resistive sensor ($R_S = 1.0\,\text{k}\Omega \ldots 1.4\,\text{k}\Omega$) shall be excited with 1 mA by a BJT current source at $V_{CC} = 12\,\text{V}$ ($V_{BE} = 0.65\,\text{V}$, $\beta = 100$).

a) The base voltage comes from $R_1 = 10\,\text{k}\Omega$, $R_2 = 2.2\,\text{k}\Omega$. Calculate V_B with and without the base current, and choose R_E from E12.

b) Calculate the voltage across the sensor at $1.0\,\text{k}\Omega$ and $1.4\,\text{k}\Omega$, and the compliance range.

c) The base divider is replaced by two red LEDs in series ($V_F = 1.8\,\text{V}$ each) fed by $4.7\,\text{k}\Omega$. Recalculate R_E.

d) Read the collector currents for $1\,\text{k}\Omega$, $4.7\,\text{k}\Omega$ and $15\,\text{k}\Omega$ in Falstad.

Solution 3.7

a) Unloaded: $V_B = 12\,\text{V} \cdot 2.2/12.2 = 2.16\,\text{V}$. $I_B \approx 10\,\mu\text{A}$ through $R_1 \parallel R_2 = 1.8\,\text{k}\Omega$ lowers it by $18\,\text{mV}$: $V_B = 2.15\,\text{V}$. $R_E = (2.15\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 1.5\,\text{k}\Omega$ (E12).

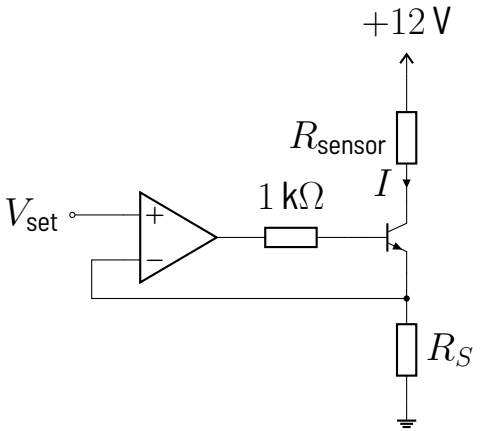
b) $V_S = 1.0\,\text{V}$ and $1.4\,\text{V}$. With $V_E = 1.5\,\text{V}$ and $V_{CE} \geq 0.3\,\text{V}$: $R_{L,\text{max}} = (12 - 1.5 - 0.3)\,\text{V}/1\,\text{mA} = 10.2\,\text{k}\Omega$; the sensor is far inside.

c) $V_B = 3.6\,\text{V}$, $R_E = (3.6\,\text{V} - 0.65\,\text{V})/1\,\text{mA} = 2.95\,\text{k}\Omega \Rightarrow$ E24 $3.0\,\text{k}\Omega$ ($0.98\,\text{mA}$). V_B no longer depends on V_{CC}.

d) ▶ Falstad: `examples/sources/bjt_current_source.txt`: $V_B = 2.15\,\text{V}$, $V_E = 1.55\,\text{V}$, $I_C = 1.023\,\text{mA}$ for $1\,\text{k}\Omega$ and $4.7\,\text{k}\Omega$ ($V_C = 7.19\,\text{V}$), $0.71\,\text{mA}$ for $15\,\text{k}\Omega$ (saturated).

3.7 Op-Amp Current Sources and Sensor Excitation

Op-amp and transistor

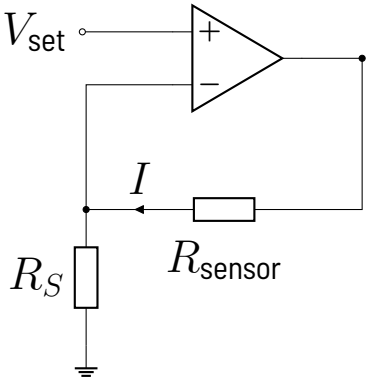


Op-amp current source

$$V_{R_S} = V_{\text{set}} \quad \Rightarrow \quad I_E = \frac{V_{\text{set}}}{R_S} \qquad I = I_C = I_E \frac{\beta}{\beta + 1}$$

- V_{BE} and its drift are inside the loop and drop out; the base current is lost ($1\,\%$ at $\beta = 100$, a MOSFET avoids it)

Floating load in the feedback path

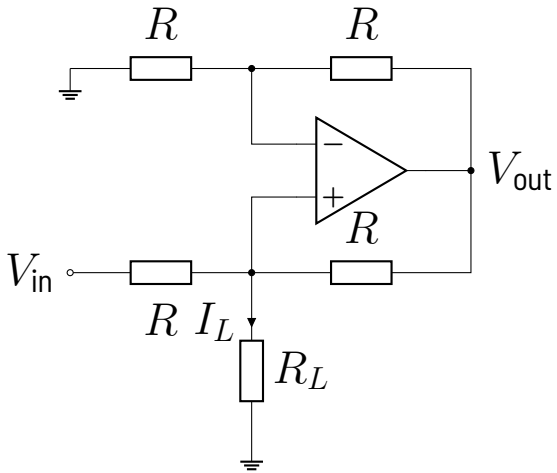


$$V_{\text{out}} = V_{\text{set}}\,(1 + R_{\text{sensor}}/R_S)$$

- non-inverting amplifier: the current V_{set}/R_S through R_S must flow through the sensor
- exact current (no base current); but neither sensor terminal is at ground
- compliance: V_{out} must stay inside the op-amp output range

▶ Falstad: `examples/sources/opamp_current_source.txt`

Howland current pump (grounded load)



- four equal resistors: $I_L = V_{\text{in}}/R$, independent of R_L (within the compliance range)
- the resistors must be matched well: a mismatch of $0.1\,\%$ limits the output resistance

Sensor excitation: why 1 mA?

- resistive temperature sensor (Pt1000 like): $R(T) = R_0\,(1 + \alpha T)$, $R_0 = 1\,\text{k}\Omega$, $\alpha = 3.85 \times 10^{-3}/\text{K}$
  - $0\,°\text{C}$: $1000\,\Omega$, $100\,°\text{C}$: $1385\,\Omega$
- with a constant current I: $V_{\text{sensor}} = I \cdot R(T)$ is linear in T: $3.85\,\text{mV/K}$ at $1\,\text{mA}$
- more current gives more signal, but **self-heating**: $\Delta T = P/G_{\text{th}}$
  - $P = I^2 R = (1\,\text{mA})^2 \cdot 1.385\,\text{k}\Omega = 1.39\,\text{mW}$; with a typical $0.4\,\text{K/mW}$: $0.55\,\text{K}$ error
- the signal offset ($1\,\text{V}$ at $0\,°\text{C}$) is removed by a **bridge**

Example: $1\,\text{mA}$ for a sensor

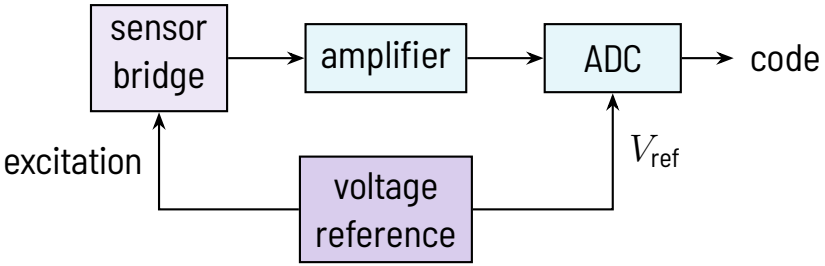
- $V_{\text{set}} = 1.00\,\text{V}$ (from a reference), $R_S = 1.00\,\text{k}\Omega$ ($0.1\,\%$): $I = 1.000\,\text{mA}$
- op-amp and transistor: sensor current $0.990\,\text{mA}$ (base current missing)
- floating sensor in the feedback path: $1.000\,\text{mA}$, $V_{\text{out}} = 1\,\text{V}\,(1 + 1.385) = 2.385\,\text{V}$ at $100\,^\circ\text{C}$
- accuracy of the current = accuracy of V_{set} and R_S

▶ Falstad: `examples/sources/opamp_current_source.txt`

4 Voltage References

4.1 Why Voltage References?

The reference sets the scale of the whole chain



- ADC: code $= 2^N \cdot V_{\text{in}}/V_{\text{ref}}$ (an ADC measures a **ratio**)
- bridge: output is a fraction of the excitation voltage
- an error of V_{ref} is a **gain error** of the complete chain

Regulator or reference?

	voltage regulator	voltage reference
purpose	supply power	define a voltage
initial accuracy	$1\,\%$ … $5\,\%$	$0.02\,\%$ … $1\,\%$
temperature coefficient	100 ppm/K … 300 ppm/K	1 ppm/K … 50 ppm/K
output current	mA … A	µA … 10 mA
noise	high	low, specified

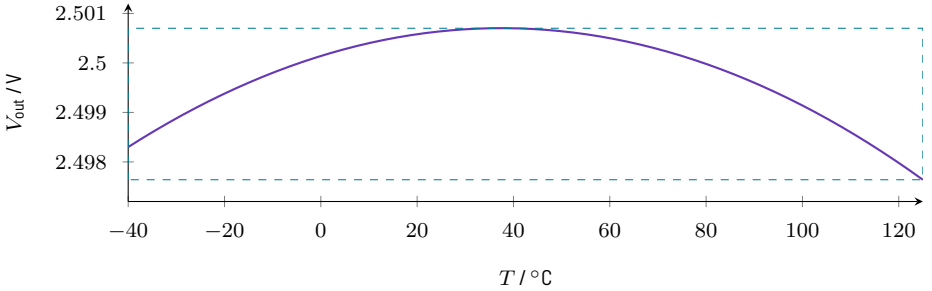
- a reference is a “voltage standard” on the board, not a power supply
- if a load needs current: buffer the reference with an op-amp

4.2 Specifications

Data sheet parameters

- **initial accuracy**: tolerance at $25\,°\text{C}$, e.g. $\pm 0.1\,\%$
- **temperature coefficient** (TC): e.g. 20 ppm/K
- **line regulation**: $\Delta V_{\text{out}}/\Delta V_{\text{in}}$, e.g. 50 µV/V
- **load regulation**: $\Delta V_{\text{out}}/\Delta I_{\text{out}}$, e.g. 100 ppm/mA
- **noise**: 0.1 Hz ... 10 Hz peak-to-peak, e.g. 10 µV pp
- **long-term drift**: e.g. 50 ppm per 1000 h
- **thermal hysteresis**: change after a temperature cycle

Temperature coefficient: the box method

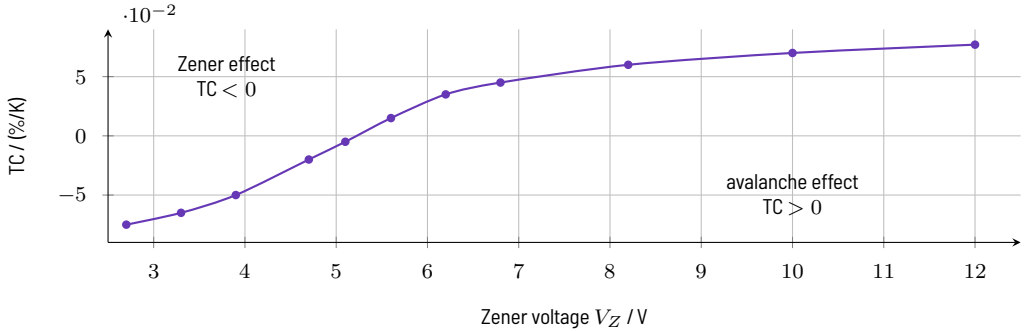


Box method

$$TC = \frac{V_{max} - V_{min}}{V_{nom}\,(T_{max} - T_{min})} \cdot 10^6 \text{ ppm}$$

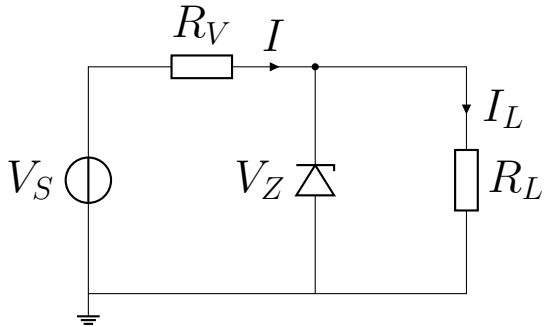
4.3 The Zener Reference

Temperature coefficient of Zener diodes



- minimum $|TC|$ near 5 V ... 6 V: preferred reference Zeners 5.6 V, 6.2 V
- 6.2 V Zener ($+2$ mV/K) in series with a forward diode (-2 mV/K): compensated reference diode (1N821 family)

Feeding a Zener through a resistor

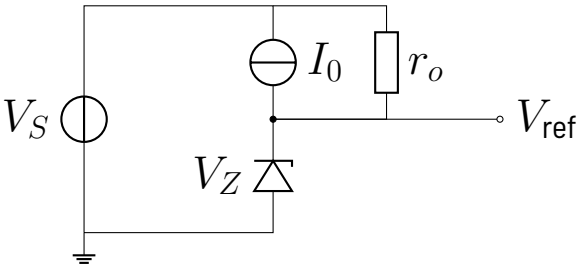


$$I_Z = I - I_L$$

- $R_V = \dfrac{V_S - V_Z}{I_Z + I_L}$, I_Z must stay above $I_{Z,\text{min}}$ for all V_S and I_L
- dynamic resistance $r_Z = \mathrm{d}V_Z/\mathrm{d}I_Z$: $5\,\Omega \ldots 50\,\Omega$
- line regulation $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{R_V + r_Z}$, load regulation $\Delta V_Z \approx -r_Z\,\Delta I_L$

▶ Falstad: `examples/references/zener_resistor.txt`

Feeding a Zener from a current source



- the Zener current no longer depends on V_S
- only the output resistance r_o of a real current source remains: $\frac{\Delta V_Z}{\Delta V_S} \approx \frac{r_Z}{r_o}$
- e.g. transistor current source from chapter 3, $r_o = 100\,\text{k}\Omega$

▶ Falstad: `examples/references/zener_current_source.txt`

Exercise 4.1: Zener reference with series resistor

A Zener diode with $V_Z = 5.6\,\text{V}$ (at $5\,\text{mA}$) is fed from $V_S = 12\,\text{V}$ through $R_V = 1.2\,\text{k}\Omega$. Use the Falstad model of the Zener, $I_Z = 5\,\text{mA} \cdot e^{(V - V_Z)/V_T}$ with $V_T = 25.9\,\text{mV}$.

a) Calculate the Zener current and the dynamic resistance r_Z.
b) How much does V_{ref} change if V_S rises to $14\,\text{V}$?
c) A load $R_L = 2.2\,\text{k}\Omega$ is connected at $12\,\text{V}$. How much does V_{ref} drop?

Solution 4.1

a) $I_Z \approx (12\,\mathrm{V} - 5.6\,\mathrm{V})/1.2\,\mathrm{k\Omega} = 5.33\,\mathrm{mA}$, so $V_{\mathrm{ref}} = V_Z + V_T \ln(5.33/5) = 5.602\,\mathrm{V}$ and $r_Z = V_T/I_Z = 25.9\,\mathrm{mV}/5.33\,\mathrm{mA} = 4.9\,\Omega$.

b) At $14\,\mathrm{V}$: $I_Z = 7.0\,\mathrm{mA}$, $\Delta V_{\mathrm{ref}} = V_T \ln(7.0/5.33) = 7.0\,\mathrm{mV}$ (small-signal estimate with the mean $r_Z \approx 4.2\,\Omega$: $2\,\mathrm{V} \cdot 4.2/1204 = 7.0\,\mathrm{mV}$). Line regulation $3.5\,\mathrm{mV/V}$, i.e. $0.06\,\%/\mathrm{V}$.

c) The load takes $I_L = 5.6\,\mathrm{V}/2.2\,\mathrm{k\Omega} = 2.55\,\mathrm{mA}$, the Zener keeps $I_Z = 5.33 - 2.55\,\mathrm{mA} = 2.79\,\mathrm{mA}$: $\Delta V_{\mathrm{ref}} = V_T \ln(2.79/5.33) = -16.6\,\mathrm{mV}$.

▶ Falstad: `examples/references/zener_resistor.txt`: the labels show `Vz12` $= 5.602\,\mathrm{V}$, `Vz14` $= 5.609\,\mathrm{V}$ ($+7.0\,\mathrm{mV}$) and `Vz12L` $= 5.585\,\mathrm{V}$ ($-16.6\,\mathrm{mV}$).

Exercise 4.2: Zener with current-source feed

The Zener of the previous exercise is fed from a current source of 5 mA with an output resistance $r_o = 100\,\text{k}\Omega$. How much does V_{ref} change when V_S rises from 12 V to 14 V? Compare with the series resistor.

Solution 4.2

a) The current through r_o changes by $2\,\text{V}/100\,\text{k}\Omega = 20\,\mu\text{A}$, the Zener current from $5.064\,\text{mA}$ to $5.084\,\text{mA}$. $\Delta V_{\text{ref}} = V_T \ln(5.084/5.064) = 0.10\,\text{mV}$, 70 times less than with $R_V = 1.2\,\text{k}\Omega$ ($7.0\,\text{mV}$).

▶ Falstad: `examples/references/zener_current_source.txt`: at $12\,\text{V}$ `VzI` $= 5.600\,\text{V}$; moving the slider to $14\,\text{V}$ changes `VzR` by $7.0\,\text{mV}$ and `VzI` by only $0.1\,\text{mV}$.

4.4 The Bandgap Reference

Two voltages with opposite temperature coefficients

- base-emitter voltage: $V_{BE} = V_T \ln(I_C/I_S)$ with $V_T = kT/q = 25.7\,\text{mV}$ at $25\,^\circ\text{C}$
  - I_S rises strongly with T: V_{BE} falls by about $2\,\text{mV/K}$ (CTAT: complementary to absolute temperature)
- two transistors with current densities in the ratio N:

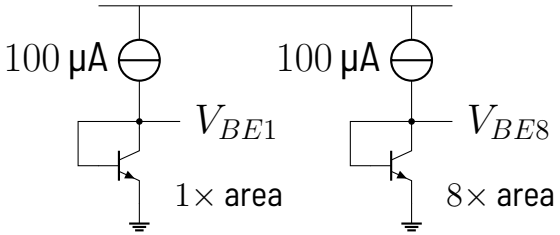
Delta-VBE

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln N = \frac{kT}{q} \ln N$$

(PTAT: proportional to absolute temperature)

  - $k/q = 86.2\,\mu\text{V/K}$; $N = 8$: $\Delta V_{BE} = 53.5\,\text{mV}$ at $25\,^\circ\text{C}$, $+0.179\,\text{mV/K}$
  - I_S cancels: ΔV_{BE} depends only on T and N

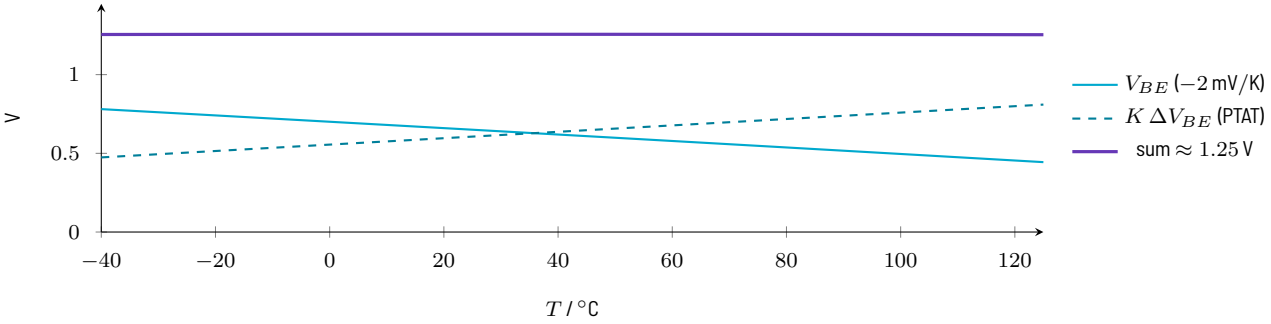
Delta-VBE in Falstad



- same current, 8 transistors in parallel: current density $1/8$
- Falstad has no temperature, but ΔV_{BE} at room temperature can be measured: $V_{BE1} = 0.536\,\text{V}$, $V_{BE1} - V_{BE8} = 53.8\,\text{mV}$
- current ratio 10: $\Delta V_{BE} = V_T \ln 10 = 59.6\,\text{mV}$ ("$60\,\text{mV}$ per decade")

▶ Falstad: `examples/references/delta_vbe.txt`

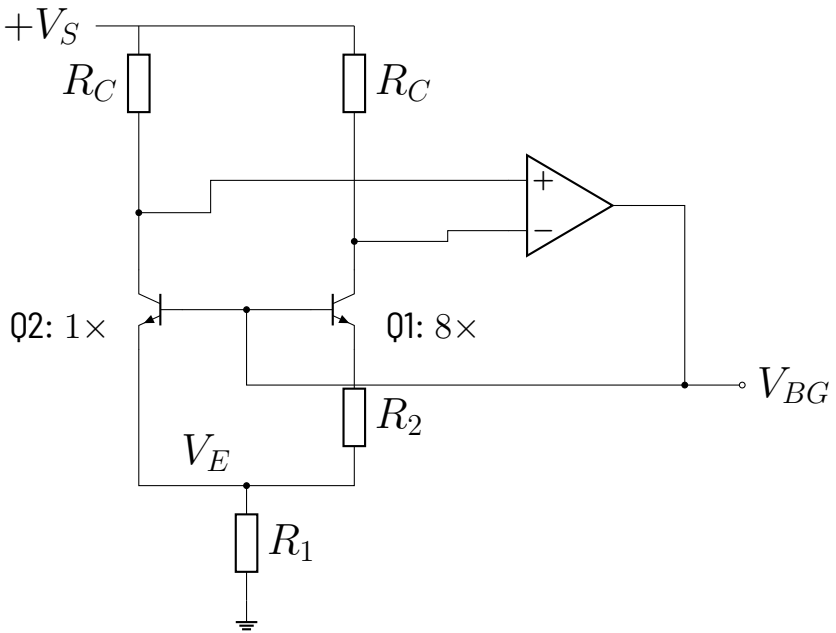
The bandgap principle



Bandgap reference

$V_{BG} = V_{BE} + K \cdot V_T \ln N \approx 1.25\,\text{V}$, K chosen so that $\mathrm{d}V_{BG}/\mathrm{d}T = 0$ at room temperature

The Brokaw cell



Brokaw cell: how it works

- the op-amp forces equal collector currents I (R_C equal, inputs at the collectors)
- current density ratio $N = 8$: $\Delta V_{BE} = V_T \ln 8$ appears across R_2
  - $I = V_T \ln 8 / R_2$ (PTAT current)
- R_1 carries both emitter currents: $V_E = 2IR_1 = 2\,\dfrac{R_1}{R_2}\,V_T \ln 8$
- output: $V_{BG} = V_{BE2} + 2\dfrac{R_1}{R_2}\,V_T \ln 8$ ($K = 2R_1/R_2$)
- Falstad, $R_2 = 1\,\text{k}\Omega$, $R_1 = 5.6\,\text{k}\Omega$: $I = 53.8\,\mu\text{A}$, $V_E = 0.602\,\text{V}$, $V_{BE2} = 0.520\,\text{V}$, $V_{BG} = 1.122\,\text{V}$

▶ Falstad: `examples/references/brokaw_cell.txt`

Exercise 4.3: Bandgap design

A Brokaw cell has an area ratio $N = 8$, $R_2 = 1\,\text{k}\Omega$ and $R_1 = 5.6\,\text{k}\Omega$. Use $V_T = 25.865\,\text{mV}$ (as Falstad).

a) Calculate ΔV_{BE}, the collector current I and V_E.
b) The Falstad transistor model gives $V_{BE2} = 0.520\,\text{V}$ at this current. Calculate V_{BG}.
c) Calculate the temperature coefficient of the PTAT part and compare it with $\text{d}V_{BE}/\text{d}T \approx -2\,\text{mV/K}$.

Solution 4.3

a) $\Delta V_{BE} = 25.865\,\text{mV} \cdot \ln 8 = 53.8\,\text{mV}$, $I = 53.8\,\text{mV}/1\,\text{k}\Omega = 53.8\,\mu\text{A}$, $V_E = 2 \cdot 53.8\,\mu\text{A} \cdot 5.6\,\text{k}\Omega = 0.602\,\text{V}$.

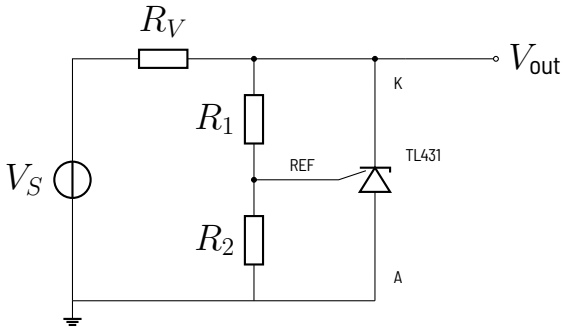
b) $V_{BG} = 0.520\,\text{V} + 0.602\,\text{V} = 1.122\,\text{V}$.

c) $\mathrm{d}V_E/\mathrm{d}T = 2\frac{R_1}{R_2}\frac{k}{q}\ln 8 = 11.2 \cdot 86.2\,\mu\text{V/K} \cdot 2.08 = +2.0\,\text{mV/K}$: it cancels the TC of V_{BE}.

▶ Falstad: `examples/references/brokaw_cell.txt`: VBG $= 1.122$ V, VE $= 0.602$ V; C1 and C2 are equal (4.47 V), the voltage across R_2 is 53.8 mV.

4.5 Shunt and Series References

The adjustable shunt reference TL431



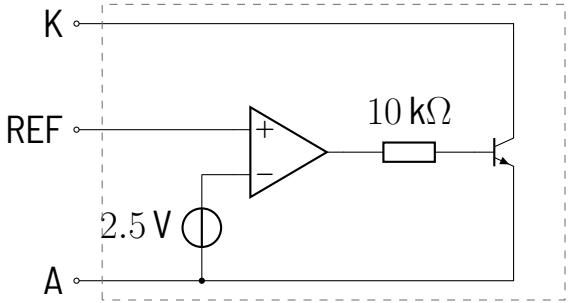
- the shunt element conducts until $V_{\text{REF}} = 2.5\,\text{V}$

TL431

$$V_{\text{out}} = V_{\text{REF}} \left(1 + \frac{R_1}{R_2}\right) + I_{\text{REF}}\, R_1, \qquad V_{\text{REF}} = 2.495\,\text{V},\ I_{\text{REF}} \approx 2\,\mu\text{A}$$

- cathode current between 1 mA and 100 mA

Inside the TL431 (behavioural model)



- error amplifier compares REF with the internal $2.5\,\text{V}$ bandgap
- V_{REF} too high $\Rightarrow$ transistor conducts more $\Rightarrow$ V_{out} falls
- $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$: $V_{\text{out}} = 2.5 \cdot 1.6 = 4.00\,\text{V}$

▶ Falstad: `examples/references/shunt_ref.txt`

Shunt or series reference?

- **shunt** (TL431, LM4040, LM385)
  - two terminals, works at any supply above V_{out}, can be used as a negative reference
  - the series resistor wastes current: R_V must deliver $I_{\text{min}} + I_{L,\text{max}}$ at the lowest supply
- **series** (REF50xx, ADR4xx, MAX6xxx)
  - three terminals, quiescent current only 0.1 mA … 1 mA
  - better for varying loads and battery operation, needs $V_{\text{in}} > V_{\text{out}} + V_{\text{dropout}}$
- both: decouple according to the data sheet (some are unstable with the wrong capacitor)

Exercise 4.4: TL431 set to 4.00 V

A TL431 ($V_{\text{REF}} = 2.495$ V, $I_{\text{REF}} = 2$ µA, $I_{K,\text{min}} = 1$ mA) shall provide 4.00 V from a supply of 6 V … 15 V.

a) Choose R_1 and R_2 from the E24 series.
b) How large is the error caused by I_{REF} and by $V_{\text{REF}} = 2.495$ V instead of 2.5 V?
c) Choose R_V such that a load of 1 mA can be supplied at 6 V.

Solution 4.4

a) $1 + R_1/R_2 = 1.6 \Rightarrow R_1/R_2 = 0.6$: $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ (both E24).

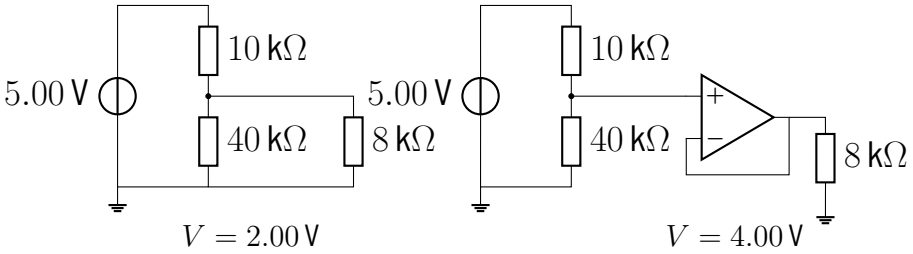
b) $I_{\text{REF}} R_1 = 2\,\mu\text{A} \cdot 12\,\text{k}\Omega = 24\,\text{mV}$ ($+0.6\,\%$); $2.495\,\text{V} \cdot 1.6 = 3.992\,\text{V}$ ($-0.2\,\%$). Together $V_{\text{out}} = 4.016\,\text{V}$: for an accurate $4.00\,\text{V}$ choose smaller divider resistors or trim.

c) At $6\,\text{V}$ R_V must carry $I_{K,\text{min}} + I_L + I_{\text{div}} = 1 + 1 + 0.125 = 2.125\,\text{mA}$: $R_V \leq (6\,\text{V} - 4\,\text{V})/2.125\,\text{mA} = 941\,\Omega$, choose $910\,\Omega$. At $15\,\text{V}$ the TL431 then carries $(15 - 4)/0.91 - 0.125 - 1 = 11\,\text{mA}$ ($44\,\text{mW}$): fine.

▶ Falstad: `examples/references/shunt_ref.txt` (ideal model with $2.5\,\text{V}$, $R_V = 1\,\text{k}\Omega$, no load): `VK` $= 4.00\,\text{V}$ at $12\,\text{V}$ and at $6\,\text{V}$, `REF` $= 2.50\,\text{V}$.

4.6 Buffering and Scaling a Reference

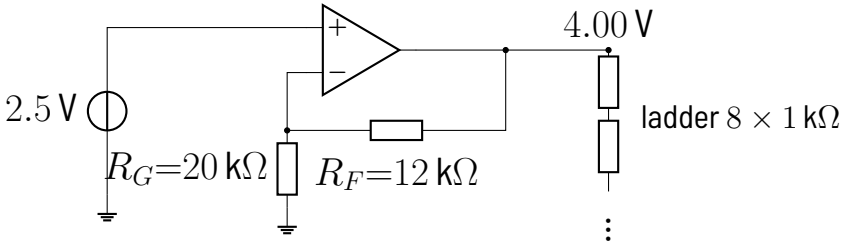
A divider is no reference



- divider $10\,\text{k}\Omega/40\,\text{k}\Omega$: 4.00 V open, but only 2.00 V with the $8\,\text{k}\Omega$ ladder
- source resistance $R_1 \parallel R_2 = 8\,\text{k}\Omega$ = load resistance
- follower: input current ≈ 0, output resistance ≈ 0: 4.00 V

▶ Falstad: `examples/references/ref_divider_loaded.txt`

Scaling with a non-inverting amplifier



- $G = 1 + R_F/R_G = 1 + 12/20 = 1.6$: $\quad 2.5\,\text{V} \cdot 1.6 = 4.00\,\text{V}$
- the op-amp delivers ladder current 0.5 mA plus 0.125 mA for $R_F + R_G$
- gain error from resistor tolerance: $\frac{\Delta G}{G} = \frac{G-1}{G}\left(\frac{\Delta R_F}{R_F} - \frac{\Delta R_G}{R_G}\right)$

▶ Falstad: `examples/references/ref_buffer_ladder.txt`

Design card: Reference for an ADC ladder

- full scale V_{FS} and LSB: LSB $= V_{\text{FS}}/2^N$ ($4.00\,\text{V}$, 3 bit: $0.5\,\text{V}$)
- ladder current $I = V_{\text{FS}}/(2^N R)$ ($8 \times 1\,\text{k}\Omega$: $0.5\,\text{mA}$)
- reference voltage V_R from a standard value, gain $G = V_{\text{FS}}/V_R$ (2.5 V, $G = 1.6$)
- non-inverting amplifier: $R_F/R_G = G - 1$, $R_F + R_G \gg$ ladder ($12\,\text{k}\Omega + 20\,\text{k}\Omega$)
- check: op-amp output current, $(G-1)/G\cdot$ resistor tolerance, $G \cdot V_{\text{offset}}$
- decouple the ladder top with $100\,\text{nF}$ ... $1000\,\text{nF}$ close to the comparators

Exercise 4.5: Reference for the 3-bit flash converter

The flash converter of the capstone needs $4.00\,\text{V}$ at the top of a ladder of $8 \times 1\,\text{k}\Omega$. A $2.5\,\text{V}$ reference is available.

a) Calculate the ladder current and the tap voltages.
b) Design a non-inverting amplifier with E24 resistors.
c) Why can the $4.00\,\text{V}$ not be taken from a divider $10\,\text{k}\Omega/40\,\text{k}\Omega$ at a $5\,\text{V}$ reference?
d) How large is the worst-case error of V_{ref} with $1\,\%$ resistors?

Solution 4.5

a) $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; taps every $0.5\,\text{V}$: $T_1 = 0.5\,\text{V}$, ..., $T_4 = 2.0\,\text{V}$, ..., $T_7 = 3.5\,\text{V}$.

b) $G = 4/2.5 = 1.6$, $R_F/R_G = 0.6$: $R_F = 12\,\text{k}\Omega$, $R_G = 20\,\text{k}\Omega$.

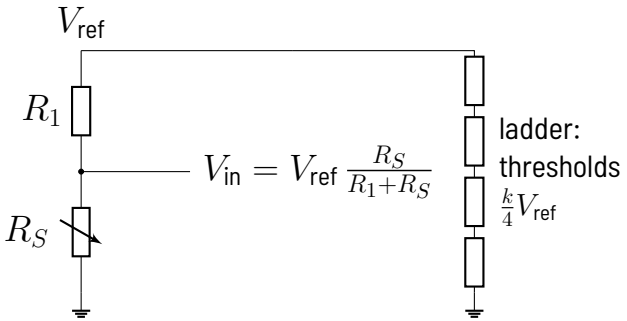
c) The divider has a source resistance of $10\,\text{k}\Omega \parallel 40\,\text{k}\Omega = 8\,\text{k}\Omega$, equal to the ladder. The loaded voltage is $5 \cdot \frac{40 \parallel 8}{10+40 \parallel 8} = 5 \cdot \frac{6.67}{16.67} = 2.00\,\text{V}$.

d) $\Delta G/G = 0.375 \cdot 2 \cdot 1\,\% = 0.75\,\%$, i.e. $\pm 30\,\text{mV}$; for the 3-bit converter (LSB $0.5\,\text{V}$) irrelevant, for a 12-bit converter 30 LSB.

▶ Falstad: `examples/references/ref_buffer_ladder.txt`: `VREF4` $= 4.00\,\text{V}$, `T4` $= 2.00\,\text{V}$, `T1` $= 0.50\,\text{V}$, ladder current $0.5\,\text{mA}$;
▶ Falstad: `examples/references/ref_divider_loaded.txt`: `Vopen` $= 4.00\,\text{V}$, `Vload` $= 2.00\,\text{V}$, `Vbuf` $= 4.00\,\text{V}$.

4.7 Ratiometric Measurement

The reference cancels



Ratiometric

$$\text{code} = 2^N \frac{V_{\text{in}}}{V_{\text{ref}}} = 2^N \frac{R_S}{R_1 + R_S}$$ independent of V_{ref}

Ratiometric versus absolute

- Falstad: two 2-bit flash converters (3 comparators), sensor divider $1\,\mathrm{k\Omega}/2.7\,\mathrm{k\Omega}$: $V_{\mathrm{in}} = 0.730\,V_{\mathrm{exc}}$
- left: sensor and ladder from V_{ref}: code 2 for $V_{\mathrm{ref}} = 3.6, 4.0, 4.4\,\mathrm{V}$
- right: sensor from a fixed $4.00\,\mathrm{V}$, ladder from V_{ref}
  - $V_{\mathrm{ref}} = 4.0\,\mathrm{V}$: $V_{\mathrm{in}} = 2.92\,\mathrm{V} < 3.0\,\mathrm{V}$: code 2
  - $V_{\mathrm{ref}} = 3.6\,\mathrm{V}$: threshold $\frac{3}{4}V_{\mathrm{ref}} = 2.7\,\mathrm{V}$: code 3 (wrong)
- ratiometric works only if both see the **same** voltage at the **same** time
  - a common connection with voltage drops, or noise on one path only, breaks it

▶ Falstad: `examples/references/ratiometric.txt`

4.8 Error Budget

Error budget of the 4.00 V reference

contribution	data	error
initial accuracy	$\pm 0.2\,\%$	$0.20\,\%$
temperature	25 ppm/K, ± 25 K	$0.06\,\%$
gain resistors	$2 \times 0.1\,\%$, factor 0.375	$0.08\,\%$
op-amp offset	1 mV $\times$ 1.6	$0.04\,\%$
long-term drift	50 ppm / 1000 h	$0.01\,\%$
worst case (sum)		$0.39\,\%$
RSS		$0.23\,\%$

- compare: 3 bit, LSB $= 12.5\,\%$ of full scale: no problem at all
- 12 bit, LSB $= 0.024\,\%$: 16 LSB worst case $\Rightarrow$ calibrate or measure ratiometrically

Exercise 4.6: Ratiometric measurement and error budget

A sensor divider $R_1 = 1\,\text{k}\Omega$, $R_S = 2.7\,\text{k}\Omega$ is read by a 2-bit flash converter with a ladder of $4 \times 1\,\text{k}\Omega$.

a) The divider and the ladder are fed from the same V_{ref}. Which code results for $V_{\text{ref}} = 4.0\,\text{V}$ and $3.6\,\text{V}$?

b) The divider is fed from a fixed $4.00\,\text{V}$, the ladder from V_{ref}. Repeat a).

c) A 12-bit converter uses a reference with $\pm 0.2\,\%$ initial accuracy and $25\,\text{ppm/K}$. How many LSB can the result be off at $50\,^\circ\text{C}$ after calibration at $25\,^\circ\text{C}$?

Solution 4.6

a) $V_{\text{in}}/V_{\text{ref}} = 2.7/3.7 = 0.730$; the thresholds are at 0.25, 0.5 and $0.75\,V_{\text{ref}}$. $0.5 < 0.730 < 0.75$: code 2 for every V_{ref}.

b) $V_{\text{in}} = 2.92\,\text{V}$. $V_{\text{ref}} = 4.0\,\text{V}$: thresholds 1, 2, 3 V: code 2. $V_{\text{ref}} = 3.6\,\text{V}$: thresholds 0.9, 1.8, $2.7\,\text{V}$: code 3. A $10\,\%$ reference error changes the result by one code.

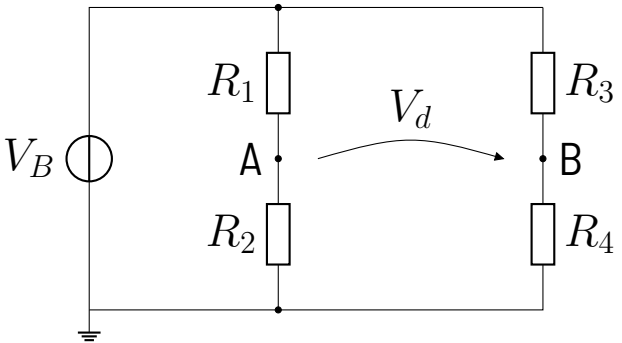
c) After calibration only the drift remains: $25\,\text{ppm/K} \cdot 25\,\text{K} = 625\,\text{ppm} = 0.000625 \cdot 4096 = 2.6$ LSB. Without calibration add $0.002 \cdot 4096 = 8.2$ LSB.

▶ Falstad: `examples/references/ratiometric.txt`: with the `VREF` slider at 4.0 V both converters show D = E = code 2 (`VIN1` $= 2.92$ V); at 3.6 V the right converter switches `E3` on (code 3), the left stays at code 2.

5 Bridge Circuits

5.1 The Wheatstone Bridge

Two voltage dividers



Bridge output

$$V_d = V_A - V_B = V_B \left(\frac{R_2}{R_1 + R_2} - \frac{R_4}{R_3 + R_4} \right)$$

Balance condition

- the bridge is **balanced** if $V_d = 0$:

Balance condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4} \quad \Leftrightarrow \quad R_1 R_4 = R_2 R_3$$

- the balance does **not** depend on V_B: supply noise and drift do not appear at the output
- null method: adjust a known resistor until $V_d = 0$, then $R_2 = R_1 R_4 / R_3$
- deflection method: fixed resistors, balanced at the reference point (e.g. $0\,°\text{C}$), measure V_d

5.2 Output Voltage and Linearity

Quarter bridge with equal resistors

- $R_1 = R_3 = R_4 = R_0$, sensor $R_2 = R_0(1 + x)$:

$$V_d = V_B \left(\frac{1+x}{2+x} - \frac{1}{2} \right) = \frac{V_B}{4} \cdot \frac{x}{1 + x/2}$$

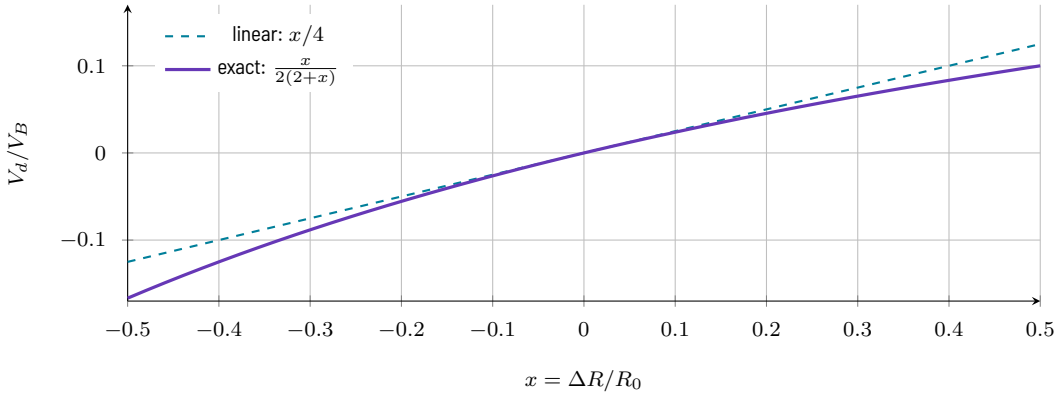
Quarter bridge, small signal

$$V_d \approx \frac{V_B}{4} \frac{\Delta R}{R_0} \qquad \text{relative error} \approx -\tfrac{x}{2}$$

- example: $V_B = 5\,\text{V}$, $x = 1\,\%$: $V_d = 12.44\,\text{mV}$ (estimate $12.5\,\text{mV}$)
- $x = 50\,\%$: $V_d = 0.500\,\text{V}$ instead of $0.625\,\text{V}$

▶ Falstad: `examples/bridges/wheatstone.txt`

The bridge is not linear



- positive x: output too small, negative x: too large
- strain gauges ($x < 0.5\,\%$): negligible; Pt100 over 100 K ($x = 38.5\,\%$): $16\,\%$

Unequal arms: trade sensitivity for linearity

- $R_1 = R_3 = m\,R_0$, $R_4 = R_0$, sensor $R_0(1+x)$:

$$V_d = V_B \frac{m\,x}{(m+1)(m+1+x)} \approx V_B \frac{m}{(m+1)^2}\,x$$

- sensitivity $m/(m+1)^2$: maximum $1/4$ at $m = 1$; $m = 39$: 0.024
- relative nonlinearity $\approx -x/(m+1)$: $m = 39$ makes it 20 times smaller
- large m also limits the sensor current: $I = V_B/(R_1 + R_{\text{sensor}})$ (self-heating!)

Exercise 5.1: Wheatstone bridge

A bridge with $R_1 = R_3 = R_4 = 1\,\text{k}\Omega$ is supplied with $V_B = 5\,\text{V}$. The resistor $R_2 = R_x$ is the sensor.

a) Calculate V_d exactly and with the small-signal formula for $R_x = 1010\,\Omega$.
b) Repeat for $R_x = 1500\,\Omega$. How large is the error of the formula?
c) Check both values in Falstad.

Solution 5.1

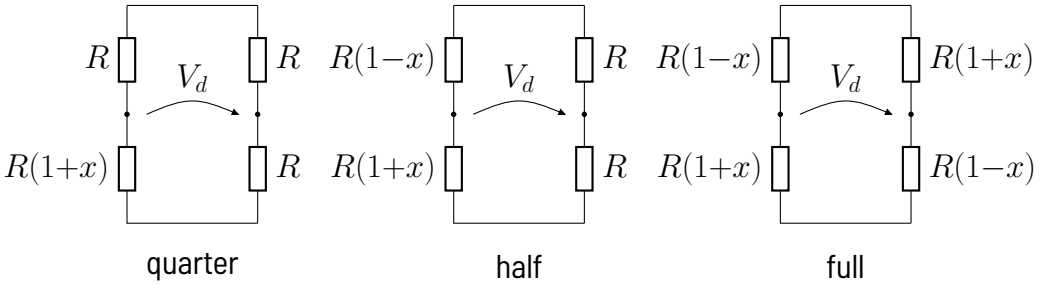
a) $V_d = 5\,(1010/2010 - 0.5) = 12.44\,\text{mV}$; formula: $\frac{5}{4} \cdot 0.01 = 12.5\,\text{mV}$ ($0.5\,\%$ too high, $\approx x/2$).

b) $V_d = 5\,(1500/2500 - 0.5) = 0.500\,\text{V}$; formula: $\frac{5}{4} \cdot 0.5 = 0.625\,\text{V}$: $25\,\%$ too high.

c) ▶ Falstad: `examples/bridges/wheatstone.txt`: the voltmeter shows $12.44\,\text{mV}$; with the slider at $1500\,\Omega$ it shows $0.500\,\text{V}$.

5.3 Half and Full Bridges: Strain Gauges

Quarter, half and full bridge



- quarter: $V_d \approx \frac{V_B}{4} x$ half: $V_d = \frac{V_B}{2} x$ full: $V_d = V_B\, x$ $(x = k\,\varepsilon)$
- half and full bridge are exactly linear: the leg current stays constant
- temperature changes all gauges equally: compensated

▶ Falstad: `examples/bridges/bridge_types.txt`

Exercise 5.2: Strain gauge bridge

Strain gauges with $R = 350\,\Omega$ and $k = 2$ are glued to a beam. The strain is $\varepsilon = 1000\,\mu\text{m/m}$, the bridge supply $V_B = 5\,\text{V}$.

a) Calculate ΔR.
b) Calculate the output of a quarter, a half and a full bridge.
c) Which gain is needed to get $2\,\text{V}$ at this strain from the full bridge?

Solution 5.2

a) $x = k\varepsilon = 2 \cdot 10^{-3}$, $\Delta R = 0.002 \cdot 350\,\Omega = 0.7\,\Omega$.

b) Quarter: $V_d = 5\,(350.7/700.7 - 0.5) = 2.4975$ mV ($\approx \frac{5}{4} \cdot 0.002 = 2.5$ mV); half: $\frac{5}{2} \cdot 0.002 = 5.00$ mV; full: $5 \cdot 0.002 = 10.0$ mV.

c) $G = 2\,\text{V}/10\,\text{mV} = 200$, with an instrumentation amplifier (chapter 6).

▶ Falstad: `examples/bridges/bridge_types.txt`:
the three voltmeters show 2.50 mV, 5.00 mV and 10.0 mV.

5.4 Resistive Temperature Sensors

Platinum sensors Pt100 and Pt1000

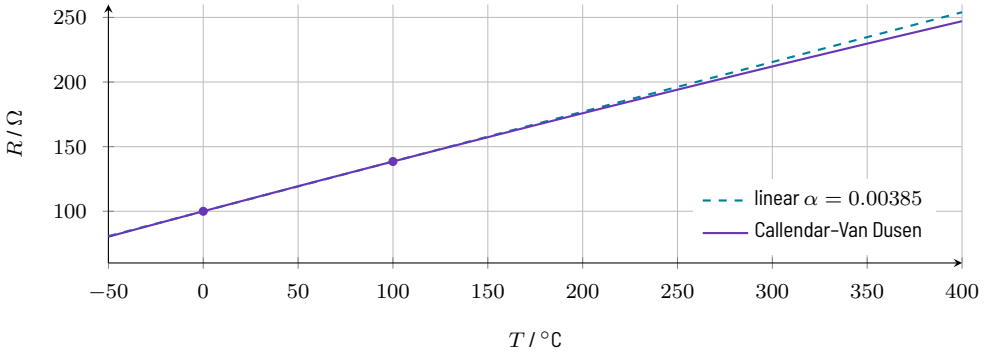
- linear model ($0\,°\text{C} \ldots 100\,°\text{C}$):

Pt100 / Pt1000

$$R(T) = R_0\,(1 + \alpha T), \quad R_0 = 100\,\Omega \,/\, 1000\,\Omega, \quad \alpha = 3.85 \times 10^{-3}/\text{K}$$

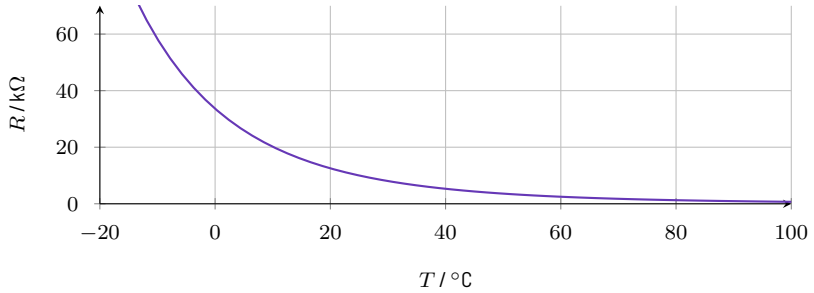
- Pt100: $100\,\Omega$ at $0\,°\text{C}$, $138.5\,\Omega$ at $100\,°\text{C}$, $0.385\,\Omega/\text{K}$
- Callendar–Van Dusen ($T \geq 0\,°\text{C}$): $R(T) = R_0(1 + AT + BT^2)$
  - $A = 3.9083 \times 10^{-3}/\text{K}$, $B = -5.775 \times 10^{-7}/\text{K}^2$
- tolerance class B: $\pm(0.3\,\text{K} + 0.005\,|T|)$, class A: $\pm(0.15\,\text{K} + 0.002\,|T|)$

Pt100 characteristic



- $0\,°\mathrm{C}$ … $100\,°\mathrm{C}$: linear model and standard agree at both ends, $0.4\,\mathrm{K}$ apart at $50\,°\mathrm{C}$
- $400\,°\mathrm{C}$: $254\,\Omega$ (linear) vs. $247\,\Omega$: $18\,\mathrm{K}$ error

NTC thermistors



- $R(T) = R_{25} \exp\left(B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)\right)$, e.g. $R_{25} = 10\,\mathrm{k\Omega}$, $B = 3950\,\mathrm{K}$ (T in kelvin)
- sensitivity at $25\,°\mathrm{C}$: $-B/T^2 = -4.4\,\%/\mathrm{K}$ (Pt: $+0.39\,\%/\mathrm{K}$)
- strongly nonlinear: linearise with a series resistor, or a table in software

Self-heating: why the sensor current is limited

- the measuring current heats the sensor: $P = I^2 R$
- self-heating coefficient E: e.g. $0.1\,\mathrm{K/mW}$ (thin film, in water) … $1\,\mathrm{K/mW}$ (in still air)

Self-heating error

$$\Delta T = E \cdot I^2 R$$

- Pt100, 1 mA: $P = 0.1$ mW, $\Delta T = 0.01\,\mathrm{K} \ldots 0.1\,\mathrm{K}$
- Pt100, 10 mA: $P = 10$ mW, $\Delta T = 1\,\mathrm{K} \ldots 10\,\mathrm{K}$
- rule: Pt100 ≤ 1 mA, Pt1000 ≤ 0.3 mA (capstone: $I \leq 1$ mA)

Exercise 5.3: Pt100 values

a) Calculate the resistance of a Pt100 at $25\,°\mathrm{C}$ and $37\,°\mathrm{C}$ with the linear model.

b) Calculate $R(100\,°\mathrm{C})$ and $R(200\,°\mathrm{C})$ with the Callendar–Van Dusen equation. Which temperature does the linear model assign to these resistances?

c) A Pt100 is read with 5 mA in still air ($E = 0.5\,\mathrm{K/mW}$). How large is the self-heating error at $0\,°\mathrm{C}$?

Solution 5.3

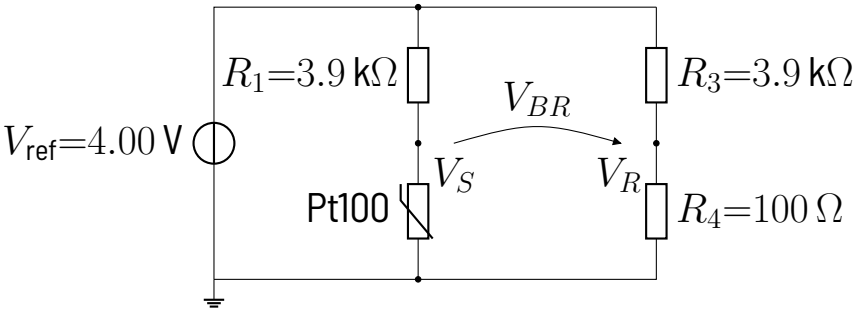
a) $R(25) = 100\,(1 + 0.00385 \cdot 25) = 109.63\,\Omega$, $R(37) = 114.25\,\Omega$.

b) $R(100) = 100\,(1 + 0.39083 - 0.005775) = 138.51\,\Omega$, linear model: $T = 0.3851/0.00385 = 100.0\,°\text{C}$. $R(200) = 100\,(1 + 0.78166 - 0.0231) = 175.86\,\Omega$ $\Rightarrow$ linear: $T = 197.0\,°\text{C}$ (3 K too low).

c) $P = (5\,\text{mA})^2 \cdot 100\,\Omega = 2.5\,\text{mW}$, $\Delta T = 1.25\,\text{K}$: four times the class B tolerance at $0\,°\text{C}$ ($0.3\,\text{K}$).

5.5 Designing the Sensor Bridge

The capstone bridge



- $R_4 = R_0 = 100\,\Omega$: balanced at $0\,°\text{C}$
- $I = V_{\text{ref}}/(R_1 + R) \leq 1\,\text{mA} \Rightarrow R_1 \geq 3.9\,\text{k}\Omega$; $R_3 = R_1$ $(m = 39)$
- $V_{BR}(T) = V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right)$

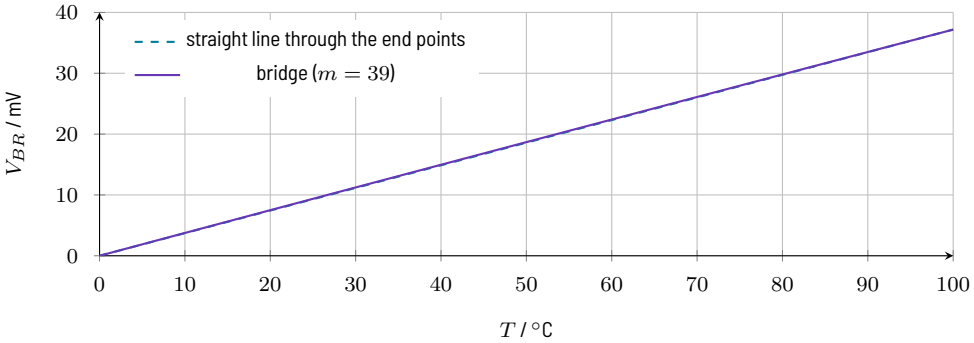
Numbers of the capstone bridge

	$0\,°C$	$50\,°C$	$100\,°C$
sensor R	$100\,\Omega$	$119.25\,\Omega$	$138.5\,\Omega$
sensor current	1.000 mA	0.995 mA	0.990 mA
V_S	100.0 mV	118.7 mV	137.2 mV
V_R	100.0 mV	100.0 mV	100.0 mV
V_{BR}	0 mV	18.68 mV	37.18 mV

- sensitivity ≈ 0.37 mV/K; nonlinearity at $50\,°C$: $+0.09$ mV ($0.25\,\%$ of full scale)
- required gain: $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- self-heating: $P = 0.1$ mW

▶ Falstad: `examples/bridges/pt100_bridge.txt`

Bridge output over temperature



- with $m = 39$ the curve is almost straight: maximum deviation $0.09\,\text{mV} \approx 0.24\,\text{K}$
- with $m = 1$ (all arms $100\,\Omega$) the deviation would be $4.4\,\text{K}$

Design card: Sensor bridge

- excitation V_B: the ADC reference (ratiometric), here $4.00\,\text{V}$
- reference arm: $R_4 = R_{\text{sensor}}(T_{\text{min}}) \Rightarrow V_d = 0$ at T_{min}
- current limit: $R_1 \geq V_B / I_{\text{max}} - R_{\text{sensor}}$, round up to E-series; $R_3 = R_1$
- full-scale output $V_d(T_{\text{max}})$ with the exact formula, gain $G = V_{\text{FS}} / V_d(T_{\text{max}})$
- check nonlinearity ($\approx x/(m+1)$), self-heating ($I^2 R \cdot E$), common-mode voltage ($\approx V_B/(m+1)$)

Exercise 5.4: Design of the capstone bridge

A Pt100 ($\alpha = 3.85 \times 10^{-3}/\mathrm{K}$) shall measure $0\,°\mathrm{C} \ldots 100\,°\mathrm{C}$. The bridge is fed from $V_{\mathrm{ref}} = 4.00\,\mathrm{V}$, the sensor current must not exceed $1\,\mathrm{mA}$.

a) Choose R_1, R_3 and R_4.
b) Calculate the bridge output at $0\,°\mathrm{C}$, $50\,°\mathrm{C}$ and $100\,°\mathrm{C}$ and the sensor currents.
c) How large is the deviation from a straight line at $50\,°\mathrm{C}$, in millivolt and in kelvin?
d) Which total amplifier gain gives $4.00\,\mathrm{V}$ at $100\,°\mathrm{C}$?

Solution 5.4

a) $R_4 = 100\,\Omega$ (balance at $0\,°C$); $R_1 \geq 4\,V/1\,mA - 100\,\Omega = 3.9\,k\Omega$ (E12), $R_3 = R_1 = 3.9\,k\Omega$.

b) $V_R = 4 \cdot 100/4000 = 100.0\,mV$. $0\,°C$: $V_{BR} = 0$, $I = 1.000\,mA$. $50\,°C$: $V_S = 4 \cdot 119.25/4019.25 = 118.68\,mV$, $V_{BR} = 18.68\,mV$. $100\,°C$: $V_S = 4 \cdot 138.5/4038.5 = 137.18\,mV$, $V_{BR} = 37.18\,mV$, $I = 0.990\,mA$.

c) Straight line: $37.18/2 = 18.59\,mV$; deviation $0.09\,mV$, with $0.372\,mV/K$: $0.24\,K$.

d) $G = 4.00\,V/37.18\,mV = 107.6$ (capstone: 10×10.76).

▶ Falstad: `examples/bridges/pt100_bridge.txt`: with the slider at $100\,\Omega$ the voltmeter shows $0\,V$ and $V_R = 100\,mV$; at $119.25\,\Omega$ $18.68\,mV$; at $138.5\,\Omega$ $37.18\,mV$.

Exercise 5.5: Alternative: Pt1000 at 5 V

A Pt1000 is used in a bridge with $V_B = 5\,\text{V}$, $R_1 = R_3 = 4.7\,\text{k}\Omega$, $R_4 = 1\,\text{k}\Omega$.

a) Calculate the sensor current and the self-heating power at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

b) Calculate the bridge output at $100\,^\circ\text{C}$ and the required gain for $4.00\,\text{V}$.

c) How large is the deviation from the straight line at $50\,^\circ\text{C}$? Compare with the Pt100 design.

Solution 5.5

a) $I = 5/5700 = 0.877\,\text{mA}$, $P = I^2 \cdot 1\,\text{k}\Omega = 0.77\,\text{mW}$; at $100\,°\text{C}$: $I = 5/6085 = 0.822\,\text{mA}$, $P = 0.94\,\text{mW}$, almost ten times the Pt100.

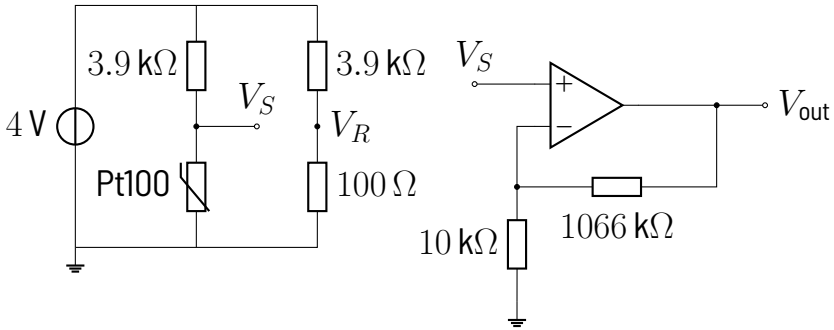
b) $V_d = 5\,(1385/6085 - 1000/5700) = 260.9\,\text{mV}$, $G = 4.00/0.2609 = 15.33$.

c) $V_d(50\,°\text{C}) = 5\,(1192.5/5892.5 - 0.17544) = 134.7\,\text{mV}$, straight line $130.4\,\text{mV}$: $4.3\,\text{mV} = 1.6\,\%$ of full scale ($m = 4.7$), almost seven times worse than the Pt100 design with $m = 39$; in exchange the signal is seven times larger.

▶ Falstad: `examples/bridges/pt1000_bridge.txt`: voltmeter $0\,\text{V}$, $134.7\,\text{mV}$ and $260.9\,\text{mV}$ at the slider positions 1000, 1192.5 and $1385\,\Omega$.

5.6 Reading the Bridge: Common-Mode Voltage

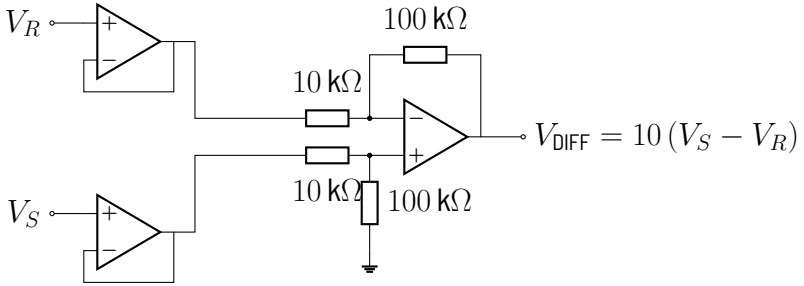
A single-ended amplifier fails



- gain 107.6 applied to $V_S = 100$ mV: $V_{\text{out}} = 10.75$ V at $0\,^\circ$C (wanted: 0 V)
- at $100\,^\circ$C: 14.76 V; the useful signal is a small change on a large offset
- any drift of V_{ref} appears amplified at the output

▶ Falstad: `examples/bridges/bridge_single_ended.txt`

Difference and instrumentation amplifier

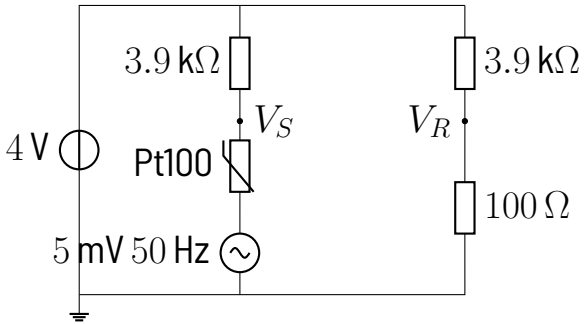


- difference amplifier: amplifies $V_S - V_R$, rejects V_{CM} (if the resistors match)
- buffers: no current from the bridge, the bridge is not loaded
- second stage $G_2 = 10.76$: $0\,\text{V}$ ... $4.00\,\text{V}$ (details: chapter 6)

▶ Falstad: `examples/bridges/bridge_ina.txt`

5.7 Interference and Lead Resistance

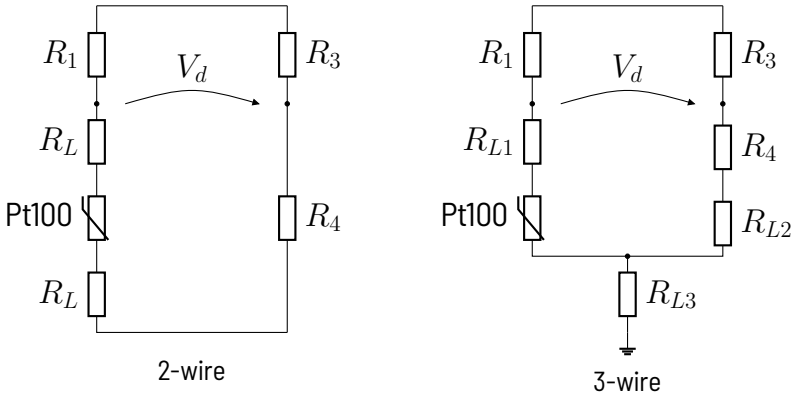
50 Hz in series with the sensor



- at V_S: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} = 5\,\text{mV} \cdot \frac{3900}{4000} = 4.88\,\text{mV}$ amplitude
- after $G = 107.6$: $0.52\,\text{V}$ amplitude: about one LSB ($0.5\,\text{V}$)
- remedies: twisted pair, shield, short leads, low-pass filter before the ADC (chapters 6, 7)

▶ Falstad: `examples/bridges/bridge_50hz.txt`

Lead resistance: 2-wire and 3-wire connection



- 2-wire: $2R_L$ in series with the sensor; $R_L = 0.5\,\Omega$: $1\,\Omega \mathrel{\hat{=}} 2.6\,\mathrm{K}$
- 3-wire: R_{L1} in the sensor arm, R_{L2} in the neighbouring arm R_4: they cancel at balance; R_{L3} only carries the supply current
- 4-wire: current source through two wires, voltage sensed by the other two (precision instruments)

▶ Falstad: `examples/bridges/bridge_3wire.txt`

Exercise 5.6: Interference and lead resistance

The capstone bridge ($V_{\text{ref}} = 4\,\text{V}$, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100) is followed by an amplifier with gain 107.6.

a) A $5\,\text{mV}$, $50\,\text{Hz}$ interference voltage is induced in series with the sensor. Calculate its amplitude at V_S and at the amplifier output. Why does the instrumentation amplifier not reject it?

b) The sensor is connected with two leads of $0.5\,\Omega$ each. Which bridge output and temperature error result at $0\,^\circ\text{C}$?

c) Repeat b) for the 3-wire connection, at $0\,^\circ\text{C}$ and $100\,^\circ\text{C}$.

Solution 5.6

a) The source sees R_1 above and the sensor below: $5\,\text{mV} \cdot 3900/4000 = 4.875\,\text{mV}$ at V_S, $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$ at the output. It changes only V_S, not V_R: it is a differential signal.

b) $R = 101\,\Omega$: $V_d = 4\,(101/4001 - 0.025) = 0.975\,\text{mV}$; with $0.372\,\text{mV/K}$: $2.6\,\text{K}$.

c) $0\,°\text{C}$: both lower arms have $100.5\,\Omega$, $V_d = 0$. $100\,°\text{C}$: $V_d = 37.16\,\text{mV}$ instead of $37.18\,\text{mV}$ (2-wire: $38.13\,\text{mV}$); the remaining error is $0.05\,\text{K}$.

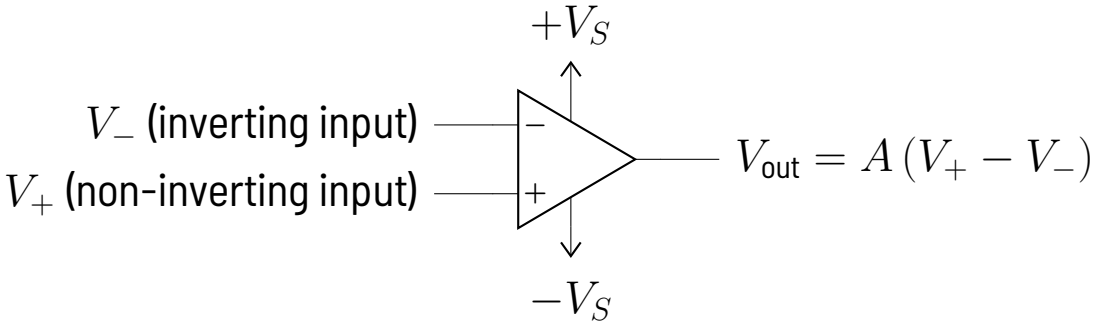
▶ Falstad: `examples/bridges/bridge_50hz.txt`: the scope shows $4.88\,\text{mV}$ amplitude at V_S and $0.52\,\text{V}$ at `Vout`;

▶ Falstad: `examples/bridges/bridge_3wire.txt`: voltmeters $0.975\,\text{mV}$ (2-wire) and $0\,\text{V}$ (3-wire) at $0\,°\text{C}$, $38.13\,\text{mV}$ and $37.16\,\text{mV}$ with both sliders at $138.5\,\Omega$.

6 Operational Amplifiers

6.1 The Ideal Operational Amplifier

The op-amp: symbol and model

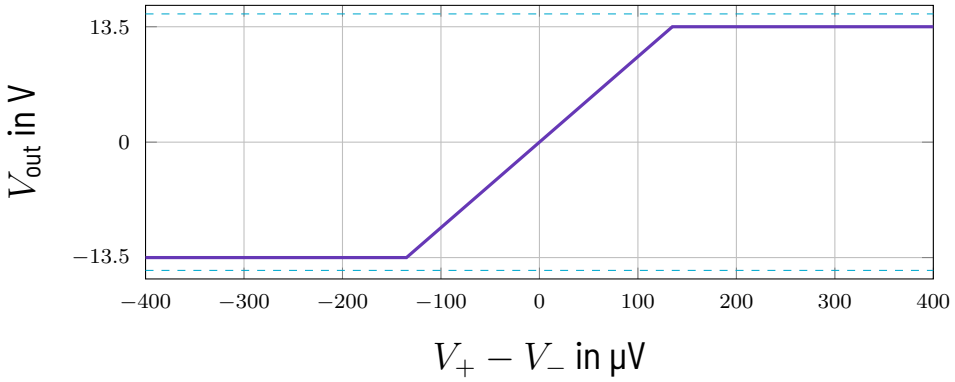


- amplifies the **difference** of its two input voltages
- open-loop gain A: typically $10^5 \ldots 10^7$ (100 dB ... 140 dB)
- supply $\pm V_S$ (or single supply V_S and ground); the output cannot leave the supply range
- supply pins are often omitted in drawings – but they are always there

Properties of the ideal op-amp

property	ideal	real (typical)
open-loop gain A	∞	$10^5 \ldots 10^7$
input resistance	∞	1 MΩ ...1 TΩ
input currents	0	pA (CMOS) ...100 nA (bipolar)
output resistance	0	$10\,\Omega$ ... $100\,\Omega$ (open loop)
offset voltage	0	10 µV ...5 mV
bandwidth	∞	GBW 1 MHz ...100 MHz

Open loop: the op-amp as a comparator



- with $A = 10^5$ and ± 13.5 V output swing the linear range is only ± 135 µV
- any larger difference drives the output to a rail: V_{out} tells only **which input is higher**
- this is a **comparator** – the building block of the flash ADC (chapter 9)

Negative feedback and the golden rules

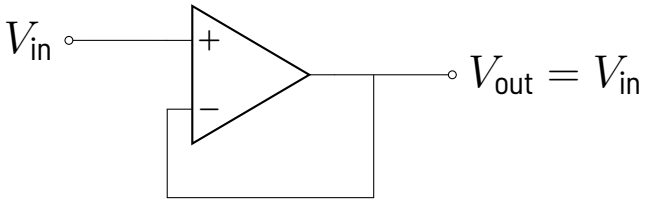
- **negative feedback**: a part of V_{out} is fed back to the **inverting** input
- if V_{out} rises, V_- rises, the difference $V_+ - V_-$ shrinks and pulls V_{out} back
- the circuit settles where $V_+ - V_- = V_{\text{out}}/A \approx 0$

Rule 6.1 ***Golden rules*** *for an op-amp with negative feedback:*

1. *The output does whatever is necessary to make the voltage difference between the inputs zero:* $V_+ = V_-$.
2. *The inputs draw no current:* $I_+ = I_- = 0$.

6.2 Basic Amplifier Circuits

Voltage follower (buffer)



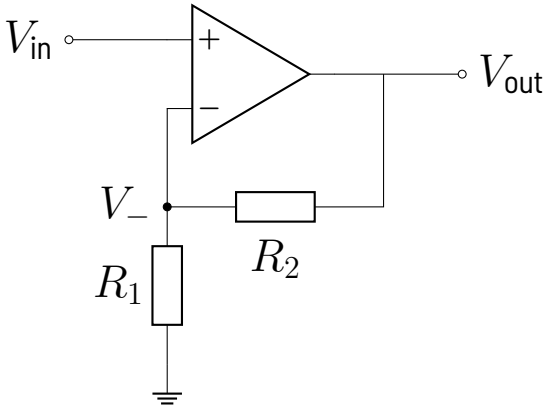
- rule 1: $V_- = V_+$, and $V_- = V_{\text{out}} \Rightarrow V_{\text{out}} = V_{\text{in}}$, gain 1
- input current ≈ 0: the source is **not loaded**
- output: low resistance, can drive mA into a load

Why buffering matters

- divider $100\,\text{k}\Omega/100\,\text{k}\Omega$ at $10\,\text{V}$, load $R_L = 10\,\text{k}\Omega$
- directly: $V_{\text{out}} = 10\,\text{V} \cdot \frac{R_2 \parallel R_L}{R_1 + R_2 \parallel R_L} = 0.833\,\text{V}$ instead of $5\,\text{V}$
- behind a follower: $5.00\,\text{V}$, the divider does not notice the load
- uses in this course
  - bridge taps $\rightarrow$ instrumentation amplifier inputs
  - reference ladder of the flash ADC, hold capacitor of the S&H

▶ Falstad: `examples/opamps/opamp_follower.txt`

Non-inverting amplifier

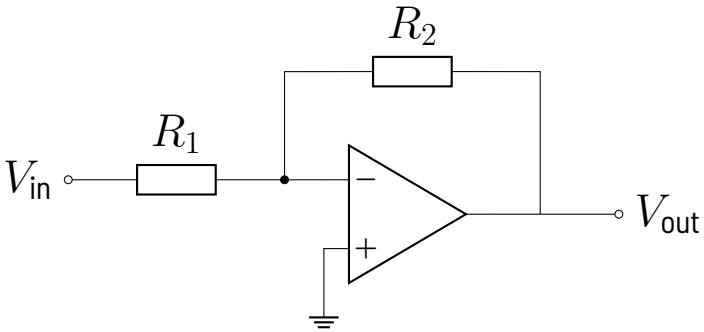


- R_1, R_2 form a divider: $V_- = V_{\text{out}} \frac{R_1}{R_1+R_2}$; rule 1: $V_- = V_{\text{in}}$

Non-inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = 1 + \frac{R_2}{R_1}$$ input resistance $\approx \infty$

Inverting amplifier

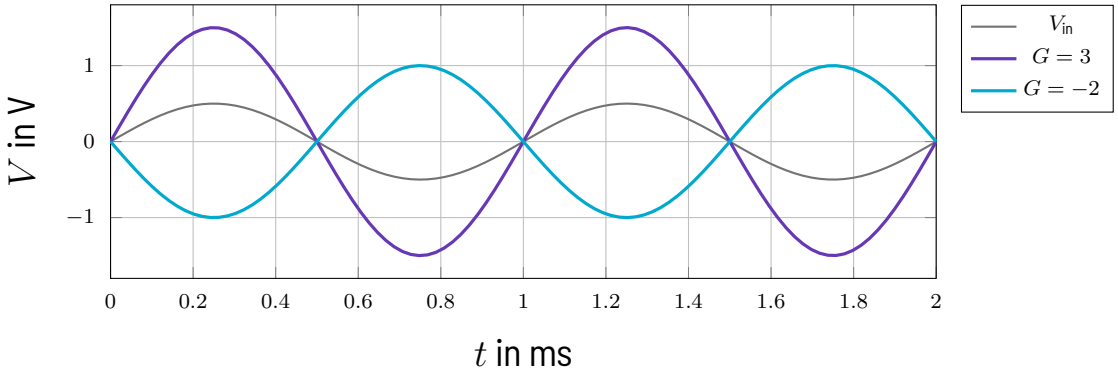


- rule 1: $V_- = V_+ = 0$, the inverting input is a **virtual ground**
- rule 2: the current V_{in}/R_1 continues through R_2: $V_{\text{out}} = -R_2 \cdot V_{\text{in}}/R_1$
- input resistance $= R_1$ (it loads the source!)

Inverting amplifier

$$G = \frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2}{R_1}$$

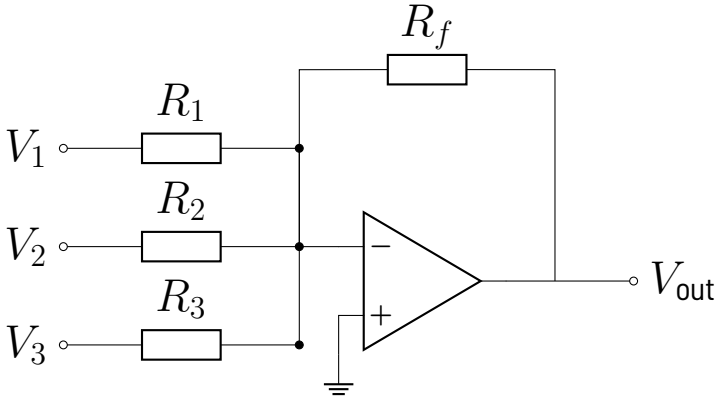
Non-inverting and inverting in Falstad



- V_{in}: 0.5 V amplitude, 1 kHz; $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ in both circuits
- non-inverting: $G = 1 + 2 = 3$, 1.5 V in phase; inverting: $G = -2$, 1.0 V, phase $180°$

▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`

Summing amplifier



- the virtual ground decouples the inputs; the currents add up in R_f

Inverting summing amplifier

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

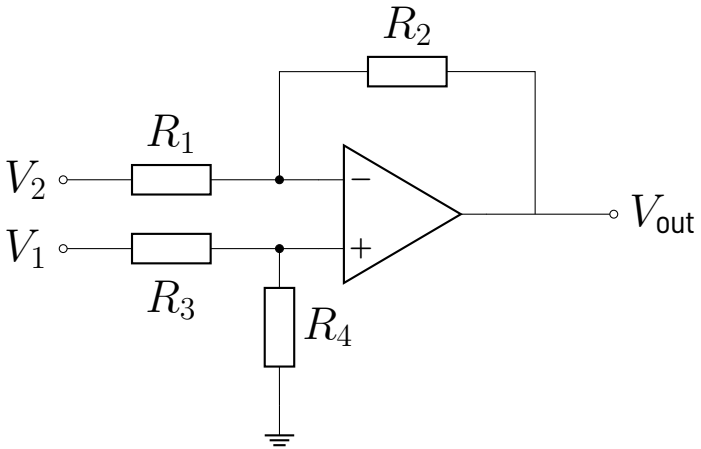
6.3 Difference and Instrumentation Amplifiers

Differential and common-mode signal

- two input voltages V_1, V_2 can be written as
  - differential-mode voltage $V_D = V_1 - V_2$ (the measurement signal)
  - common-mode voltage $V_{CM} = (V_1 + V_2)/2$ (not wanted)
- symmetric bridge at $5\,\text{V}$: V_D some mV, $V_{CM} \approx 2.5\,\text{V}$; capstone bridge: $V_D = 0\,\text{mV} \ldots 37\,\text{mV}$, $V_{CM} \approx 0.1\,\text{V}$
- ideal: $V_{\text{out}} = A_D\, V_D$; real: $V_{\text{out}} = A_D\, V_D + A_{CM}\, V_{CM}$

Definition 6.1 *The **common-mode rejection ratio** is the ratio of differential gain to common-mode gain:* $\text{CMRR} = A_D/A_{CM}$, *in decibels* $20 \log_{10}(A_D/A_{CM})$.

Difference amplifier



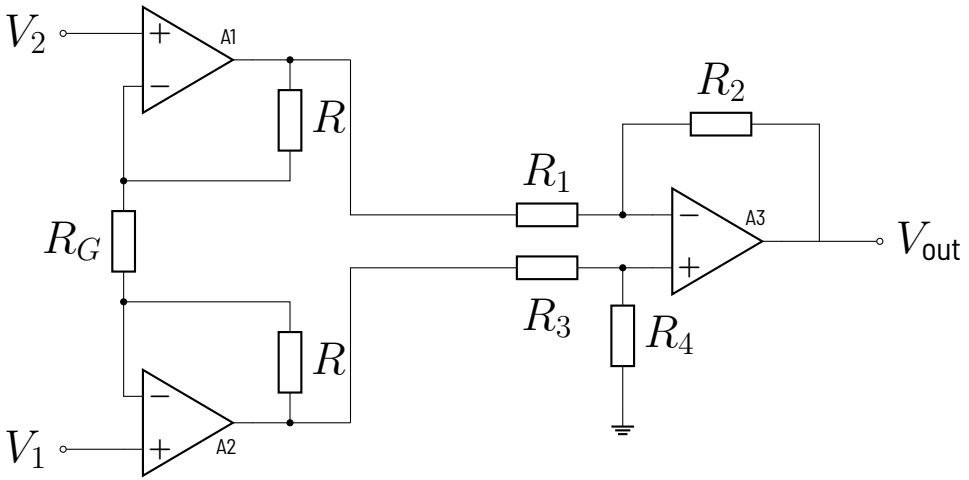
- superposition: $V_{\text{out}} = V_1 \dfrac{R_4}{R_3 + R_4}\left(1 + \dfrac{R_2}{R_1}\right) - V_2 \dfrac{R_2}{R_1}$
- with $R_4/R_3 = R_2/R_1$: $V_{\text{out}} = \dfrac{R_2}{R_1}(V_1 - V_2)$

Resistor matching limits the CMRR

- the common-mode rejection depends only on how well R_4/R_3 equals R_2/R_1
- example $G = 10$, $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$: ideal $V_\text{out} = 50.0\,\text{mV}$
- R_4 only $1\,\%$ too high: $V_\text{out} = 72.6\,\text{mV}$ – an error of $45\,\%$!
- worst case with resistor tolerance ε: CMRR $\approx \dfrac{1+G}{4\varepsilon}$
  - $1\,\%$ resistors, $G = 10$: CMRR ≈ 275 ($49\,\text{dB}$)
  - remedy: $0.1\,\%$ resistors, matched resistor networks, a trimmer, or an integrated difference amplifier (e.g. INA105, AD8276)

▶ Falstad: `examples/opamps/opamp_diffamp.txt`

Instrumentation amplifier (three op-amps)



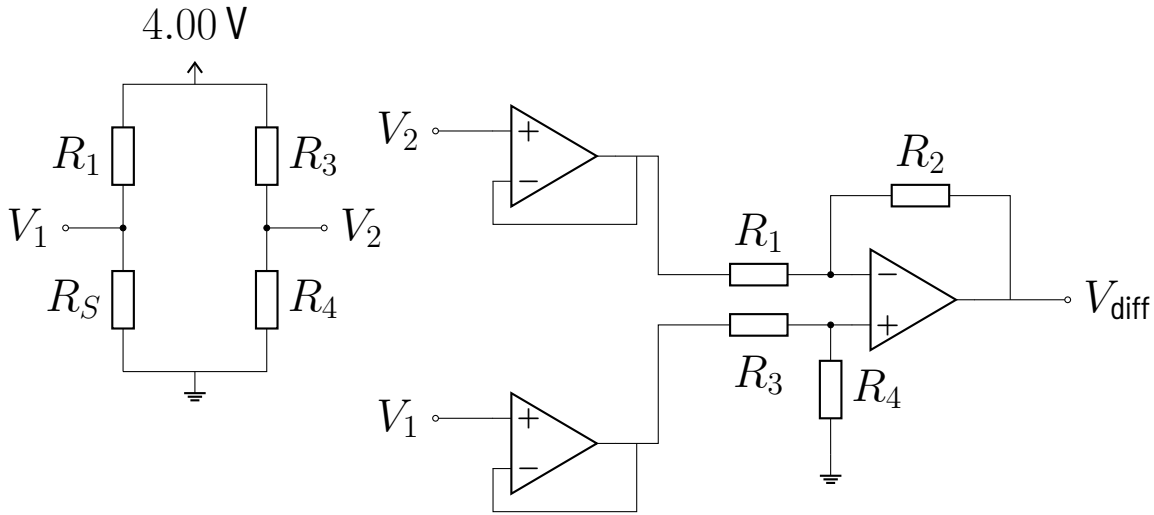
Instrumentation amplifier

$$V_{\text{out}} = \left(1 + \frac{2R}{R_G}\right) \frac{R_2}{R_1} (V_1 - V_2)$$

How the three op-amp INA works

- A1, A2 are non-inverting: the inputs draw no current, the bridge is not loaded
- rule 1 puts V_1 and V_2 across R_G: current $(V_1 - V_2)/R_G$ flows through R, R_G, R
- difference at the outputs of A1/A2: $(V_1 - V_2)(1 + 2R/R_G)$; the common mode passes with gain 1
- A3 is a difference amplifier with gain R_2/R_1 and removes the common mode
- one resistor R_G sets the gain; integrated INAs (INA128, AD620, AD8221) contain everything except R_G

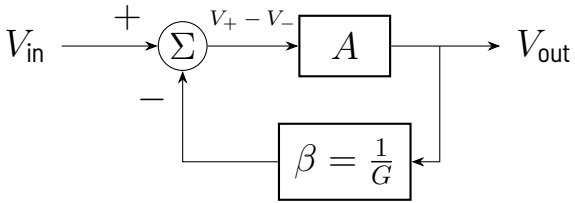
The lab version: two buffers and a difference amplifier



- followers isolate the bridge; the difference stage sets the gain $G_1 = R_2/R_1$
- simpler than the three op-amp version, but all gain must come from R_2/R_1

6.4 Limits of Real Op-Amps

Finite open-loop gain



- $V_{\text{out}} = A\,(V_{\text{in}} - \beta\,V_{\text{out}})$, with the feedback factor $\beta = 1/G$ (non-inverting: $\beta = R_1/(R_1 + R_2)$)
- loop gain $A\beta = A/G$: the larger, the closer the circuit comes to the ideal gain G

Closed-loop gain with finite open-loop gain A

$$G_{\text{real}} = \frac{A}{1 + A\beta} = \frac{G}{1 + G/A} = \frac{1}{1/G + 1/A} \qquad \text{relative error} \approx -\frac{G}{A}$$

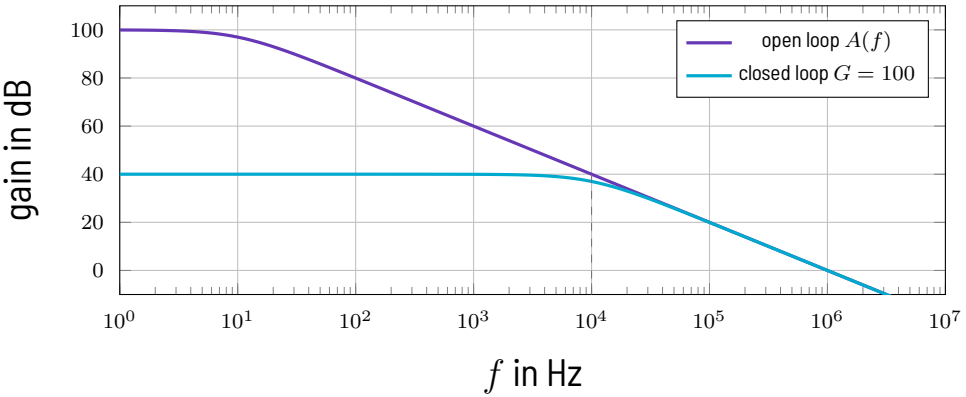
How large is the error?

ideal gain G	G_{real} ($A = 10^5$)	error	V_{out} for 1 mV
10	9.999	$-0.01\,\%$	10.0 mV
100	99.90	$-0.1\,\%$	99.9 mV
1000	990.1	$-1\,\%$	0.990 V
10000	9091	$-9.1\,\%$	9.09 V

- the error grows proportionally to G: one stage with $G > 1000$ shows a visible error
- **split the gain**: $G = 10 \cdot 100$ gives 998.9 ($-0.11\,\%$), $G = 100 \cdot 100$ gives 9980 ($-0.2\,\%$)
- rule of the lab: aim for $G < 100$ per stage

▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

Gain-bandwidth product



- above a low corner frequency (here 10 Hz) A falls by 20 dB per decade: $A(f) \approx \text{GBW}/f$
- closed-loop bandwidth $f_c \approx \text{GBW}/G$ (here 1 MHz/100 = 10 kHz)

Consequences of the GBW

- on the falling slope gain and frequency are inversely proportional:

Gain-bandwidth product

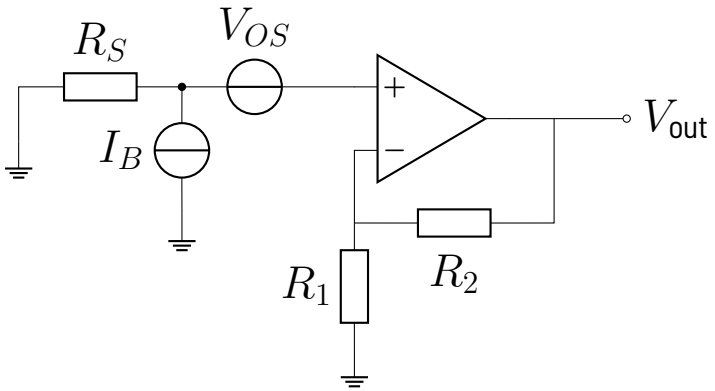
GBW $= A(f) \cdot f = G \cdot f_c$ (constant above the corner frequency)

- at 1 kHz an op-amp with GBW = 1 MHz has only $A = 1000$: a $G = 100$ stage then has $10\,\%$ error
- splitting the gain also helps here: two stages $G = 10$ each have 100 kHz bandwidth
- our sensor signal is slow (a few Hz): the GBW is no problem in the capstone
- **Falstad does not simulate the GBW**: its op-amp has a constant $A = 10^5$ at all frequencies

Output swing and rail-to-rail

- the output cannot exceed the supply; a classic op-amp stays $1\,\mathrm{V} \ldots 2\,\mathrm{V}$ away from the rails
  - LM358 at $5\,\mathrm{V}$: output from about $0\,\mathrm{V}$ to $3.5\,\mathrm{V}$
- **rail-to-rail output**: reaches the rails within some $10\,\mathrm{mV}$ (light load) to a few $100\,\mathrm{mV}$ (mA load)
- the **input common-mode range** is limited as well: not every op-amp works with inputs near the rails (**rail-to-rail input**); the capstone bridge sits at only $0.1\,\mathrm{V}$
- consequence for the capstone: an output of exactly $0\,\mathrm{V}$ is not possible on a single $5\,\mathrm{V}$ supply – use a negative supply, or a small offset and a range of e.g. $0.1\,\mathrm{V} \ldots 4.1\,\mathrm{V}$

Input offset voltage and bias currents



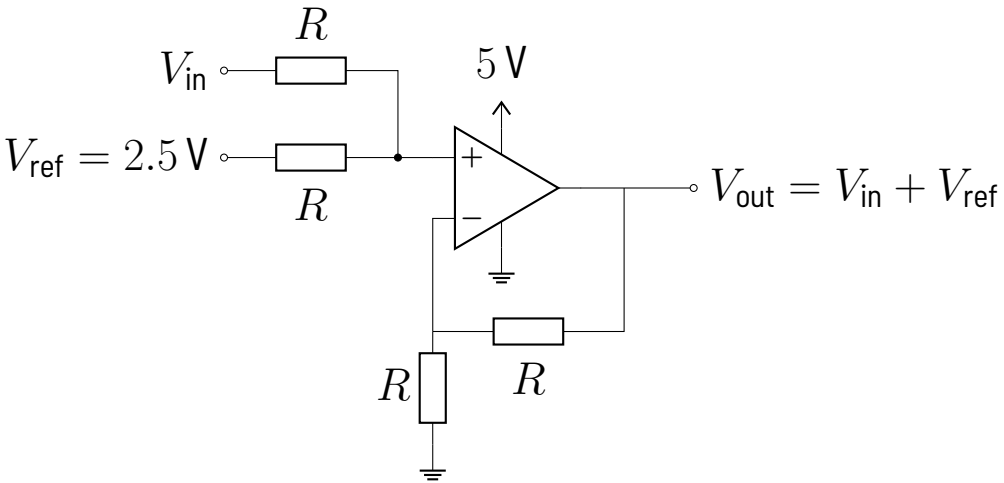
- offset V_{OS}: a small voltage source in series with an input, amplified by $1 + R_2/R_1$
- bias current I_B flows through the source resistance R_S and adds $R_S I_B$
- $V_{\text{out}} = (1 + R_2/R_1)\,(V_{OS} + R_S I_B)$ with $V_{\text{in}} = 0$

Offset after a high gain

- typical values
  - general purpose bipolar (LM358): $V_{OS} \approx 2\,\text{mV}$, $I_B \approx 20\,\text{nA}$
  - CMOS (MCP6002, TLV9002): $V_{OS} \approx 1\,\text{mV}$, $I_B \approx 1\,\text{pA}$
  - zero-drift / chopper (OPA2333, MCP6V02): $V_{OS} < 10\,\mu\text{V}$
- Falstad example with $G = 101$: $V_{OS} = 1\,\text{mV}$ gives $0.101\,\text{V}$; $I_B = 100\,\text{nA}$ through $100\,\text{k}\Omega$ gives $-1.01\,\text{V}$
- capstone: $2\,\text{mV}$ offset $\times$ 107.6 = $0.22\,\text{V}$ – $5\,\%$ of the full scale, almost half an LSB!
- remedies: zero-drift op-amps, CMOS inputs, low source resistances, **offset trimming**

▶ Falstad: `examples/opamps/opamp_offset.txt`

Single-supply operation and level shifting

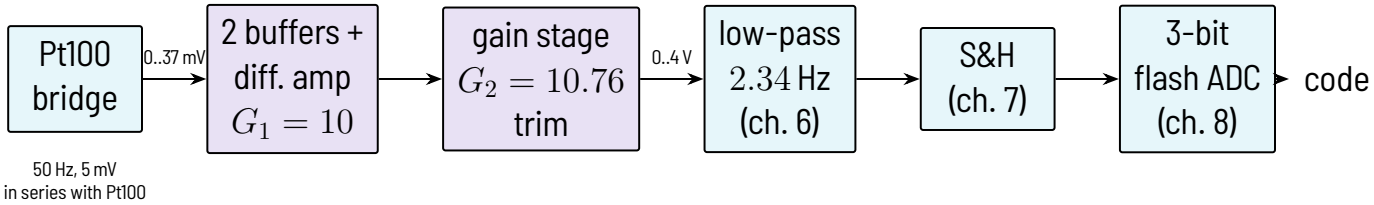


- single supply: the output can only be positive; bipolar signals need a **bias point** (often $V_S/2$)
- here $V_+ = (V_{\text{in}} + V_{\text{ref}})/2$, gain 2: $V_{\text{in}} = \pm 2\,\text{V}$ becomes $0.5\,\text{V} \ldots 4.5\,\text{V}$
- a follower on the same supply clips the negative half wave at $0\,\text{V}$

▶ Falstad: `examples/opamps/opamp_level_shift.txt`

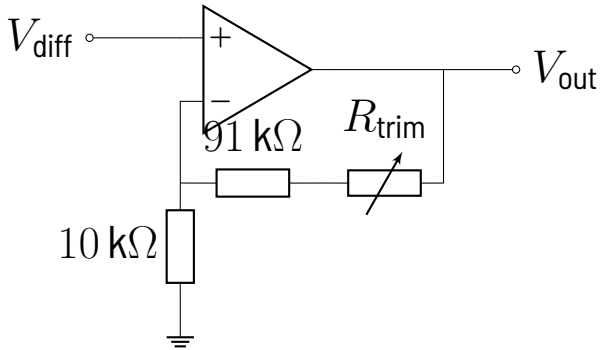
6.5 Designing the Measurement Amplifier

The signal chain of the capstone



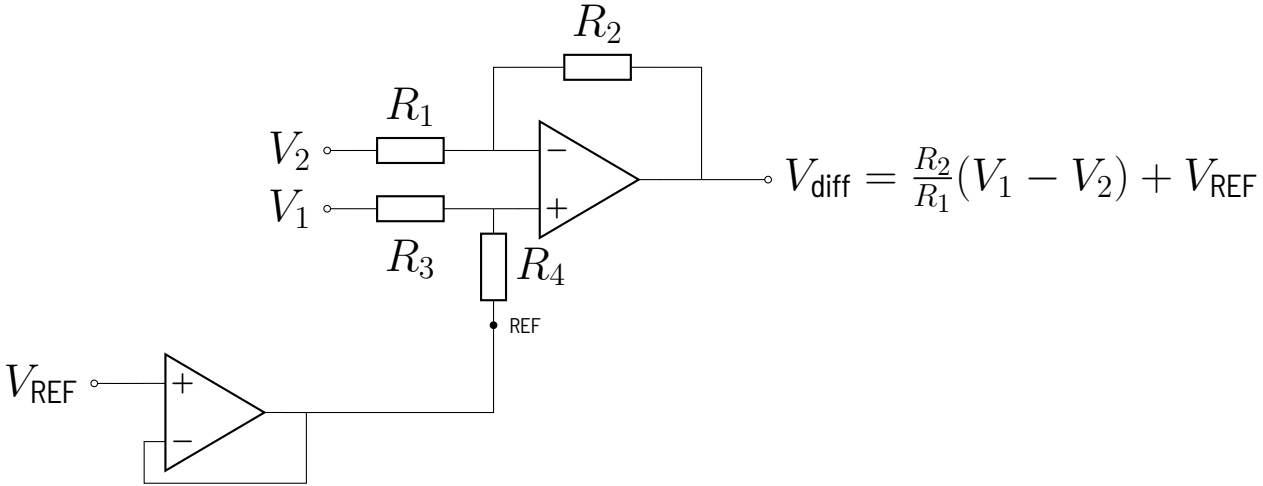
- bridge from $4.00\,\text{V}$: $V_{BR} = V_{\text{ref}} \left(\frac{R}{R_1+R} - \frac{R_4}{R_3+R_4} \right) = 0\,\text{mV} \ldots 37.18\,\text{mV}$ for $R = 100\,\Omega \ldots 138.5\,\Omega$
- total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: difference amplifier $G_1 = 100\,\text{k}\Omega/10\,\text{k}\Omega = 10$, non-inverting stage $G_2 = 10.76$ – both < 100
- zero: set by the bridge ($R_4 = R(0\,°\text{C})$); span: one resistor R_F in the second stage

Gain stage with trimmer



- required: $G_2 = 4.00\,\text{V}/(10 \cdot 37.18\,\text{mV}) = 10.758 \Rightarrow R_F = 97.58\,\text{k}\Omega$
- fixed E96 value: $R_F = 97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$ (used in Falstad)
- in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer: $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V} \ldots 4.13\,\text{V}$; setting $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- the fixed resistor carries most of the value: fine trimmer resolution, enough range for the tolerances

Zero adjustment at the reference input



- the lower end of R_4 is the **reference input**: V_{REF} adds directly to the output
- it must be driven from a low impedance (follower), otherwise it changes R_4 and the CMRR
- in the capstone the zero is set by the bridge ($R_4 = R(0\,^\circ\text{C})$); a REF adjustment of a few mV is needed only if op-amp offsets or resistor tolerances shift it

Trimming procedure

- two adjustments: zero (offset) first, then span (gain)

Design card: Measurement amplifier (zero and span)

1. Calculate the bridge span $V_{BR,\text{max}}$ and the total gain $G = V_{\text{FS}}/V_{BR,\text{max}}$ (here $4.00\,\text{V}/37.18\,\text{mV} = 107.6$).
2. Split the gain: input stage (buffers + difference amplifier) $G_1 = 10$, second stage $G_2 = G/G_1 = 10.76$, each stage $G < 100$.
3. Use a fixed resistor for most of R_F and a trimmer for the rest ($91\,\text{k}\Omega$ + $10\,\text{k}\Omega$).
4. **Zero**: set the sensor to the bottom of the range ($100\,\Omega$, or a precise resistor) and check $V_{\text{out}} = 0.00\,\text{V}$; correct with R_4 or V_{REF} if necessary.
5. **Span**: set the sensor to the top of the range ($138.5\,\Omega$), adjust R_{trim} until $V_{\text{out}} = 4.00\,\text{V}$.
6. Check the zero again.

The complete front end in Falstad

- bridge 4.00 V, $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$, Pt100 with a slider from $100\,\Omega$ to $138.5\,\Omega$
- two followers, difference amplifier $10\,\text{k}\Omega/100\,\text{k}\Omega$, gain stage $1 + (91\,\text{k}\Omega + R_{\text{trim}})/10\,\text{k}\Omega$, $R_{\text{trim}} = 6.6\,\text{k}\Omega$
- readings: $V_R = 0.100$ V, $V_{BR} = 37.18$ mV, $V_{\text{diff}} = 372$ mV, $V_{\text{out}} = 4.00$ V at $138.5\,\Omega$; $V_{\text{out}} = 0.00$ V at $100\,\Omega$
- the 50 Hz interference of 5 mV in series with the Pt100 is a **differential** signal: 4.9 mV at the tap, $\times 107.6 = 0.52$ V amplitude
  - no CMRR helps here – the low-pass filter of chapter 7 has to remove it
  - the output must have headroom: $4\,\text{V} + 0.52\,\text{V}$ peak

▶ Falstad: `examples/opamps/opamp_ina.txt`

6.6 Exercises

Exercise 6.1: Non-inverting and inverting amplifier

A sine voltage with 0.5 V amplitude and 1 kHz is to be amplified. Resistors: $R_1 = 10\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$.

a) Calculate gain and output amplitude of the non-inverting amplifier.

b) Calculate gain and output amplitude of the inverting amplifier. What is the phase shift?

c) What input resistance does the source see in both circuits?

d) Which resistor would you change to get an inverting gain of -4.7?

e) Check a) and b) in Falstad.

Solution 6.1

a) $G = 1 + R_2/R_1 = 1 + 20/10 = 3$; output amplitude $3 \cdot 0.5\,\text{V} = 1.5\,\text{V}$, in phase with the input.

b) $G = -R_2/R_1 = -2$; output amplitude $1.0\,\text{V}$, phase shift $180°$.

c) Non-inverting: practically infinite (the op-amp input, MΩ to TΩ). Inverting: $R_1 = 10\,\text{k}\Omega$, because the inverting input is a virtual ground.

d) $R_2 = 4.7 \cdot R_1 = 47\,\text{k}\Omega$ (E12 value).

e) ▶ Falstad: `examples/opamps/opamp_noninv_inv.txt`
the scope shows `Vout_ni` with $\pm 1.50\,\text{V}$ and `Vout_inv` with $\pm 1.00\,\text{V}$, inverted.

Exercise 6.2: Summing amplifier

An inverting summing amplifier has $R_f = 10\,\text{k}\Omega$ and three inputs: $V_1 = 1\,\text{V}$ via $R_1 = 10\,\text{k}\Omega$, $V_2 = 2\,\text{V}$ via $R_2 = 20\,\text{k}\Omega$, $V_3 = -0.5\,\text{V}$ via $R_3 = 10\,\text{k}\Omega$.

a) Calculate V_{out} and the voltage at the inverting input.
b) Calculate the current through R_f.
c) How must R_3 be changed so that V_3 is weighted with -0.25?
d) Check a) in Falstad.

Solution 6.2

a) $V_{\text{out}} = -10\,\text{k}\Omega\left(\frac{1\,\text{V}}{10\,\text{k}\Omega} + \frac{2\,\text{V}}{20\,\text{k}\Omega} + \frac{-0.5\,\text{V}}{10\,\text{k}\Omega}\right) = -(1 + 1 - 0.5)\,\text{V} = -1.50\,\text{V}$. The inverting input is a virtual ground: $V_- = 0\,\text{V}$.

b) $I = 0.1\,\text{mA} + 0.1\,\text{mA} - 0.05\,\text{mA} = 0.15\,\text{mA}$, which gives $V_{\text{out}} = -10\,\text{k}\Omega \cdot 0.15\,\text{mA} = -1.5\,\text{V}$.

c) Weight $-R_f/R_3 = -0.25 \Rightarrow R_3 = 40\,\text{k}\Omega$ (e.g. $39\,\text{k}\Omega + 1\,\text{k}\Omega$).

d) ▶ Falstad: `examples/opamps/opamp_summing.txt`

`Vout` $= -1.50\,\text{V}$, `V_sum_node` $= 0.00\,\text{V}$.

Exercise 6.3: Difference amplifier and resistor matching

A difference amplifier with $R_1 = R_3 = 10\,\text{k}\Omega$ and $R_2 = R_4 = 100\,\text{k}\Omega$ amplifies a bridge signal: $V_1 = 2.505\,\text{V}$, $V_2 = 2.500\,\text{V}$.

a) Calculate differential and common-mode voltage and the output voltage.
b) R_4 is $1\,\%$ too high ($101\,\text{k}\Omega$). Calculate the output voltage.
c) Calculate the common-mode gain and the CMRR for b).
d) Estimate the worst-case CMRR with $1\,\%$ resistors.
e) Check a) and b) in Falstad.

Solution 6.3

a) $V_D = 5\,\text{mV}$, $V_{CM} = 2.5025\,\text{V}$; $V_{\text{out}} = 10 \cdot 5\,\text{mV} = 50.0\,\text{mV}$.

b) $V_+ = 2.505\,\text{V} \cdot \frac{101}{111} = 2.279\,32\,\text{V}$, $V_{\text{out}} = V_+\,(1+10) - 2.5\,\text{V} \cdot 10 = 25.0726\,\text{V} - 25\,\text{V} = 72.6\,\text{mV}$: $45\,\%$ too high.

c) $A_{CM} = 11 \cdot \frac{101}{111} - 10 = 0.0090$ (input both at $2.5\,\text{V}$ gives $22.5\,\text{mV}$); $A_D \approx 10$; $\text{CMRR} = 10/0.009 \approx 1100$ ($61\,\text{dB}$).

d) $\text{CMRR} \approx (1+G)/(4\varepsilon) = 11/0.04 = 275$ ($49\,\text{dB}$): with $2.5\,\text{V}$ common mode up to $9\,\text{mV}$ error at the input, i.e. almost half of the full bridge span. Matched resistors or an integrated difference amplifier are needed.

e) ▶ Falstad: `examples/opamps/opamp_diffamp.txt`

`Vout_matched` $= 50.0\,\text{mV}$, `Vout_mismatch` $= 72.6\,\text{mV}$.

Exercise 6.4: Finite open-loop gain

A signal of 1 mV DC is amplified with non-inverting stages ($R_1 = 1\,\text{k}\Omega$). The op-amps have $A = 10^5$ (as in Falstad).

a) Calculate V_{out} for one stage with $G = 1000$ and one stage with $G = 10000$. Give R_2 and the errors.

b) Calculate V_{out} for $G = 10 \cdot 100$ and $G = 100 \cdot 100$ in two stages.

c) How large is the error of the $G = 100$ stage at 1 kHz if the op-amp has GBW $= 1$ MHz?

d) Check a) and b) in Falstad.

Solution 6.4

a) $G = 1000$: $R_2 = 999\,\text{k}\Omega$, $G_\text{real} = 1000/(1+0.01) = 990.1$, $V_\text{out} = 0.990\,\text{V}$ ($-1\,\%$). $G = 10000$: $R_2 = 9.999\,\text{M}\Omega$, $G_\text{real} = 10000/1.1 = 9091$, $V_\text{out} = 9.09\,\text{V}$ ($-9.1\,\%$).

b) $10/(1+10^{-4})\cdot 100/(1+10^{-3}) = 9.999\cdot 99.90 = 998.9$: $V_\text{out} = 0.999\,\text{V}$ ($-0.11\,\%$). $(100/1.001)^2 = 9980$: $V_\text{out} = 9.98\,\text{V}$ ($-0.2\,\%$).

c) $A(1\,\text{kHz}) = 1\,\text{MHz}/1\,\text{kHz} = 1000$, error $\approx G/A = 10\,\%$ (and a phase shift). Falstad does not show this, because it does not simulate the GBW.

d) ▶ Falstad: `examples/opamps/opamp_finite_gain.txt`

`Vout_1000` $= 0.990\,\text{V}$, `Vout_10x100` $= 0.999\,\text{V}$,
`Vout_10000` $= 9.09\,\text{V}$, `Vout_100x100` $= 9.98\,\text{V}$.

Exercise 6.5: Offset voltage and bias current

A non-inverting amplifier has $R_1 = 1\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$; its input is at $0\,\text{V}$.

a) The op-amp has an offset voltage of $1\,\text{mV}$. Calculate V_{out}.

b) The input is connected to ground through $R_S = 100\,\text{k}\Omega$ and a bias current of $100\,\text{nA}$ flows into the non-inverting input (no offset). Calculate V_{out}.

c) The capstone amplifier has $G = 107.6$ and the op-amps have $2\,\text{mV}$ offset (referred to the input). Compare the output error with the full scale of $4\,\text{V}$ and with one LSB of the 3-bit ADC.

d) Check a) and b) in Falstad.

Solution 6.5

a) $V_{\text{out}} = (1 + 100) \cdot 1\,\text{mV} = 0.101\,\text{V}$.

b) $V_+ = -R_S I_B = -100\,\text{k}\Omega \cdot 100\,\text{nA} = -10\,\text{mV}$, $V_{\text{out}} = 101 \cdot (-10\,\text{mV}) = -1.01\,\text{V}$. A CMOS op-amp with $I_B = 1\,\text{pA}$ would give only $10\,\mu\text{V}$.

c) $107.6 \cdot 2\,\text{mV} = 0.215\,\text{V}$: $5.4\,\%$ of the full scale, 0.43 LSB (LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$). The zero must be checked and trimmed, or a zero-drift op-amp used.

d) ▶ Falstad: `examples/opamps/opamp_offset.txt`

`Vout_offset` $= 0.101\,\text{V}$, `V_plus_b` $= -10.0\,\text{mV}$, `Vout_bias` $= -1.01\,\text{V}$.

Exercise 6.6: Capstone: instrumentation amplifier with 4.00 V full scale

The Pt100 of the capstone ($100\,\Omega$ at $0\,°C$, $138.5\,\Omega$ at $100\,°C$) sits in a bridge fed from the 4.00 V reference: R_1 ($3.9\,k\Omega$) above the Pt100 gives V_S, $R_3 = 3.9\,k\Omega$ above $R_4 = 100\,\Omega$ gives V_R. The amplifier consists of two followers, a difference amplifier and a non-inverting stage. Its output shall be 0 V at $0\,°C$ and exactly 4.00 V at $100\,°C$.

a) Calculate V_R, V_S and the bridge output $V_{BR} = V_S - V_R$ at $0\,°C$ and $100\,°C$. What is the common-mode voltage?

b) Calculate the total gain. Why is it split into two stages?

c) Choose $G_1 = 10$ for the difference amplifier and give its resistors.

d) Dimension the second stage with $R_g = 10\,k\Omega$. Give the exact R_F, the nearest E96 value and a version with $91\,k\Omega$ plus a $10\,k\Omega$ trimmer (setting and full-scale range).

e) The three op-amp INA could replace the followers. Which R_G gives an input stage gain of 10 with $R = 10\,k\Omega$?

f) A 50 Hz interference of 5 mV amplitude is in series with the Pt100. How large is it at the output? What follows for the supply voltage and the next stage?

g) Check the design in Falstad.

Solution 6.6

a) $V_R = 4\,\text{V} \cdot 100/4000 = 0.100\,\text{V}$. $0\,°\text{C}$: $V_S = 0.100\,\text{V}$, $V_{BR} = 0$ (sensor current $1.000\,\text{mA}$). $100\,°\text{C}$: $V_S = 4\,\text{V} \cdot 138.5/4038.5 = 0.137\,18\,\text{V}$, $V_{BR} = 37.18\,\text{mV}$. The common mode is only about $0.1\,\text{V}$: the op-amp inputs must work close to the negative rail.

b) $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$. One stage would be possible (error $G/A = 0.11\,\%$), but two stages of about 10 keep each error at $0.01\,\%$ and the bandwidth high, follow the lab rule $G < 100$, and separate zero (bridge) and span (R_F).

c) $R_1 = R_3 = 10\,\text{k}\Omega$, $R_2 = R_4 = 100\,\text{k}\Omega$ ($0.1\,\%$ or matched). $V_{\text{diff}} = 10 \cdot 37.18\,\text{mV} = 371.8\,\text{mV}$ at $100\,°\text{C}$.

d) $G_2 = 4.00\,\text{V}/371.8\,\text{mV} = 10.758$, $R_F = 9.758 \cdot 10\,\text{k}\Omega = 97.58\,\text{k}\Omega$. E96: $97.6\,\text{k}\Omega$, $G_2 = 10.76$, full scale $4.0005\,\text{V}$. Trimmer version: $91\,\text{k}\Omega$ + $R_{\text{trim}} = 6.6\,\text{k}\Omega$; range $G_2 = 10.1 \ldots 11.1$, full scale $3.76\,\text{V}$ … $4.13\,\text{V}$.

e) $1 + 2R/R_G = 10 \Rightarrow R_G = 2R/9 = 2.22\,\text{k}\Omega$; then the difference stage can have gain 1 and the second stage stays as it is.

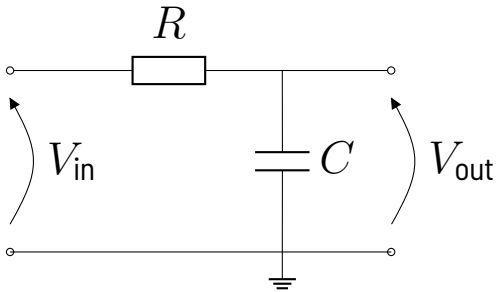
f) At the tap: $5\,\text{mV} \cdot \frac{3.9\,\text{k}\Omega}{3.9\,\text{k}\Omega + 100\,\Omega} = 4.875\,\text{mV}$, at the output $\times 107.6 = 0.52\,\text{V}$ amplitude: the output swings up to $4.52\,\text{V}$. The op-amps need headroom (e.g. $\pm 15\,\text{V}$ as in Falstad); the low-pass of chapter 7 must attenuate the interference below $\text{LSB}/2 = 0.25\,\text{V}$.

g) ▶ Falstad: `examples/opamps/opamp_ina.txt`
with the Pt100 at $138.5\,\Omega$ the labels show `VB_minus` $= 0.100\,\text{V}$, `V_diff` $= 372\,\text{mV}$ and `Vout` $= 4.00\,\text{V}$; with the slider at $100\,\Omega$ `Vout` $= 0.00\,\text{V}$.

7 Active Filters

7.1 Passive RC Filters

RC low-pass



- low frequency: the capacitor is open, $V_{\text{out}} = V_{\text{in}}$
- high frequency: the capacitor is a short, $V_{\text{out}} \to 0$
- a frequency-dependent voltage divider with the impedance $Z_C = \dfrac{1}{j\omega C}$

RC low-pass

$$\underline{H}(j\omega) = \frac{\underline{V}_{\text{out}}}{\underline{V}_{\text{in}}} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$$

Magnitude, phase and cutoff frequency

- with $f_c = \dfrac{1}{2\pi RC}$: $\quad \underline{H} = \dfrac{1}{1 + jf/f_c}$
- magnitude (amplitude ratio) and phase:

First-order low-pass

$$|\underline{H}| = \frac{1}{\sqrt{1 + (f/f_c)^2}} \qquad \varphi = -\arctan(f/f_c) \qquad f_c = \frac{1}{2\pi RC}$$

- at $f = f_c$: $|\underline{H}| = 1/\sqrt{2} = 0.707, \varphi = -45°$
- far above f_c: $|\underline{H}| \approx f_c/f$ – ten times the frequency, one tenth of the amplitude

Decibels

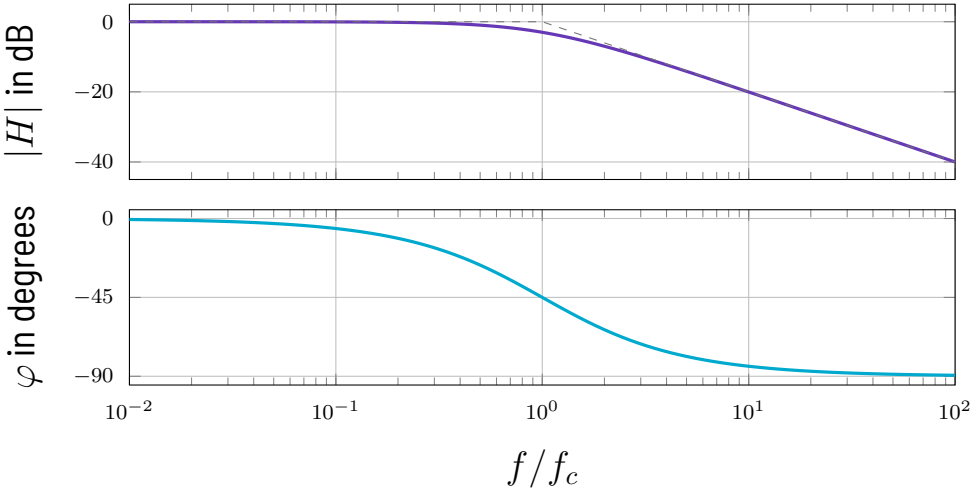
- gains and attenuations span many decades: a logarithmic measure is convenient

Definition 7.1 *The gain in decibels is* $|H|_{\text{dB}} = 20 \log_{10} |H|$ *(voltage or current ratios).*

$\lvert H\rvert$	dB	$\lvert H\rvert$	dB
1	0 dB	2	6 dB
$1/\sqrt{2} = 0.707$	-3 dB	10	20 dB
0.5	-6 dB	100	40 dB
0.1	-20 dB	1000	60 dB
0.01	-40 dB	10^5	100 dB

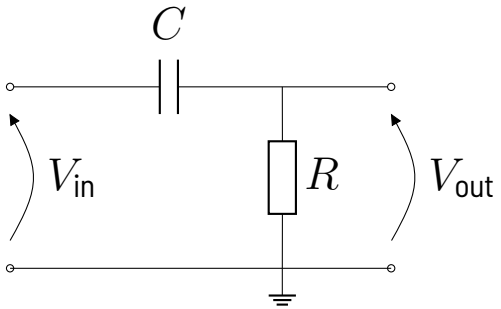
- gains of cascaded stages **multiply**; in dB they **add**
- a decade is a factor 10 in frequency, an octave a factor 2

Bode plot of the RC low-pass



- logarithmic frequency axis, magnitude in dB: the curve becomes two straight asymptotes
- 0 dB below f_c, -20 dB per decade above; the corner is at f_c (-3 dB)

RC high-pass



First-order high-pass

$$\underline{H} = \frac{jf/f_c}{1+jf/f_c} \qquad |\underline{H}| = \frac{f/f_c}{\sqrt{1+(f/f_c)^2}} \qquad \varphi = 90° - \arctan(f/f_c)$$

- blocks DC (coupling capacitor), 20 dB per decade below f_c
- same cutoff frequency $f_c = 1/(2\pi RC)$, phase $45°$ at f_c

RC filters in Falstad

- $R = 1.6\,\text{k}\Omega$, $C = 100\,\text{nF}$: $f_c = 995\,\text{Hz}$; three copies driven with $1\,\text{V}$ at $100\,\text{Hz}$, $1\,\text{kHz}$, $10\,\text{kHz}$

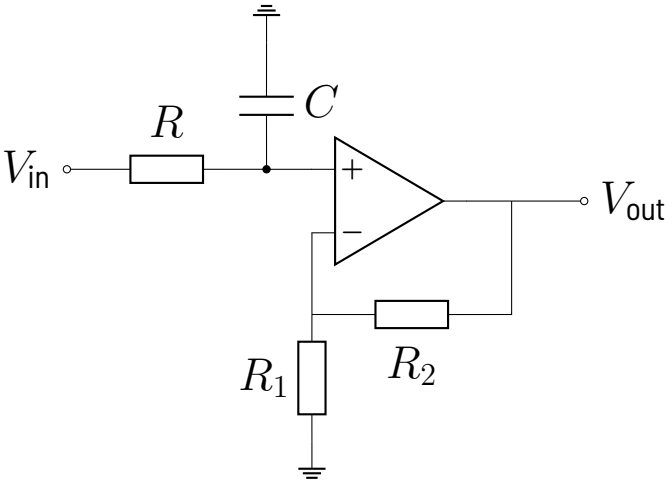
	$100\,\text{Hz}$	$1\,\text{kHz}$	$10\,\text{kHz}$
low-pass amplitude	$0.995\,\text{V}$	$0.705\,\text{V}\ (-45°)$	$0.099\,\text{V}$
high-pass amplitude	$0.100\,\text{V}$	$0.709\,\text{V}\ (45°)$	$0.995\,\text{V}$

▶ Falstad: `examples/filters/filter_rc_lowpass.txt` ▶ Falstad: `examples/filters/filter_rc_highpass.txt`

- limits of passive filters: the load changes f_c (it is in parallel to C or R), no gain, steep filters need inductors
- remedy: an op-amp buffers the filter and adds gain $\rightarrow$ **active filter**

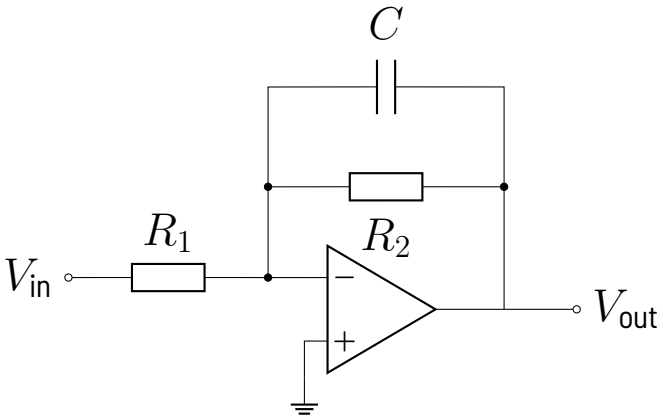
7.2 First-Order Active Filters

Non-inverting first-order low-pass



- RC low-pass, then a non-inverting amplifier (or a follower for $G = 1$)
- $\underline{H} = \left(1 + \frac{R_2}{R_1}\right) \frac{1}{1 + j\omega RC}$, $\quad f_c = \frac{1}{2\pi RC}$
- the op-amp input does not load C; the output can drive the next stage

Inverting first-order low-pass

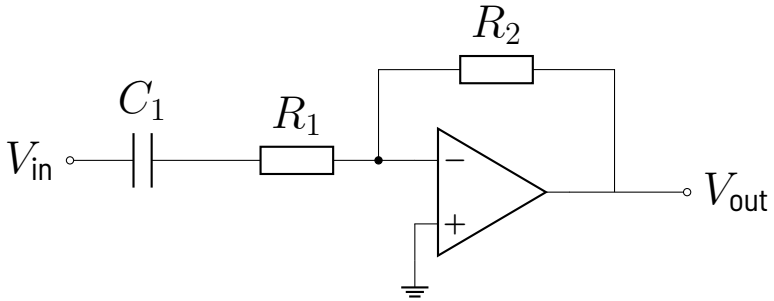


- feedback impedance $Z_2 = R_2 \parallel \frac{1}{j\omega C}$, gain $-Z_2/R_1$

Inverting first-order low-pass

$$\underline{H} = -\frac{R_2}{R_1}\frac{1}{1+j\omega R_2 C} \qquad \text{DC gain } -\frac{R_2}{R_1}, \qquad f_c = \frac{1}{2\pi R_2 C}$$

First-order high-pass (inverting)



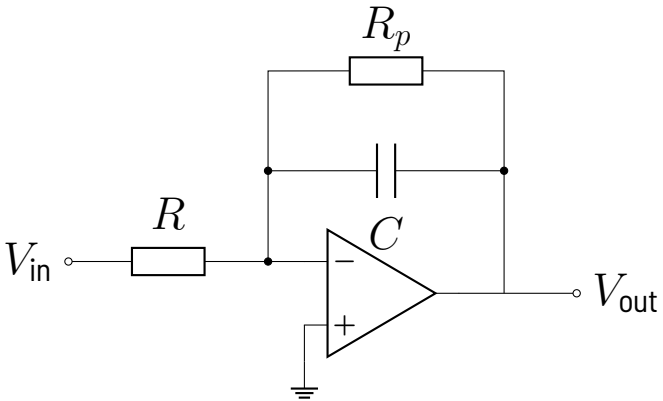
Inverting first-order high-pass

$$\underline{H} = -\frac{R_2}{R_1}\frac{j\omega R_1 C_1}{1 + j\omega R_1 C_1} \qquad \text{HF gain } -\frac{R_2}{R_1}, \qquad f_c = \frac{1}{2\pi R_1 C_1}$$

- Falstad: all three circuits with $f_c = 995$ Hz, input 0.1 V at 1 kHz
  - non-inverting LP ($G = 2$): 0.141 V; inverting LP (-10): 0.705 V; inverting HP (-10): 0.709 V

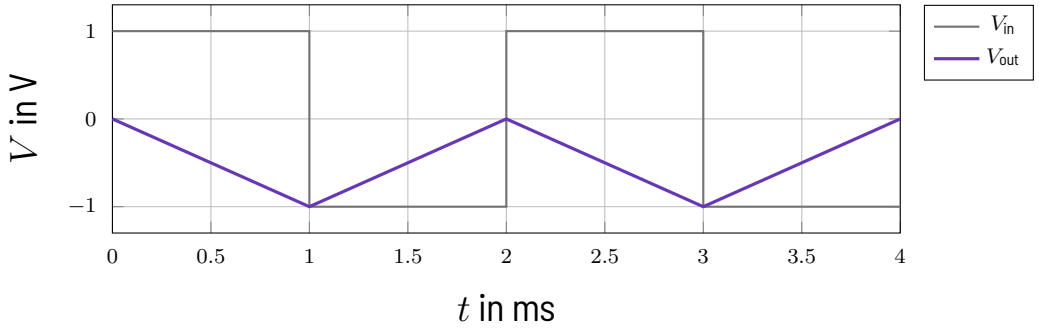
▶ Falstad: `examples/filters/filter_active_first_order.txt`

Integrator



- the current V_{in}/R charges C: $V_{\text{out}}(t) = -\frac{1}{RC}\int V_{\text{in}}\,dt$, $\underline{H} = -\frac{1}{j\omega RC}$
- -20 dB per decade at all frequencies; DC gain ∞: any offset drives it into saturation
- practical integrator: R_p limits the DC gain to $-R_p/R$ (it is then a low-pass with $f_c = 1/(2\pi R_p C)$)

Integrator: square wave in, triangle out

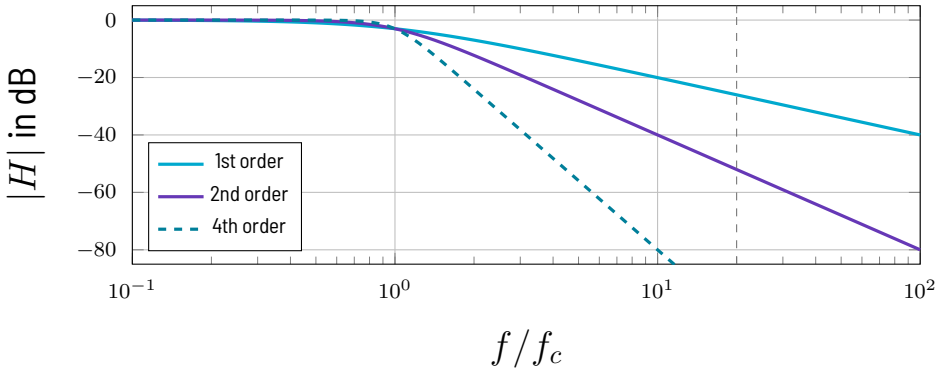


- $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$, $R_p = 1\,\text{M}\Omega$; $V_{in} = \pm 1\,\text{V}$, $500\,\text{Hz}$
- slope $V_{in}/(RC) = 1000\,\text{V/s}$: $1\,\text{V}$ in $1\,\text{ms}$, triangle $1.0\,\text{V}$ peak-to-peak
- the triangle starts between $0\,\text{V}$ and $-1\,\text{V}$; R_p centres it slowly ($\tau = R_pC = 0.1\,\text{s}$)

▶ Falstad: `examples/filters/filter_integrator.txt`

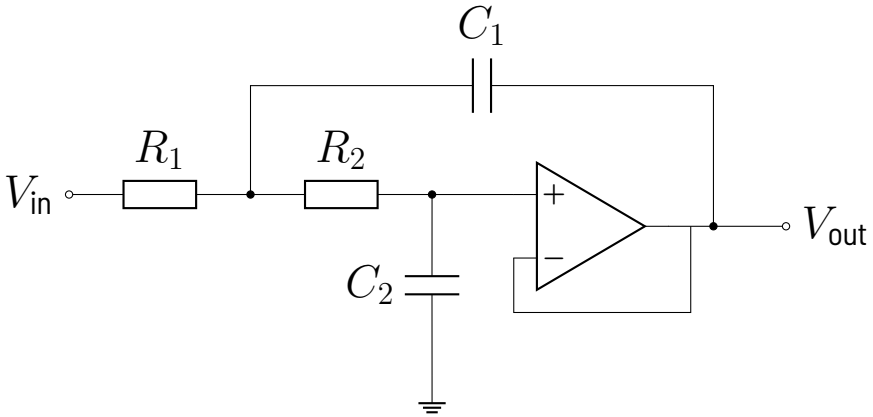
7.3 Second-Order Filters

Filter order and attenuation



- Butterworth filter of order n: $|\underline{H}| = 1/\sqrt{1+(f/f_c)^{2n}}$; slope $-20n$ dB per decade
- dashed line: $f = 20\ f_c$ (50 Hz for $f_c = 2.5$ Hz): -26 dB, -52 dB, -104 dB

Sallen-Key low-pass (unity gain)



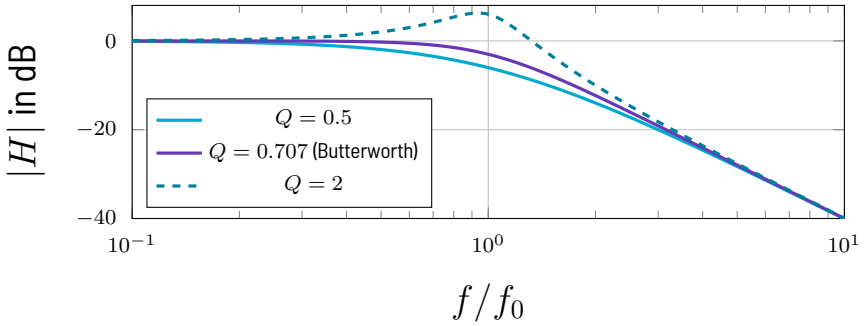
- the follower makes $V_{\text{out}} = V_+$; C_1 feeds the output back to the middle node (positive feedback near f_0)
- $\underline{H} = \dfrac{1}{1 + j\omega C_2(R_1 + R_2) - \omega^2 R_1 R_2 C_1 C_2}$

Natural frequency and quality factor

- comparing with $1/(1 + j\frac{f}{Qf_0} - (f/f_0)^2)$ gives f_0 and Q:

Unity-gain Sallen-Key low-pass

$$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \qquad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{C_2\,(R_1 + R_2)} \overset{R_1=R_2}{=} \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$



- $|\underline{H}(f_0)| = Q$; Butterworth $Q = 1/\sqrt{2}$: maximally flat, -3 dB at f_0; higher Q: peaking and ringing

Design of a Butterworth Sallen-Key low-pass

- capacitors come in coarse steps: choose them first, then calculate the resistors

Design card: Unity-gain Sallen-Key low-pass, $Q = 0.707$

1. Choose $R_1 = R_2 = R$. Then $Q = 0.707$ requires $C_1 = 2\,C_2$ (C_1 goes to the output).
2. Choose C_2 from the E6 series, $C_1 \approx 2C_2$ (e.g. 10 nF and 22 nF: $Q = 0.74$).
3. Calculate $R = \dfrac{1}{2\pi f_0 \sqrt{C_1 C_2}}$ and round to the E24 or E96 series.
4. Check f_0 and Q with the rounded values. Use film or C0G capacitors (stable, low leakage).

- example $f_0 = 1$ kHz: $C_2 = 10$ nF, $C_1 = 22$ nF, $R = 10.7\,\text{k}\Omega$ (E96): $f_0 = 1003$ Hz, $Q = 0.742$

First order vs. second order in Falstad

	1 kHz	10 kHz
first order RC ($f_c = 995$ Hz)	0.705 V	0.099 V (-20 dB)
Sallen-Key ($f_0 = 1003$ Hz, $Q = 0.742$)	0.744 V	0.0101 V (-40 dB)

- input 1 V amplitude; one decade above the corner the second-order filter attenuates ten times more
- near f_0 both behave similarly

▶ Falstad: `examples/filters/filter_sallen_key.txt`

7.4 The Anti-Aliasing Filter of the Capstone

Why filter before sampling? (preview of chapter 9)

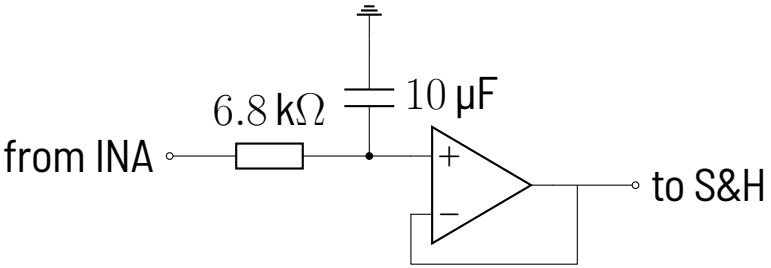
- sampling theorem (Nyquist, Shannon [10]): only frequencies below $f_s/2$ are reproduced correctly
- with $f_s = 5\,\text{Hz}$: $f_s/2 = 2.5\,\text{Hz} \Rightarrow$ cutoff of the low-pass $f_c \approx 2.5\,\text{Hz}$
- a component above $f_s/2$ is not removed by sampling – it appears at a wrong, low frequency (**aliasing**)
  - $50\,\text{Hz} = 10 \cdot f_s$: the samples see the interference always at the same phase – a constant, random-looking error
  - if the mains frequency drifts to $50.1\,\text{Hz}$, the error becomes a slow $0.1\,\text{Hz}$ wave in the data
- after sampling nothing can separate the alias from the signal: the filter must act **before** the S&H

How much attenuation do we get at 50 Hz?

- first order, $f_c = 2.5\,\text{Hz}$: $|\underline{H}(50\,\text{Hz})| = \dfrac{1}{\sqrt{1+(50/2.5)^2}} = \dfrac{1}{\sqrt{401}} = 0.050$ $\hat{=}\ -26\,\text{dB}$
- second order (Butterworth): $\dfrac{1}{\sqrt{1+(50/2.5)^4}} = 0.0025\ \hat{=}\ -52\,\text{dB}$
- capstone filter $6.8\,\text{k}\Omega$, $10\,\mu\text{F}$: $f_c = 2.34\,\text{Hz}$, $|\underline{H}| = 1/\sqrt{1+(50/2.34)^2} = 0.047\ \hat{=}\ -26.6\,\text{dB}$
- interference before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$

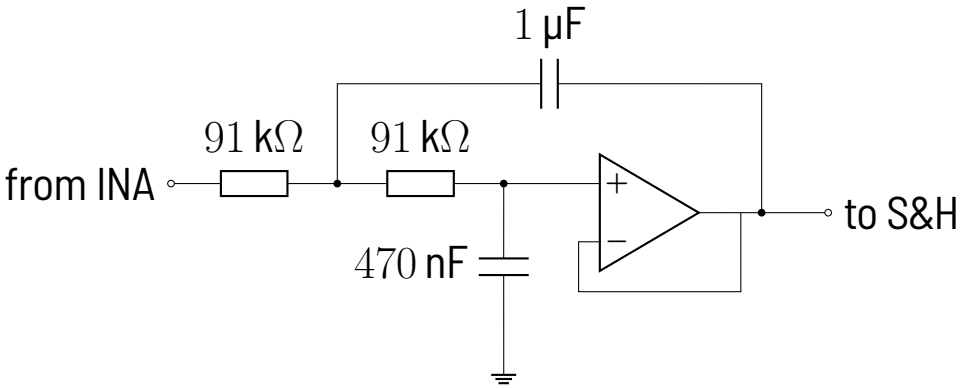
	amplitude	in LSB ($0.5\,\text{V}$)	
no filter	$0.52\,\text{V}$	1 LSB	useless
first order, $2.34\,\text{Hz}$	$25\,\text{mV}$	0.05 LSB	sufficient
Sallen-Key, $2.55\,\text{Hz}$	$1.4\,\text{mV}$	0.003 LSB	large margin

First-order anti-aliasing filter with E-series parts



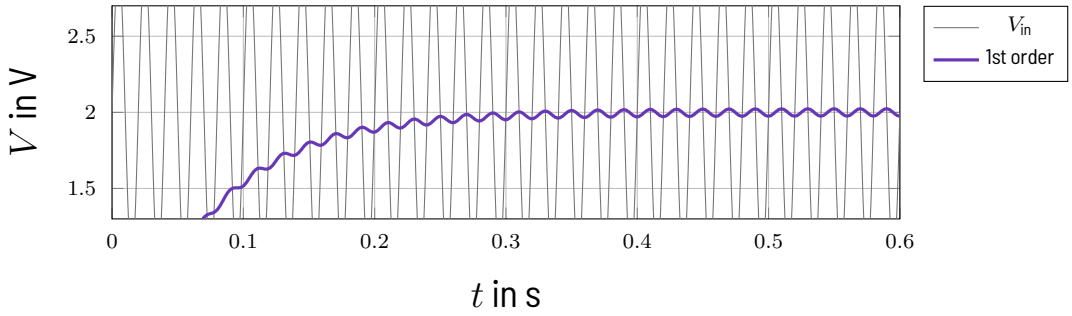
- $RC = 1/(2\pi \cdot 2.5\,\text{Hz}) = 63.7\,\text{ms}$; choose $C = 10\,\mu\text{F} \Rightarrow R = 6.37\,\text{k}\Omega$
- round so that $f_c \leq f_s/2$: $6.8\,\text{k}\Omega$ (E12) $\Rightarrow f_c = 2.34\,\text{Hz}$, $-26.6\,\text{dB}$ at $50\,\text{Hz}$
- also usable: $62\,\text{k}\Omega$ with $1\,\mu\text{F}$ ($f_c = 2.57\,\text{Hz}$, slightly above $f_s/2$, still $-26\,\text{dB}$ at $50\,\text{Hz}$)
- the follower decouples the filter from the S&H input
- step response: $\tau = 68\,\text{ms}$; settling to $1\,\%$ takes $4.6\,\tau = 0.31\,\text{s}$

Second-order version (Sallen-Key)



- $C_1 = 1\,\mu\text{F}, C_2 = 470\,\text{nF}$: $Q = \frac{1}{2}\sqrt{1/0.47} = 0.729$
- $R = 1/(2\pi \cdot 2.5\,\text{Hz} \cdot \sqrt{1\,\mu\text{F} \cdot 470\,\text{nF}}) = 92.8\,\text{k}\Omega \rightarrow$ E24 $91\,\text{k}\Omega$: $f_0 = 2.55\,\text{Hz}$

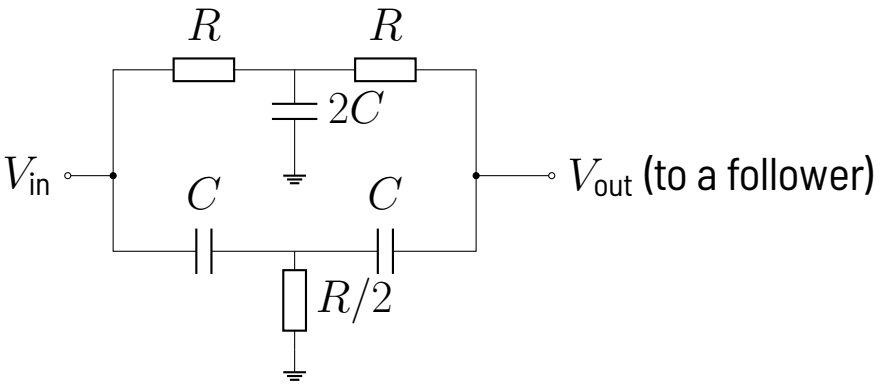
Both filters in Falstad



- input: 2 V signal + 0.52 V, 50 Hz interference (= 4.875 mV $\times$ 107.6)
- after settling: both outputs 2.00 V DC; remaining 50 Hz: 25 mV (1st order), 1.4 mV (Sallen-Key)

▶ Falstad: `examples/filters/filter_antialias.txt`

Alternative: twin-T notch at 50 Hz



- zero transmission at $f_0 = 1/(2\pi RC)$: $R = 31.8\,\text{k}\Omega$, $C = 100\,\text{nF} \Rightarrow 50\,\text{Hz}$
- Falstad: 50 Hz reduced from 1 V to 0.45 mV; but 5 Hz is also attenuated (0.927)
- needs precise parts, removes only one frequency – it does **not** replace the anti-aliasing low-pass

▶ Falstad: `examples/filters/filter_twin_t.txt`

Design card: Anti-aliasing low-pass

1. Cutoff frequency from the sample rate: $f_c \leq f_s/2$ (capstone: $f_s = 5$ Hz, $f_c = 2.34$ Hz).
2. Required attenuation: interference amplitude at the ADC input compared with LSB$/2$; with $f/f_c = 20$ a first-order filter gives -26 dB, a second-order filter -52 dB (capstone: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < 250\,\text{mV}$).
3. First order: $RC = 1/(2\pi f_c)$; choose C, calculate R, round to the E-series so that $f_c \leq f_s/2$ (capstone: 6.8 k, $10\,\mu$F). Buffer the output.
4. Second order: unity-gain Sallen-Key with $C_1 \approx 2C_2$, $R = 1/(2\pi f_0 \sqrt{C_1 C_2})$.
5. Check: settling time of a step ($\approx 5\tau$) against the sample period, and headroom of the amplifier before the filter (signal + interference must not clip).

7.5 Exercises

Exercise 7.1: RC low-pass

An RC low-pass has $R = 1.6\,\text{k}\Omega$ and $C = 100\,\text{nF}$.

a) Calculate the cutoff frequency.
b) Calculate magnitude (also in dB) and phase at 100 Hz, 1 kHz and 10 kHz.
c) What changes if a load of $1.6\,\text{k}\Omega$ is connected to the output?
d) Check b) in Falstad.

Solution 7.1

a) $f_c = 1/(2\pi \cdot 1.6\,\text{k}\Omega \cdot 100\,\text{nF}) = 995\,\text{Hz}$.

b) $100\,\text{Hz}$: $f/f_c = 0.1005$, $|H| = 1/\sqrt{1.0101} = 0.995$ (-0.04 dB), $\varphi = -5.7°$.
$1\,\text{kHz}$: $f/f_c = 1.005$, $|H| = 0.705$ (-3.0 dB), $\varphi = -45.2°$.
$10\,\text{kHz}$: $f/f_c = 10.05$, $|H| = 0.099$ (-20.1 dB), $\varphi = -84.3°$.

c) The load is in parallel to C. Thevenin: $V_\text{th} = V_\text{in}/2$, $R_\text{th} = 0.8\,\text{k}\Omega$: the DC gain drops to 0.5 and f_c doubles to $1.99\,\text{kHz}$. A follower avoids this.

d) ▶ Falstad: `examples/filters/filter_rc_lowpass.txt`
amplitudes $0.995\,\text{V}$, $0.705\,\text{V}$ (lagging by $45°$, i.e. $125\,\mu\text{s}$) and $0.099\,\text{V}$.

Exercise 7.2: First-order active filters

a) Design an inverting low-pass with DC gain -10, input resistance $10\,\text{k}\Omega$ and $f_c = 1\,\text{kHz}$.
b) Calculate the output amplitude for $0.1\,\text{V}$ at $1\,\text{kHz}$ with C rounded to the E-series.
c) Design a non-inverting low-pass with $G = 2$ and the same f_c, using $C = 10\,\text{nF}$.
d) Design an inverting high-pass with HF gain -10, $R_1 = 10\,\text{k}\Omega$, $f_c \approx 1\,\text{kHz}$.
e) Check b) to d) in Falstad (input $0.1\,\text{V}$, $1\,\text{kHz}$).

Solution 7.2

a) $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $C = 1/(2\pi \cdot 100\,\text{k}\Omega \cdot 1\,\text{kHz}) = 1.59\,\text{nF}$.

b) $C = 1.6\,\text{nF}$ (E24): $f_c = 995\,\text{Hz}$; $0.1\,\text{V} \cdot 10 \cdot 0.705 = 0.705\,\text{V}$.

c) $R = 1/(2\pi \cdot 1\,\text{kHz} \cdot 10\,\text{nF}) = 15.9\,\text{k}\Omega \rightarrow 16\,\text{k}\Omega$; gain stage $R_1 = R_2 = 10\,\text{k}\Omega$. Output: $0.1\,\text{V} \cdot 2 \cdot 0.705 = 0.141\,\text{V}$.

d) $R_2 = 100\,\text{k}\Omega$, $C_1 = 1/(2\pi \cdot 10\,\text{k}\Omega \cdot 1\,\text{kHz}) = 15.9\,\text{nF} \rightarrow 16\,\text{nF}$: $f_c = 995\,\text{Hz}$, output $0.1\,\text{V} \cdot 10 \cdot 0.709 = 0.709\,\text{V}$.

e) ▶ Falstad: `examples/filters/filter_active_first_order.txt`

`Vout_a` $= 0.141\,\text{V}$, `Vout_b` $= 0.705\,\text{V}$, `Vout_c` $= 0.709\,\text{V}$.

Exercise 7.3: Integrator

An integrator has $R = 10\,\text{k}\Omega$, $C = 100\,\text{nF}$ and $R_p = 1\,\text{M}\Omega$ parallel to C. The input is a square wave of $\pm 1\,\text{V}$ and $500\,\text{Hz}$, starting with $+1\,\text{V}$ at $t = 0$, $V_{\text{out}}(0) = 0$.

a) Calculate the slope of the output voltage and sketch $V_{\text{out}}(t)$ for the first two periods.

b) Calculate the peak-to-peak value of the triangle.

c) What is the purpose of R_p? What happens without it if the op-amp has an offset of $1\,\text{mV}$?

d) Above which frequency does the circuit behave as an integrator?

e) Check a) and b) in Falstad.

Solution 7.3

a) $dV_{\text{out}}/dt = -V_{\text{in}}/(RC) = -1\,\text{V}/1\,\text{ms} = -1000\,\text{V/s}$ while $V_{\text{in}} = +1\,\text{V}$. Triangle: $0 \to -1\,\text{V}$ in the first $1\,\text{ms}$, back to $0\,\text{V}$ at $2\,\text{ms}$, and so on.

b) $1\,\text{V}$ per half period: $1.0\,\text{V}$ peak-to-peak.

c) R_p limits the DC gain to $-R_p/R = -100$. Without it, $1\,\text{mV}$ offset causes a current of $0.1\,\mu\text{A}$ that charges C at $1\,\text{V/s}$: the output saturates after some seconds. With R_p: DC error $0.1\,\text{V}$.

d) Above $f_c = 1/(2\pi R_p C) = 1.6\,\text{Hz}$; at $500\,\text{Hz}$ the error is negligible.

e) ▶ Falstad: `examples/filters/filter_integrator.txt`
the output reaches about $-1\,\text{V}$ after $1\,\text{ms}$ ($-0.98\,\text{V}$ in the simulation, R_p and the time step cost a little), the triangle has $1.0\,\text{V}$ peak-to-peak and drifts slowly towards a symmetric position.

Exercise 7.4: Sallen-Key low-pass

Design a unity-gain Sallen-Key low-pass with $f_0 = 1$ kHz and Butterworth characteristic.

a) Start with $C_2 = 10$ nF and $R_1 = R_2$. Choose C_1 from the E6 series and calculate R (E96).
b) Calculate Q and f_0 with the chosen values.
c) Calculate the amplitude for 1 V input at 1 kHz and 10 kHz. Compare with a first-order filter.
d) Check c) in Falstad.

Solution 7.4

a) $Q = 0.707$ needs $C_1 = 2C_2 = 20\,\text{nF}$; E6: $22\,\text{nF}$.
$R = 1/(2\pi \cdot 1\,\text{kHz} \cdot \sqrt{22\,\text{nF} \cdot 10\,\text{nF}}) = 10.73\,\text{k}\Omega \to 10.7\,\text{k}\Omega$.

b) $Q = \frac{1}{2}\sqrt{22/10} = 0.742$; $f_0 = 1/(2\pi \cdot 10.7\,\text{k}\Omega \cdot 14.83\,\text{nF}) = 1003\,\text{Hz}$.

c) $|H| = 1/\sqrt{(1-x^2)^2 + (x/Q)^2}$ with $x = f/f_0$. 1 kHz: $x = 0.997$, $|H| = 0.744$ ($0.744\,\text{V}$; first order: $0.705\,\text{V}$). 10 kHz: $x = 9.97$, $|H| = 1/\sqrt{98.4^2 + 13.4^2} = 0.0101$ (-40 dB; first order: 0.099, -20 dB).

d) ▶ Falstad: `examples/filters/filter_sallen_key.txt`
`VSK_1kHz` $= 0.744\,\text{V}$, `VSK_10kHz` $= 10.1\,\text{mV}$,
`V1st_1kHz` $= 0.705\,\text{V}$, `V1st_10kHz` $= 0.099\,\text{V}$.

Exercise 7.5: Capstone: anti-aliasing low-pass

The capstone samples with $f_s = 5$ Hz. The amplifier has a gain of 107.6 and its output carries a 50 Hz interference (4.875 mV at the bridge tap). The ADC has 3 bits and 4 V full scale.

a) Choose the cutoff frequency of the anti-aliasing filter and justify it.

b) Design a first-order filter (follower behind an RC) with $C = 10$ µF and an E12 resistor.

c) Calculate the attenuation at 50 Hz in dB for an ideal first-order and second-order filter with $f_c = 2.5$ Hz, and for your filter.

d) Calculate the remaining interference at the ADC input with your filter and compare it with LSB$/2$.

e) How long does the filter output need to settle within $1\,\%$ after a step of the sensor signal?

f) Check the result in Falstad (signal 2 V DC with 0.52 V at 50 Hz).

Solution 7.5

a) Nyquist: only frequencies below $f_s/2 = 2.5\,\text{Hz}$ can be sampled without aliasing, so $f_c \leq 2.5\,\text{Hz}$. The temperature changes much more slowly than that.

b) $R = 1/(2\pi{\cdot}2.5\,\text{Hz}{\cdot}10\,\mu\text{F}) = 6.37\,\text{k}\Omega \to 6.8\,\text{k}\Omega$ (rounded up, so that f_c stays below $f_s/2$): $f_c = 1/(2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}) = 2.34\,\text{Hz}$. ($62\,\text{k}\Omega$ with $1\,\mu\text{F}$, $f_c = 2.57\,\text{Hz}$, slightly above $f_s/2$, would also be acceptable.)

c) Ideal first order: $1/\sqrt{1+20^2} = 0.0499$, $-26\,\text{dB}$; second order: $1/\sqrt{1+20^4} = 0.0025$, $-52\,\text{dB}$. With $f_c = 2.34\,\text{Hz}$: $1/\sqrt{1+(50/2.34)^2} = 0.047$, $-26.6\,\text{dB}$.

d) Before the filter: $4.875\,\text{mV} \cdot 107.6 = 0.52\,\text{V}$; after it: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$ (0.05 LSB). Sufficient.

e) $\tau = RC = 68\,\text{ms}$; 1 %: $t = \tau \ln 100 = 4.6\,\tau = 0.31\,\text{s}$, i.e. about 1.5 sample periods.

f) ▶ Falstad: `examples/filters/filter_antialias.txt`

after about $0.6\,\text{s}$ `Vout_1st` shows $2.00\,\text{V}$ with $25\,\text{mV}$ of $50\,\text{Hz}$ ripple; the Sallen-Key output ($91\,\text{k}\Omega$, $1\,\mu\text{F}$, $470\,\text{nF}$) leaves only $1.4\,\text{mV}$.

Exercise 7.6: Twin-T notch

a) Dimension a twin-T notch for 50 Hz with $C = 100$ nF.

b) Its transfer function is $\underline{H} = \dfrac{1 - x^2}{1 - x^2 + j4x}$ with $x = f/f_0$. Calculate $|H|$ at 5 Hz and at 50.5 Hz.

c) Compare with the first-order low-pass with $f_c = 2.34$ Hz. Which filter would you use in the capstone?

d) Check a) and b) in Falstad.

Solution 7.6

a) $R = 1/(2\pi \cdot 50\,\text{Hz} \cdot 100\,\text{nF}) = 31.8\,\text{k}\Omega$, $R/2 = 15.9\,\text{k}\Omega$, $2C = 200\,\text{nF}$.

b) $5\,\text{Hz}$: $x = 0.1$, $|H| = 0.99/\sqrt{0.99^2 + 0.4^2} = 0.927$.
$50.5\,\text{Hz}$: $x = 1.01$, $|H| = 0.0201/\sqrt{0.0201^2 + 4.04^2} = 0.0050$.

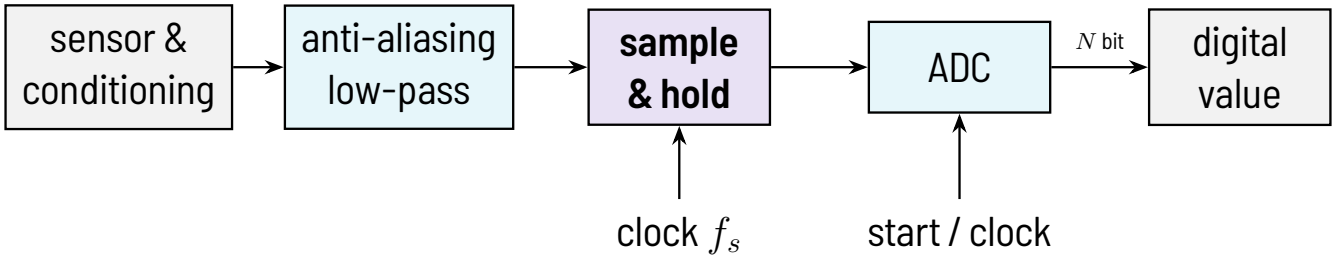
c) At $50\,\text{Hz}$ the notch is much better ($-46\,\text{dB}$ at $50.5\,\text{Hz}$ and more at exactly $50\,\text{Hz}$), but it attenuates the signal at $5\,\text{Hz}$ as well, it needs precise components, and it does nothing against other frequencies above $f_s/2$. The anti-aliasing low-pass is required anyway; the notch is at most an addition.

d) ▶ Falstad: `examples/filters/filter_twin_t.txt`
`Vout_5Hz` $= 0.927\,\text{V}$, `Vout_50Hz` $= 0.45\,\text{mV}$ (input $1\,\text{V}$).

8 Sample and Hold

8.1 Why Sample and Hold?

Position in the signal chain



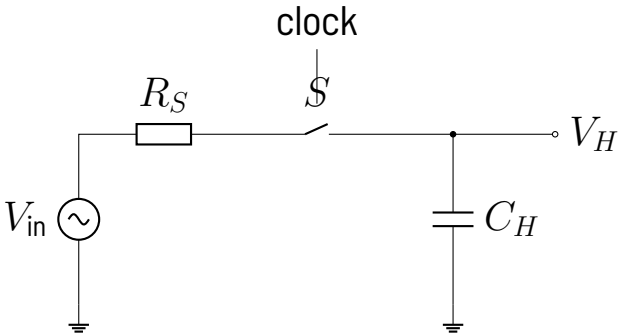
- the S&H stage **defines the sampling instant** $t_k = k\,T_s$, $T_s = 1/f_s$
- it keeps the voltage **constant while the ADC converts**
- capstone: $f_s = 5$ Hz, 3-bit flash converter, full scale 4 V

What goes wrong without S&H

- the input changes during the conversion time t_{conv}
- the error must stay below $\frac{1}{2}$ LSB: $\left|\frac{dV}{dt}\right|_{\text{max}} t_{\text{conv}} < \frac{1}{2}\,\text{LSB}$
- sine $V(t) = \hat{V}\sin(2\pi f t)$: maximum slope $2\pi f \hat{V}$
- example: 12-bit ADC, 4 V full scale (LSB $= 0.98$ mV), $t_{\text{conv}} = 10$ µs, $\hat{V} = 2$ V
  - $f < \frac{\text{LSB}/2}{2\pi \hat{V} t_{\text{conv}}} = 3.9$ Hz without S&H
  - with S&H only the much shorter **aperture time** counts (nanoseconds)
- a **flash** converter decides in one step, but the digital outputs still have to be read and displayed at a defined instant

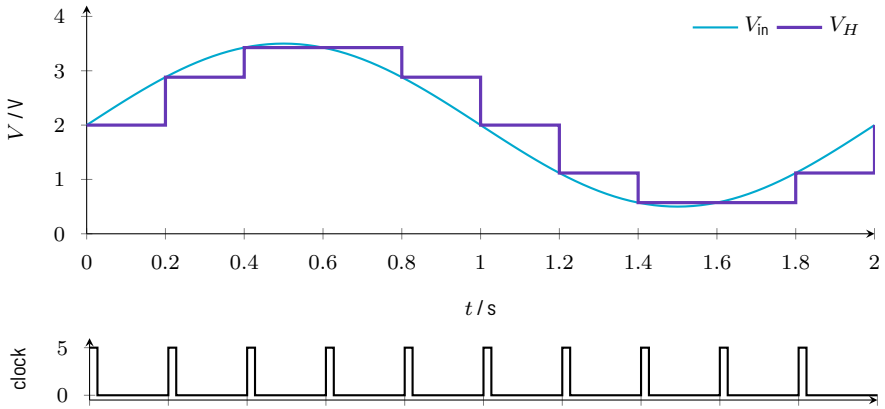
8.2 The Track-and-Hold Principle

Switch and hold capacitor



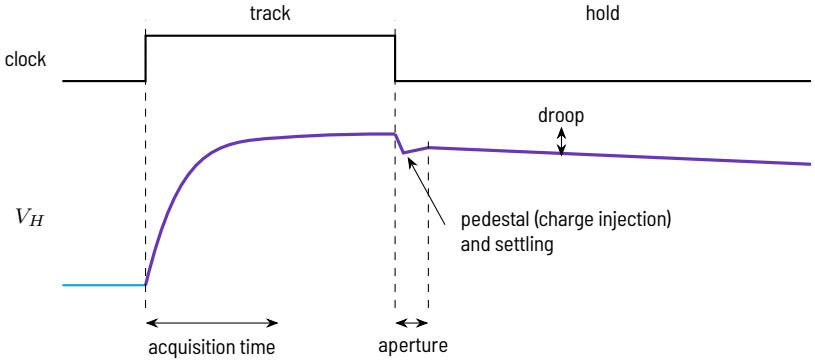
- clock high: switch closed, C_H follows V_{in} (**track**)
- clock low: switch open, C_H keeps its charge (**hold**)
- capstone clock: 5 Hz, duty cycle $10\,\%$: track 20 ms, hold 180 ms

Timing diagram



- the output is a **staircase**: one step per sampling period
- the value of each step is V_{in} at the **end** of the track phase

Terms



- **acquisition time**: track time needed until V_H is within the error band (e.g. $\frac{1}{2}$ LSB)
- **aperture time**: delay between the hold command and the real opening of the switch; its variation is the **aperture jitter**
- **pedestal**: step at the transition to hold; **droop**: slow change during hold

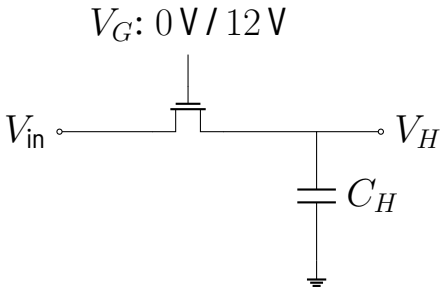
Falstad: basic S&H

- $V_{\text{in}} = 2\,\text{V} + 1.5\,\text{V}\ \sin(2\pi \cdot 0.5\,\text{Hz} \cdot t)$, $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$
- clock rail: square wave 5 Hz, 0 V … 5 V, duty cycle 0.1
- held values: 2.094 V, 2.956 V, 3.452 V, 3.394 V, 2.804 V
- the scope shows V_{in}, V_H (label `Vhold`) and the clock

▶ Falstad: `examples/samplehold/sh_basic.txt`

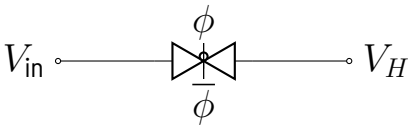
8.3 Analog Switches

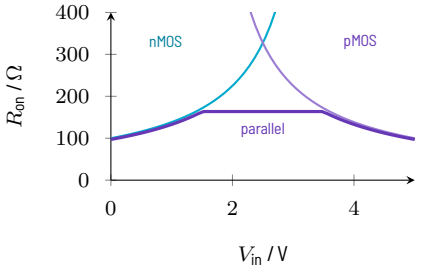
The MOSFET as a switch



- n-channel MOSFET: conducts if $V_{GS} = V_G - V_{\text{in}} > V_{th}$
- on-resistance $R_{\text{on}} \approx \frac{1}{k\,(V_{GS} - V_{th})}$: depends on V_{in}!
- near $V_{\text{in}} \approx V_G - V_{th}$ the switch does not close any more
- off: leakage currents in the pA ...nA range

CMOS transmission gate





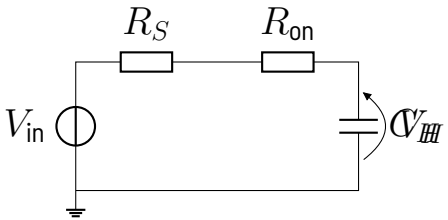
- nMOS and pMOS in parallel, driven with complementary signals
- nMOS conducts well near 0 V, pMOS near V_{DD}: together almost constant R_{on}
- integrated as CD4066, 74HC4066 (4 switches, $R_{\text{on}} \approx 50\,\Omega \ldots 300\,\Omega$), DG4xx

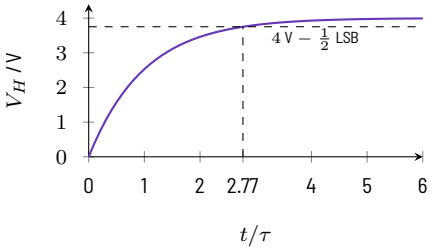
The analog switch in Falstad

- element **Analog Switch** (models a FET switch / transmission gate)
  - closed if the control input is above 2.5 V
  - $R_{\text{on}} = 20\,\Omega$, $R_{\text{off}} = 10$ GΩ (defaults, edit with a double click)
- no charge injection, no leakage: add them as extra elements if you want to see their effect
- in our circuits: $R_S = 1$ kΩ in series
  - limits the current from the op-amp output into the empty capacitor
  - isolates the op-amp from the large capacitive load (stability)
  - real lab switches have R_{on} of some $100\,\Omega$ anyway

8.4 Designing the Hold Capacitor

Acquisition: charging in the track window





- $V_H(t) = V_{\text{in}}\,(1 - e^{-t/\tau})$ with $\tau = (R_S + R_{\text{on}})\,C_H$
- error after time t: $V_{\text{in}}\,e^{-t/\tau}$; worst case: step over the full scale V_{FS}

Settling to $\frac{1}{2}$ LSB

- requirement: $V_{\text{FS}}\, e^{-t/\tau} \leq \frac{1}{2}\,\text{LSB} = \frac{V_{\text{FS}}}{2^{N+1}}$

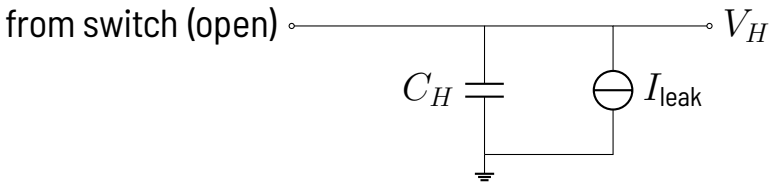
Acquisition time for N bit

$$t_{\text{acq}} = \tau\, \ln\left(2^{N+1}\right) = (N+1)\, \ln 2 \cdot \tau \approx 0.69\,(N+1)\,\tau$$

- $N = 3$: $t_{\text{acq}} = 2.77\,\tau$; $N = 8$: $6.2\,\tau$; $N = 12$: $9.0\,\tau$; $N = 16$: $11.8\,\tau$
- design rule: $t_{\text{acq}} \leq t_{\text{track}} \quad \Rightarrow \quad C_H \leq \dfrac{t_{\text{track}}}{(R_S + R_{\text{on}})\, \ln 2^{N+1}}$
- example: $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$, $C_H = 1\,\mu\text{F}$: $\tau = 1.02\,\text{ms}$, $t_{\text{acq}} = 2.83\,\text{ms} \ll 20\,\text{ms}$

▶ Falstad: `examples/samplehold/sh_acquisition.txt`

Droop: losing charge in the hold phase



- leakage currents: switch (off), capacitor dielectric, input bias current of the buffer, board surface

Droop

$$\frac{dV_H}{dt} = \frac{I_{\text{leak}}}{C_H} \quad \Rightarrow \quad \Delta V_{\text{droop}} = \frac{I_{\text{leak}}\, t_{\text{hold}}}{C_H} \quad \Rightarrow \quad C_H \geq \frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}}$$

- t_{hold}: until the conversion is finished; worst case the whole period $T_s - t_{\text{track}}$

Too small, right, too large

- Falstad: $V_{\text{in}} = 3\,\text{V}$, $R_S = 1\,\text{k}\Omega$, leakage $I_{\text{leak}} = 50\,\text{nA}$ (exaggerated)

	$C_H = 10\,\text{nF}$	$C_H = 1\,\mu\text{F}$	$C_H = 100\,\mu\text{F}$
τ	$10\,\mu\text{s}$	$1.02\,\text{ms}$	$102\,\text{ms}$
charged in $20\,\text{ms}$?	yes	yes	only $18\,\%$ per sample
droop in $180\,\text{ms}$	$0.9\,\text{V}$	$9\,\text{mV}$	$0.09\,\text{mV}$
Falstad	$3.00\,\text{V} \to 2.10\,\text{V}$	$3.00\,\text{V} \to 2.99\,\text{V}$	$0.534\,\text{V}$, ..., $1.876\,\text{V}$ (5th)

▶ Falstad: `examples/samplehold/sh_capsize.txt`

Choosing C_H: the window

- upper limit from the acquisition, lower limit from the droop:

Hold capacitor

$$\frac{I_{\text{leak}}\, t_{\text{hold}}}{\Delta V_{\text{max}}} \leq C_H \leq \frac{t_{\text{track}}}{(R_S + R_{\text{on}}) \ln 2^{N+1}}$$

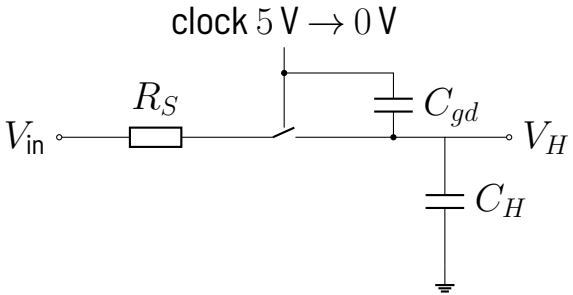
- capstone values: $I_{\text{leak}} = 10\,\text{nA}$, $\Delta V_{\text{max}} = \frac{1}{10}\,\text{LSB} = 50\,\text{mV}$, $t_{\text{hold}} = 180\,\text{ms}$, $t_{\text{track}} = 20\,\text{ms}$, $R_S + R_{\text{on}} = 1.02\,\text{k}\Omega$
  - $36\,\text{nF} \leq C_H \leq 7.1\,\mu\text{F} \quad \Rightarrow \quad C_H = 1\,\mu\text{F}$ (geometric middle, film capacitor)
- choose the capacitor type: polypropylene or polystyrene film (low leakage, low dielectric absorption), C0G ceramic for small values; no electrolytic capacitors

Design card: Sample and Hold

- structure: input follower $\rightarrow$ R_S + switch $\rightarrow$ C_H to ground $\rightarrow$ output follower
- clock: f_s, short duty cycle; track $t_{\text{track}} = D\,T_s$ (capstone: 5 Hz, $10\,\%$: 20 ms)
- charging: $\tau = (R_S + R_{\text{on}})\,C_H$, settled to $\frac{1}{2}$ LSB after $\tau \ln 2^{N+1}$ ($N = 3$: $2.77\,\tau$) $\leq t_{\text{track}}$
- droop: $\Delta V = I_{\text{leak}}\,t_{\text{hold}}/C_H \leq$ fraction of an LSB
- pedestal: $\Delta V = C_{gd}\,\Delta V_{\text{clk}}/(C_H + C_{gd})$; larger C_H helps
- capstone choice: $R_S = 1\,\text{k}\Omega$, $C_H = 1\,\mu$F film: $t_{\text{acq}} = 2.8$ ms, droop 1.8 mV at 10 nA

8.5 Errors of a Real Sample and Hold

Charge injection (pedestal)



- the gate of the switch is coupled to the hold node through C_{gd} (and the channel charge)
- when the switch opens, the clock step ΔV_{clk} is divided by C_{gd} and C_H:

$$\Delta V_{\text{ped}} = -\Delta V_{\text{clk}} \frac{C_{gd}}{C_{gd} + C_H} \approx -\frac{Q_{\text{inj}}}{C_H}$$

- $C_{gd} = 10\,\text{pF}$, $5\,\text{V}$: $C_H = 1\,\text{nF}$: $-49.5\,\text{mV}$; $C_H = 100\,\text{nF}$: $-0.5\,\text{mV}$

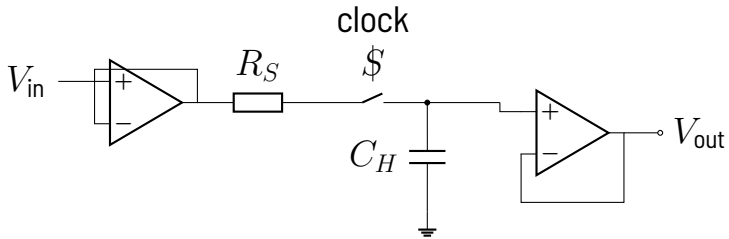
▶ Falstad: `examples/samplehold/sh_pedestal.txt`

Feedthrough, aperture and jitter

- **feedthrough**: the open switch has a capacitance C_{ds}; the input still reaches C_H: $\Delta V_H = \Delta V_{\text{in}} \frac{C_{\text{ds}}}{C_{\text{ds}}+C_H}$
- **aperture delay** t_a: the switch opens a little after the clock edge; constant, only shifts the sampling instant
- **aperture jitter** t_j: random variation of the sampling instant
  - error $\Delta V = \frac{dV}{dt}\, t_j$; sine: $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j$
  - requirement $\Delta V_{\text{max}} < \frac{1}{2}$ LSB: $t_j < \frac{1}{2\pi f\, 2^{N+1}} \cdot \frac{V_{\text{FS}}}{\hat{V}}$
  - example: 12 bit, full-scale sine of 1 MHz: $t_j < 19$ ps
- **dielectric absorption**: held voltage creeps back towards the previous value

8.6 The Buffered Sample and Hold

Follower – switch – capacitor – follower



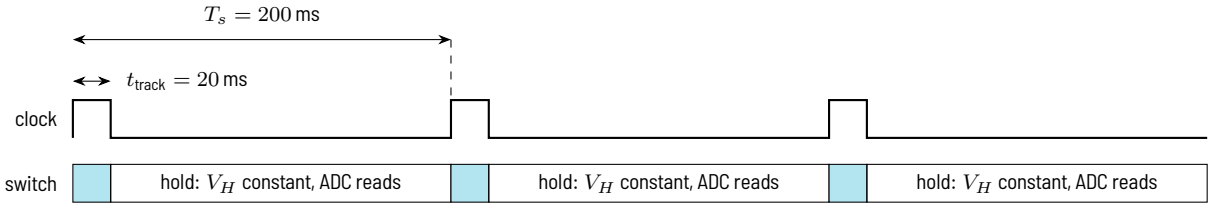
- **input buffer**: the charging current comes from the op-amp, not from the signal source
- **output buffer**: draws (almost) no current from C_H; only its input bias current adds to the leakage

Why both buffers are needed

- without buffers: source $R_i = 10\,\text{k}\Omega$, load $R_L = 100\,\text{k}\Omega$ (e.g. ADC input), $V_\text{in} = 3\,\text{V}$
  - track: $\tau = (R_i \parallel R_L)\, C_H = 9.1\,\text{ms}$, final value $3\,\text{V} \cdot \frac{100}{110} = 2.73\,\text{V}$
  - after $20\,\text{ms}$: $2.425\,\text{V}$
  - hold: discharge through R_L, $\tau = 100\,\text{ms}$: $0.401\,\text{V}$ after $180\,\text{ms}$
- with buffers: $3.000\,\text{V}$ during the whole period

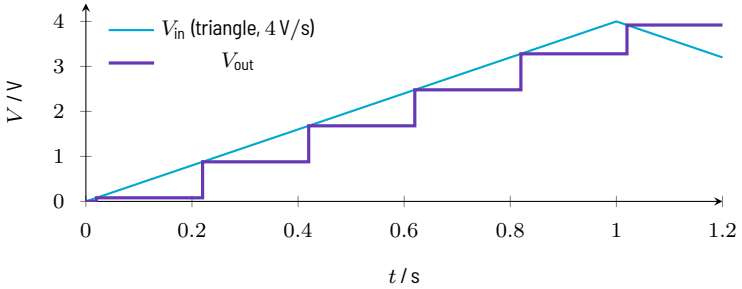
▶ Falstad: `examples/samplehold/sh_buffered.txt`

The sampling clock



- square wave with a short duty cycle $D = t_{\text{track}}/T_s$
- Falstad: rail with waveform **square**, 5 Hz, amplitude 2.5 V, offset 2.5 V, duty cycle 0.1
- in hardware: 555 timer or microcontroller output; 0/5 V logic level drives the switch
- the ADC result is read in the hold phase (e.g. in the middle or at the end)

Sampling a ramp



- buffered S&H, triangle 0 V … 4 V at 0.5 Hz: steps of $4\,\text{V/s} \cdot 0.2\,\text{s} = 0.8$ V
- Falstad: 0.08 V, 0.88 V, 1.68 V, 2.48 V, 3.28 V

▶ Falstad: `examples/samplehold/sh_staircase.txt`

Practical hints

- op-amps: rail-to-rail output for the full range $0\,\mathrm{V} \ldots 4\,\mathrm{V}$ on a single $5\,\mathrm{V}$ supply, FET input for the output buffer
- guard ring around the hold node on the PCB: surface leakage
- integrated S&H amplifiers (LF398, AD783) contain both buffers and the switch; only C_H is external
- in most modern ADCs (SAR, sigma-delta) the S&H is built in: a switched capacitor array at the input
  - the driving op-amp must recharge this capacitor in the acquisition time: the same calculation as above

8.7 Exercises

Exercise 8.1: Choosing the hold capacitor for $f_s = 5\,\text{Hz}$

A buffered S&H stage samples at $f_s = 5$ Hz with a clock duty cycle of $10\,\%$. The charging path consists of $R_S = 1\,\text{k}\Omega$ and a switch with $R_{\text{on}} = 20\,\Omega$. The total leakage current at the hold node (switch, capacitor, buffer bias) is at most $I_{\text{leak}} = 10$ nA. The stage drives a 3-bit converter with $V_{\text{FS}} = 4$ V.

a) Calculate the track and hold times.

b) The capacitor must settle to $\frac{1}{2}$ LSB for a full-scale step within the track phase. Find the upper limit for C_H.

c) The droop during the hold phase must stay below $\frac{1}{10}$ LSB. Find the lower limit for C_H.

d) Choose C_H and calculate acquisition time and droop for your choice.

e) Check the charging curve in Falstad.

Solution 8.1

a) $T_s = 1/f_s = 200\,\text{ms}$, $t_\text{track} = 0.1 \cdot T_s = 20\,\text{ms}$, $t_\text{hold} = 180\,\text{ms}$.

b) LSB $= 4\,\text{V}/8 = 0.5\,\text{V}$. Settling to $\frac{1}{2}$ LSB needs $t = \tau \ln 2^4 = 2.77\,\tau \le 20\,\text{ms}$, so $\tau \le 7.21\,\text{ms}$ and $C_H \le 7.21\,\text{ms}/1020\,\Omega = 7.1\,\mu\text{F}$.

c) $\Delta V_\text{max} = 50\,\text{mV}$: $C_H \ge I_\text{leak}\, t_\text{hold}/\Delta V_\text{max} = 10\,\text{nA} \cdot 0.18\,\text{s}/50\,\text{mV} = 36\,\text{nF}$.

d) The window is $36\,\text{nF}$ …$7.1\,\mu\text{F}$; the geometric middle is about $0.5\,\mu\text{F}$. We choose $C_H = 1\,\mu\text{F}$ (film capacitor). Then $\tau = 1.02\,\text{ms}$, $t_\text{acq} = 2.77\,\tau = 2.83\,\text{ms}$ (seven times faster than required) and $\Delta V_\text{droop} = 10\,\text{nA} \cdot 0.18\,\text{s}/1\,\mu\text{F} = 1.8\,\text{mV}$ (28 times smaller than allowed).

e) ▶ Falstad: `examples/samplehold/sh_acquisition.txt`

With a $4\,\text{V}$ input, `Vhold` shows $2.53\,\text{V}$ after $1.02\,\text{ms}$, $3.75\,\text{V}$ after $2.83\,\text{ms}$ and $4.00\,\text{V}$ at the end of the track phase. (Pause the simulation and use the scope, or reduce the simulation speed.)

Exercise 8.2: Droop and a too large capacitor

The S&H stage of exercise 8.1 has an (exaggerated) leakage current of 50 nA. The input is a constant 3 V.

a) Calculate the droop rate and the voltage at the end of the first hold phase for $C_H = 10$ nF and for $C_H = 1$ µF.

b) With $C_H = 100$ µF, which fraction of the remaining difference is charged in one track window? Calculate the held voltage after the first and after the fifth sample (leakage negligible).

c) Compare with Falstad.

Solution 8.2

a) $C_H = 10\,\text{nF}$: $dV/dt = 50\,\text{nA}/10\,\text{nF} = 5\,\text{V/s}$, droop in $180\,\text{ms}$: $0.9\,\text{V}$, the held voltage falls from $3.00\,\text{V}$ to $2.10\,\text{V}$. $C_H = 1\,\mu\text{F}$: $50\,\text{mV/s}$, droop $9\,\text{mV}$.

b) $\tau = 1.02\,\text{k}\Omega \cdot 100\,\mu\text{F} = 102\,\text{ms}$, fraction $a = 1 - e^{-20/102} = 0.178$. After n samples $V_n = 3\,\text{V}\,(1-(1-a)^n)$: $V_1 = 0.534\,\text{V}$, $V_5 = 3\,\text{V}\,(1-0.822^5) = 1.876\,\text{V}$.

c) ▶ Falstad: `examples/samplehold/sh_capsize.txt`

`V_10n` drops from $3.00\,\text{V}$ to $2.10\,\text{V}$ in every hold phase, `V_1u` loses $9\,\text{mV}$ ($3.00\,\text{V} \rightarrow 2.99\,\text{V}$), `V_100u` climbs in steps: $0.534\,\text{V}$ after the first, $1.876\,\text{V}$ after the fifth sample.

Exercise 8.3: Unbuffered sample and hold

A switch and $C_H = 1\,\mu\text{F}$ are connected directly between a source ($V_{\text{in}} = 3\,\text{V}$, internal resistance $R_i = 10\,\text{k}\Omega$) and a load $R_L = 100\,\text{k}\Omega$. Clock: $5\,\text{Hz}$, $10\,\%$. Neglect R_{on}.

a) Draw the equivalent circuit for the track phase and calculate its final value and time constant.
b) Calculate V_H at the end of the first track phase and at the end of the first hold phase.
c) How does the buffered circuit change the result? Check in Falstad.

Solution 8.3

a) Thevenin equivalent seen by C_H: $V_{\text{th}} = 3\,\text{V} \cdot \frac{100}{110} = 2.727\,\text{V}$, $R_{\text{th}} = R_i \parallel R_L = 9.09\,\text{k}\Omega$, $\tau = 9.09\,\text{ms}$.

b) End of track: $V_H = 2.727\,\text{V}\,(1 - e^{-20/9.09}) = 2.425\,\text{V}$. Hold: C_H discharges through R_L with $\tau = 100\,\text{ms}$: $V_H = 2.425\,\text{V} \cdot e^{-1.8} = 0.401\,\text{V}$.

c) The input follower removes the divider and the slow charging, the output follower removes the discharge: $V_{\text{out}} = 3.000\,\text{V}$ throughout.

▶ Falstad: `examples/samplehold/sh_buffered.txt`

`Vout_unbuf` shows $2.425\,\text{V}$ and $0.401\,\text{V}$, `Vout_buf` $3.000\,\text{V}$.

Exercise 8.4: Pedestal error

The gate of the switch has a coupling capacitance $C_{gd} = 10$ pF to the hold node. The clock falls from 5 V to 0 V when the switch opens.

a) Calculate the pedestal for $C_H = 1$ nF and $C_H = 100$ nF.
b) Which C_H is needed for a pedestal below $\frac{1}{2}$ LSB of a 12-bit converter with $V_{\text{FS}} = 4$ V?
c) Is the pedestal a problem for the 3-bit capstone with $C_H = 1$ µF?

Solution 8.4

a) $\Delta V = -5\,\text{V} \cdot \frac{10\,\text{pF}}{1.01\,\text{nF}} = -49.5\,\text{mV}$; with $100\,\text{nF}$: $-0.50\,\text{mV}$.
▶ Falstad: `examples/samplehold/sh_pedestal.txt`: $2\,\text{V}$ input, held values $1.9505\,\text{V}$ and $1.9995\,\text{V}$.

b) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{13} = 0.49\,\text{mV}$: $C_H \geq 5\,\text{V} \cdot 10\,\text{pF}/0.49\,\text{mV} = 0.1\,\mu\text{F}$.

c) $\Delta V = 50\,\mu\text{V}$, i.e. 10^{-4} of an LSB of $0.5\,\text{V}$: no problem.

Exercise 8.5: Ramp input and step size

A buffered S&H stage ($f_s = 5$ Hz) samples a triangle that rises from 0 V to 4 V in 1 s.

a) Calculate the height of the steps of the output staircase.

b) The sampling instants are at the end of the track windows, $t_k = 20\,\text{ms} + k \cdot 200\,\text{ms}$. List the first five held values.

c) What is the maximum slope of the input if consecutive samples may differ by at most one LSB of the 3-bit converter? What does this mean for a sine with an amplitude of 2 V?

Solution 8.5

a) $4\,\mathrm{V/s} \cdot 0.2\,\mathrm{s} = 0.8\,\mathrm{V}$.

b) $V(t_k) = 4\,\mathrm{V/s} \cdot t_k$: $0.08\,\mathrm{V}$, $0.88\,\mathrm{V}$, $1.68\,\mathrm{V}$, $2.48\,\mathrm{V}$, $3.28\,\mathrm{V}$.

▶ Falstad: `examples/samplehold/sh_staircase.txt`: `Vout` shows the same values (4 mV lower, because C_H lags by $\tau \cdot 4\,\mathrm{V/s}$).

c) $0.5\,\mathrm{V}/0.2\,\mathrm{s} = 2.5\,\mathrm{V/s}$. Sine: $2\pi f \cdot 2\,\mathrm{V} \leq 2.5\,\mathrm{V/s}$ gives $f \leq 0.2\,\mathrm{Hz}$. Faster signals are still sampled correctly as long as $f < f_s/2$ (next chapter), but the codes of neighbouring samples jump.

Exercise 8.6: Aperture jitter

The 5 Hz clock of the capstone comes from a microcontroller whose software has a timing uncertainty of $t_j = 1$ ms. The input is a sine with $\hat{V} = 2$ V and $f = 2.5$ Hz (the highest frequency the anti-aliasing filter passes).

a) Calculate the maximum sampling error caused by the jitter.

b) Compare with $\frac{1}{2}$ LSB of the 3-bit converter.

c) Which jitter would a 16-bit converter tolerate for the same signal ($V_{\text{FS}} = 4$ V)?

Solution 8.6

a) $\Delta V_{\text{max}} = 2\pi f \hat{V}\, t_j = 2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V} \cdot 1\,\text{ms} = 31\,\text{mV}$.

b) $\frac{1}{2}\,\text{LSB} = 250\,\text{mV}$: the jitter error is eight times smaller, harmless.

c) $\frac{1}{2}\,\text{LSB} = 4\,\text{V}/2^{17} = 30.5\,\mu\text{V}$: $t_j < 30.5\,\mu\text{V}/(2\pi \cdot 2.5\,\text{Hz} \cdot 2\,\text{V}) = 0.97\,\mu\text{s}$. A software-timed clock is no longer good enough; a hardware timer is needed.

9 Analog-to-Digital Conversion

9.1 Sampling and Aliasing

The sampling theorem

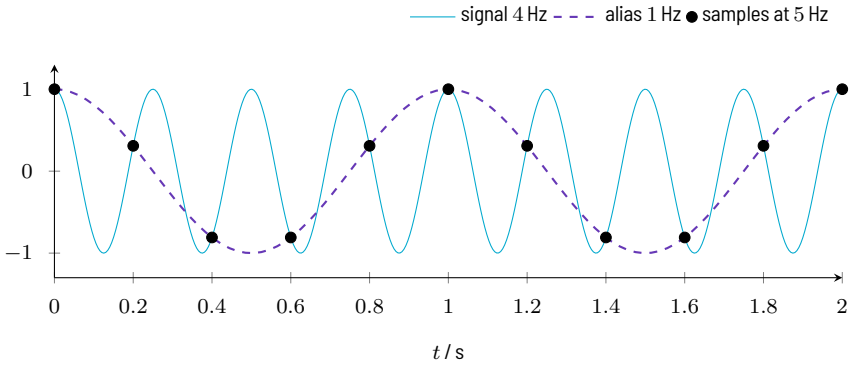
- a signal is sampled at the instants $t_k = k\,T_s$ with the sampling frequency $f_s = 1/T_s$

Sampling theorem (Nyquist–Shannon)

A signal that contains no frequencies at or above $f_s/2$ is completely determined by its samples taken at the rate f_s. $f_s/2$ is called the **Nyquist frequency**.

- the condition is $f_{\text{max}} < f_s/2$, i.e. more than two samples per period of the fastest component
- it applies to **every** component of the input, also to noise and interference
- capstone: $f_s = 5$ Hz $\Rightarrow$ only components below 2.5 Hz are allowed

Aliasing

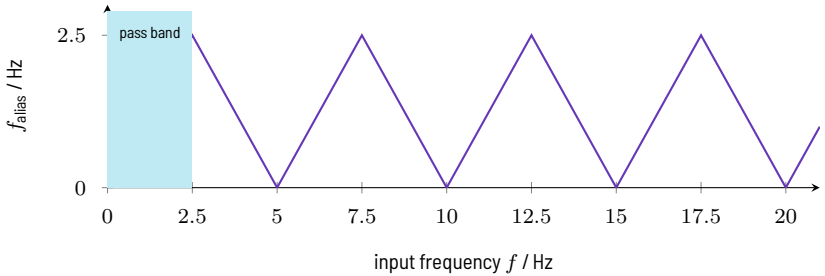


- a 4 Hz cosine sampled at 5 Hz gives the same samples as a 1 Hz cosine

Alias frequency

$f_{\text{alias}} = |f - k\, f_s|$, with the integer k that makes $f_{\text{alias}} \leq f_s/2$

Folding



- every frequency is folded into the range $0 \ldots f_s/2$
- $f_s = 5$ Hz: 4 Hz $\rightarrow 1$ Hz; 12 Hz $\rightarrow 2$ Hz; 49 Hz $\rightarrow 1$ Hz; 50 Hz $\rightarrow 0$ Hz
- after sampling an alias cannot be distinguished from a real signal component

The 50 Hz problem

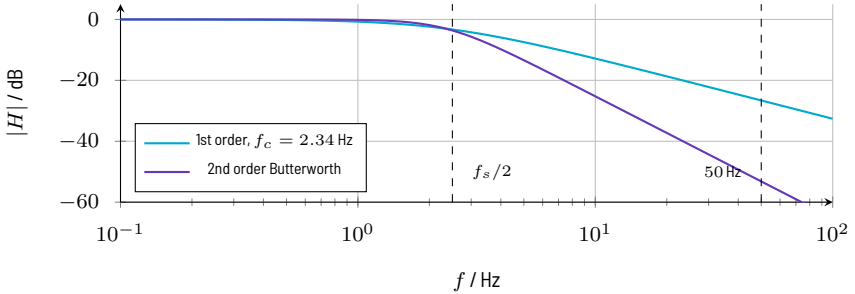
- mains interference (50 Hz) is coupled into every sensor cable
- capstone: $f_s = 5\,\text{Hz} \Rightarrow f_{\text{alias}} = |50\,\text{Hz} - 10 \cdot 5\,\text{Hz}| = 0\,\text{Hz}$
  - every sample sees the 50 Hz sine at the same phase
  - the result is a **constant DC offset** that biases every reading
  - its size depends on the (random) phase: anywhere between $-\hat{V}$ and $+\hat{V}$
  - nothing on the display looks wrong: the error is invisible
- with $f_s = 4.9\,\text{Hz}$ the alias is $|50\,\text{Hz} - 10 \cdot 4.9\,\text{Hz}| = 1\,\text{Hz}$: a slow beat
- **the anti-aliasing low-pass in front of the S&H stage is mandatory**

Falstad: sampling 50 Hz hum

- input: 2 V DC (the wanted signal) plus 0.5 V at 50 Hz
- two S&H stages ($R = 1\,\text{k}\Omega$, $C_H = 100$ nF, track window 1 ms)
  - $f_s = 5$ Hz: every sample is 2.14 V: an offset of 0.14 V
  - $f_s = 4.9$ Hz: 2.14 V, 2.50 V, 2.14 V, 1.58 V, ...: a 1 Hz sine
- change the phase of the 50 Hz source: the offset at 5 Hz changes between 1.5 V and 2.5 V

▶ Falstad: `examples/adc/alias50.txt`

Anti-aliasing filter



- pass the wanted signal, attenuate everything above $f_s/2$ below the resolution
- at 50 Hz: 1st order -26.6 dB (factor 21), 2nd order -53 dB (factor 460)
- the filter must be **before** the S&H: after sampling the damage is done

Design card: Nyquist & Anti-aliasing

- sampling theorem: all input components must be below $f_s/2$ (Nyquist frequency)
- alias: $f_{\text{alias}} = |f - kf_s|$, nearest multiple kf_s; result lies in $0 \ldots f_s/2$
- capstone: $f_s = 5$ Hz, 50 Hz hum $\rightarrow |50 - 10 \cdot 5|$ Hz $= 0$ Hz: constant offset on every reading
- low-pass before the S&H: $f_c \approx f_s/2$ (capstone 2.5 Hz), e.g. RC: $f_c = 1/(2\pi RC)$
- required attenuation: interference amplitude $\times |H(f)| < \frac{1}{2}$ LSB (better $\frac{1}{4}$ LSB)
- 1st order: $|H| \approx f_c/f$ (-20 dB/decade), 2nd order: $(f_c/f)^2$ (-40 dB/decade)
- higher f_s (oversampling) relaxes the filter: more distance between signal band and $f_s/2$

9.2 Quantization

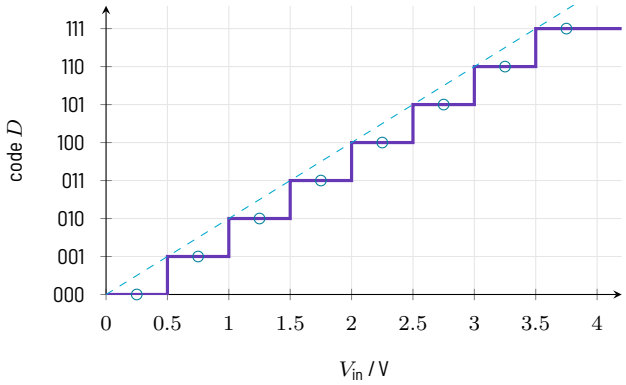
Resolution and LSB

LSB

$$\text{LSB} = \frac{V_{\text{FS}}}{2^N} \qquad (V_{\text{FS}}\text{: full-scale range, } N\text{: number of bits})$$

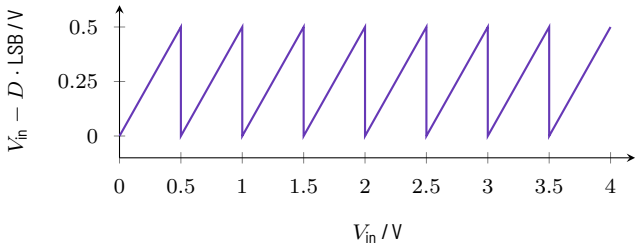
- capstone: $N = 3$, $V_{\text{FS}} = 4$ V: LSB $= 0.5$ V, 8 codes 000 …111
- 10 bit, 3.3 V (microcontroller): 3.2 mV; 16 bit, 5 V: 76 µV
- code for an input voltage (our flash converter):
  $D = \left\lfloor \frac{V_{\text{in}}}{\text{LSB}} \right\rfloor$, limited to $0 \ldots 2^N - 1$
- back to a voltage: $V_D = D \cdot \text{LSB}$ (lower edge) or $(D + \frac{1}{2})$ LSB (bin centre)

Transfer characteristic



- staircase: thresholds at $k \cdot$ LSB ($k = 1 \ldots 7$), the code changes at 0.5 V, 1.0 V, ..., 3.5 V
- circles: bin centres 0.25 V, 0.75 V, ..., 3.75 V: safe test points

Quantization error



- with thresholds at $k \cdot$ LSB: error $0 \ldots 1$ LSB (truncation), mean $\frac{1}{2}$ LSB
- shift all thresholds by $\frac{1}{2}$ LSB (ladder: $R/2$ at the bottom, $3R/2$ at the top): error $\pm\frac{1}{2}$ LSB
- or read the code as the bin centre $(D + \frac{1}{2})$ LSB: also $\pm\frac{1}{2}$ LSB

Quantization noise and SNR

- for a busy signal the error behaves like noise, uniformly distributed over one LSB
  - rms value $V_{q,\text{rms}} = \dfrac{\text{LSB}}{\sqrt{12}}$
- full-scale sine: $V_{\text{rms}} = \dfrac{V_{\text{FS}}}{2\sqrt{2}} = \dfrac{2^N\,\text{LSB}}{2\sqrt{2}}$

Signal-to-noise ratio of an ideal N-bit converter

$$\text{SNR} = 20\log_{10}\frac{2^N\sqrt{12}}{2\sqrt{2}}\,\text{dB} = 6.02\,N + 1.76\,\text{dB}$$

- 3 bit: 19.8 dB; 8 bit: 49.9 dB; 12 bit: 74.0 dB; 16 bit: 98.1 dB
- every additional bit: 6 dB more (factor 2 in amplitude)

Falstad: quantization made visible

- 3-bit flash converter (next sections) followed by a 3-bit R-2R DAC
  - the DAC turns the code back into $V_q = D \cdot 0.5\,\text{V}$
- input: triangle $0\,\text{V} \ldots 4\,\text{V}$, $0.25\,\text{Hz}$
- the scope shows V_{in} and the staircase V_q; the difference is always between $0\,\text{V}$ and $0.5\,\text{V}$
- example readings: $V_{\text{in}} = 1.8\,\text{V} \rightarrow V_q = 1.5\,\text{V}$; $V_{\text{in}} = 3.98\,\text{V} \rightarrow V_q = 3.5\,\text{V}$

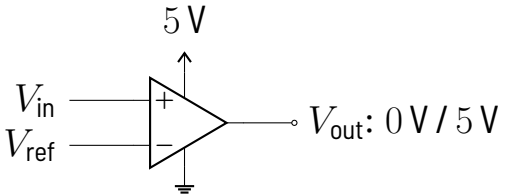
▶ Falstad: `examples/adc/quant_error.txt`

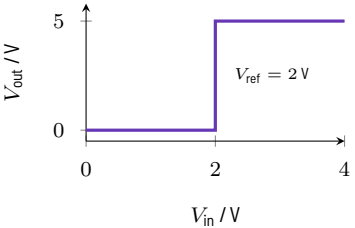
Design card: Reference Ladder & Quantization

- LSB $= V_{\text{FS}}/2^N$; capstone: $4\,\text{V}/8 = 0.5\,\text{V}$
- ladder of 2^N equal resistors from $V_{\text{ref}} = V_{\text{FS}}$ to ground: taps $V_k = k \cdot V_{\text{ref}}/2^N$, $k = 1 \ldots 2^N - 1$
- capstone: $8 \times 1\,\text{k}\Omega$ from $4\,\text{V}$: taps $0.5\,\text{V}$, $1.0\,\text{V}$, ..., $3.5\,\text{V}$; $I = 0.5\,\text{mA}$
- code $D = \lfloor V_{\text{in}}/\text{LSB} \rfloor$; quantization error $0 \ldots 1$ LSB; with $R/2$ at the bottom and $3R/2$ at the top: $\pm\frac{1}{2}$ LSB
- test points: bin centres $(D + \frac{1}{2})$ LSB = $0.25\,\text{V}$, $0.75\,\text{V}$, ..., $3.75\,\text{V}$
- ladder current $\gg$ comparator input currents; tap resistance $\leq R_{\text{total}}/4 = 2\,\text{k}\Omega$
- reference: stiff voltage source (reference IC + buffer); its error scales all thresholds
- SNR of an ideal converter: $6.02\,N + 1.76\,\text{dB}$

9.3 The Comparator

Open-loop op-amp as comparator





- $V_{out} = V_{OH}$ if $V_{in} > V_{ref}$, otherwise V_{OL}
- single 5 V supply: output levels fit 5 V logic directly (rail-to-rail output)
- Falstad: op-amp with **max output** 5 V, **min output** 0 V

Real comparators

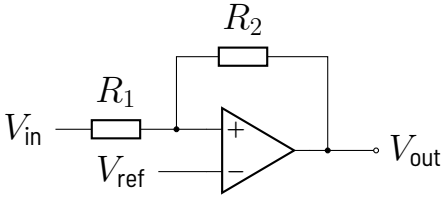
- op-amps are slow when used open loop: recovery from saturation, slew rate (µs)
- comparator ICs (LM393, LM339, TLV3501): fast, no frequency compensation, logic outputs
  - LM393/LM339: **open-collector** output, needs a pull-up resistor (e.g. $4.7\,\mathrm{k\Omega}$ to $5\,\mathrm{V}$)
- input offset voltage (mV): shifts the threshold; must be $\ll$ LSB
- input bias current: flows through the ladder resistors, small threshold shift
- never use a comparator IC as an op-amp: it has no compensation and oscillates with feedback

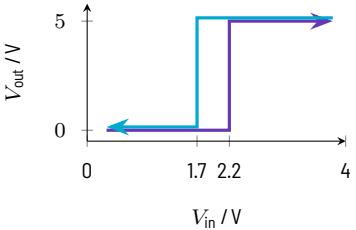
Chatter at slow, noisy inputs

- a slowly changing input with noise crosses the threshold several times
- the output switches back and forth (**chatter**): a counter counts wrong, an LED flickers
- remedy: **hysteresis**: two thresholds, V_{T+} for rising and V_{T-} for falling inputs
  - positive feedback (Schmitt trigger)
  - the hysteresis $V_{T+} - V_{T-}$ must be larger than the peak-to-peak noise
- Falstad: triangle 0 V … 4 V at 1 Hz plus 0.2 V at 50 Hz
  - plain comparator: 10 edges in one period; Schmitt trigger: exactly 2

▶ Falstad: `examples/adc/comparator_hyst.txt`

Schmitt trigger (non-inverting)





- switching when $V_+ = V_{ref}$, with $V_+ = V_{in}\frac{R_2}{R_1+R_2} + V_{out}\frac{R_1}{R_1+R_2}$:

  $$V_T = V_{ref}\left(1 + \frac{R_1}{R_2}\right) - V_{out}\frac{R_1}{R_2}$$

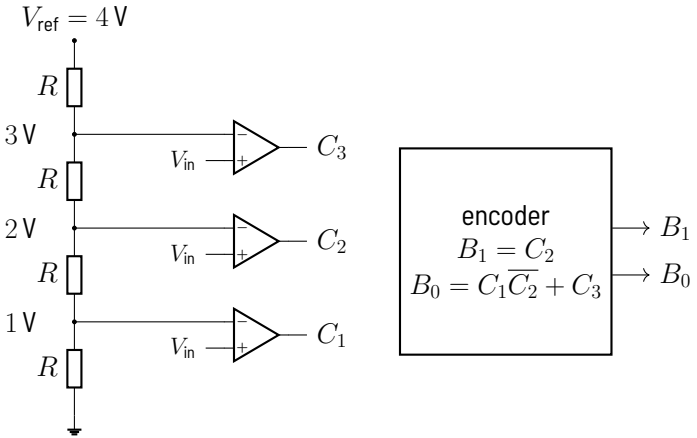
- V_{T+} (output low): $V_{ref}(1 + R_1/R_2)$; V_{T-} (output high): $V_{T+} - V_{OH}R_1/R_2$
- $R_1 = 10\,\text{k}\Omega$, $R_2 = 100\,\text{k}\Omega$, $V_{ref} = 2\,\text{V}$: $V_{T+} = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$, hysteresis $0.5\,\text{V}$

Design card: Comparator

- $V_{\text{out}} =$ high if $V_+ > V_-$, else low; op-amp open loop or comparator IC
- logic compatible: supply $0/5$ V, rail-to-rail output (or open collector + pull-up $4.7\,\text{k}\Omega$)
- Falstad: op-amp, max output 5 V, min output 0 V; logic inputs switch at 2.5 V
- offset voltage $\ll$ LSB; input current $\times$ source resistance $\ll$ LSB
- hysteresis against chatter: R_1 from V_{in} to V_+, R_2 from V_{out} to V_+, V_{ref} at V_-
- $V_{T+} = V_{\text{ref}}(1 + R_1/R_2)$, $V_{T-} = V_{T+} - V_{\text{OH}} R_1/R_2$, hysteresis $V_{\text{OH}} R_1/R_2 >$ noise (peak-to-peak)
- in a flash ADC behind an S&H: no hysteresis needed (the held input is constant), hysteresis would shift the thresholds

9.4 The Flash Converter

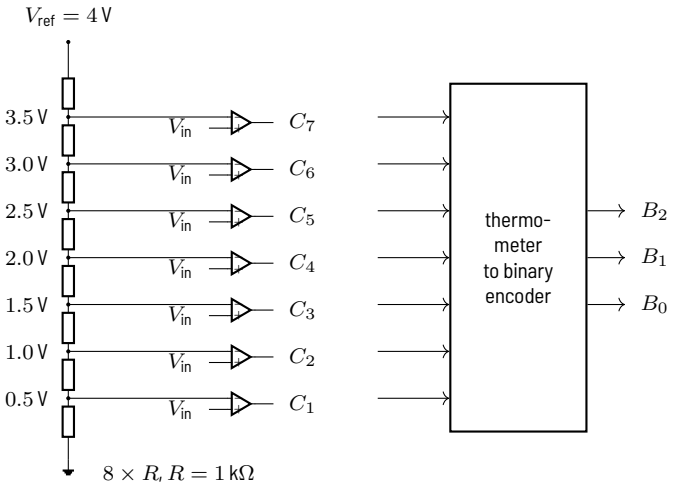
2-bit flash converter



- LSB $= 1\,\text{V}$; thresholds $1\,\text{V}$, $2\,\text{V}$, $3\,\text{V}$; 3 comparators
- $C_k = 1$ if $V_{\text{in}} > k \cdot \text{LSB}$

▶ Falstad: `examples/adc/flash2.txt`

3-bit flash converter (capstone)



- 7 comparators, $C_k = (V_{\text{in}} > k \cdot 0.5\,\text{V})$; ladder current $4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$

Thermometer code

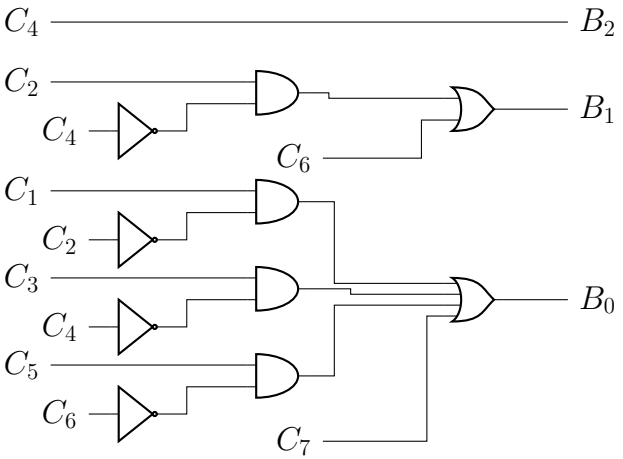
V_{in} / V	D	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.00 ...0.50	0	0	0	0	0	0	0	0	0	0	0
0.50 ...1.00	1	0	0	0	0	0	0	1	0	0	1
1.00 ...1.50	2	0	0	0	0	0	1	1	0	1	0
1.50 ...2.00	3	0	0	0	0	1	1	1	0	1	1
2.00 ...2.50	4	0	0	0	1	1	1	1	1	0	0
2.50 ...3.00	5	0	0	1	1	1	1	1	1	0	1
3.00 ...3.50	6	0	1	1	1	1	1	1	1	1	0
3.50 ...4.00	7	1	1	1	1	1	1	1	1	1	1

- like the column of a thermometer: all comparators below the input level are 1
- the number of ones is the code D; only 2^N of the 2^{2^N-1} patterns are valid

Deriving the encoder

- the top of the column: $H_k = C_k \overline{C_{k+1}}$ is 1 only for $D = k$ (one-hot code, $C_8 := 0$)
- each output bit is the OR of the H_k whose code has this bit set:
  - $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$
  - $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_3} + C_3\overline{C_4} + C_6\overline{C_7} + C_7 = C_2\overline{C_4} + C_6$
  - $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- simplification uses the thermometer property: $C_{k+1} = 1 \Rightarrow C_k = 1$
  - $C_2\overline{C_3} + C_3\overline{C_4} = C_2\overline{C_4}$: “between level 2 and level 4”
  - B_2: four ones in a row means $D \geq 4$

Encoder logic



- 3 NOT, 4 AND, 2 OR gates: the same gates as in the Falstad model

Falstad: 3-bit flash converter with display

- input: slider 0 V … 4 V (label `Vin`); ladder 8×1 kΩ, 4 V
- comparator outputs and encoder signals are labeled nodes `C1`…`C7`, `nC2`, `a24`, …, `B0`…`B2`
- 7 LEDs as bar graph, 3 LEDs for $B_2 B_1 B_0$, each with $330\,\Omega$
- checked at the bin centres 0.25 V, 0.75 V, …, 3.75 V: codes 000, 001, …, 111
- default 2.6 V: five bar LEDs on, binary 101

▶ Falstad: `examples/adc/flash3.txt`

Priority encoder and bubble errors

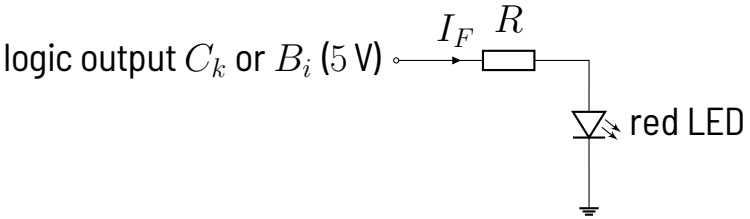
- **bubble**: a wrong comparator decision inside the column, e.g. by offset or noise near a threshold
  - true code 3: 0000111; with bubble at C_2: 0000101
  - our encoder: $B_0 = C_1\overline{C_2} = 1$, $B_1 = 0$, $B_2 = 0 \Rightarrow$ 001: error of 2 LSB
- **priority encoder** (74HC148, 8 inputs $\rightarrow$ 3 bit): outputs the number of the **highest** active input
  - ignores everything below the top: a bubble does no harm
- bubble correction with 3-input gates: $H_k = C_{k-1}\, C_k\, \overline{C_{k+1}}$
- large flash converters: ones counter (Wallace tree) or Gray-code encoding

Design card: Thermometer-to-Binary Encoder

- thermometer code: $C_k = 1$ for all $k \leq D$; the number of ones is the code D
- one-hot: $H_k = C_k \overline{C_{k+1}}$ (top of the column); $B_i =$ OR of all H_k with bit i of k set
- 3 bit (capstone): $B_2 = C_4 \quad B_1 = C_2\overline{C_4} + C_6 \quad B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$
- gates: 3 NOT ($\overline{C_2}, \overline{C_4}, \overline{C_6}$), 4 AND2, 1 OR2, 1 OR4
- check: sweep V_{in} over the bin centres 0.25 V … 3.75 V in steps of 0.5 V: 000 …111
- alternative: priority encoder (74HC148, active-low inputs and outputs), robust against bubbles

9.5 Displaying the Result

LED with series resistor



Series resistor

$$R = \frac{V_{\mathrm{OH}} - V_F}{I_F} = \frac{5\,\mathrm{V} - 2\,\mathrm{V}}{9\,\mathrm{mA}} = 333\,\Omega \quad \rightarrow \quad \text{E12: } 330\,\Omega,\ I_F = 9.1\,\mathrm{mA}$$

- red LED: $V_F \approx 1.8\,\mathrm{V} \ldots 2.0\,\mathrm{V}$; bright enough at $5\,\mathrm{mA} \ldots 10\,\mathrm{mA}$
- **one resistor per LED**: with a common resistor the current would depend on the number of LEDs that are on
- Falstad LED model: $V_F = 1.77\,\mathrm{V}$, $I_F = 9.8\,\mathrm{mA}$ with $330\,\Omega$

Current budget

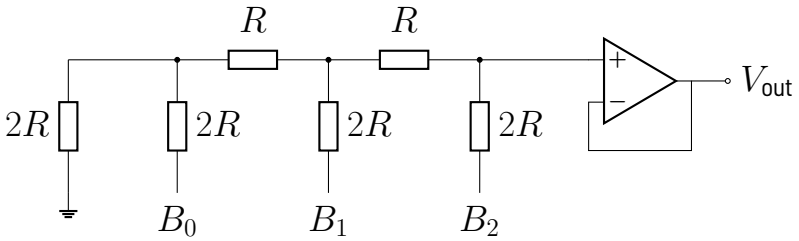
- code 7: 7 bar LEDs + 3 binary LEDs on: $10 \times 9.1\,\text{mA} \approx 91\,\text{mA}$ from the $5\,\text{V}$ supply
- each output must deliver its LED current at a high level close to $5\,\text{V}$
  - in our circuit C_4 drives its bar LED, the B_2 LED and the gate inputs: $18\,\text{mA}$
  - 74HC gates: $\pm 25\,\text{mA}$ absolute maximum per output, V_{OH} drops under load
- better: buffer gates, low-current LEDs ($2\,\text{mA}$), or a driver IC (74HC245; ULN2003 sinks)
- LED to ground (output sources current) or LED to $5\,\text{V}$ (output sinks, active low): both are fine

Design card: Falstad digital cheat-sheet

- gates: **Draw** → **Logic Gates, Input and Output**: AND, OR, NAND, NOR, XOR, Inverter; number of inputs and high level (5 V) in the edit dialog (double click)
- logic levels: input high above $V_{\text{high}}/2 = 2.5$ V; outputs are ideal 0 V/5 V sources
- comparator: op-amp with max output 5 V and min output 0 V; inputs $-$ and $+$ as drawn
- **Logic Input** (click to toggle) and **Logic Output** (shows H/L) for quick tests
- **Labeled Node**: nodes with the same name are connected; use them for $C_1 \ldots C_7$, $B_0 \ldots B_2$; hover to read the voltage
- input voltage: DC voltage source or rail with a slider (edit dialog); clock: rail with square wave, frequency, duty cycle
- analog switch: closed if control > 2.5 V, $R_{\text{on}} = 20\,\Omega$
- LED: always with a series resistor ($330\,\Omega$ at 5 V); lights above a few mA
- scope: right click on a node or element → **View in Scope**; set the time step to 0.1 ms for 5 Hz clocks

9.6 Other Converter Architectures

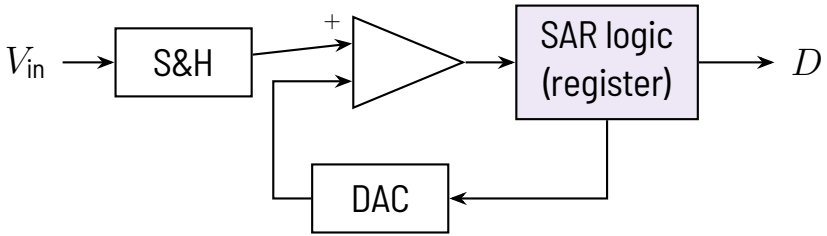
R-2R ladder DAC



- bit inputs: logic levels $0\,\text{V}$ or V_{ref}; output resistance of the ladder: always R
- $V_{\text{out}} = V_{\text{ref}} \left(\frac{B_2}{2} + \frac{B_1}{4} + \frac{B_0}{8} \right) = V_{\text{ref}} \frac{D}{2^N}$
- only two resistor values, any number of bits: easy to integrate
- Falstad: $R = 10\,\text{k}\Omega$, $5\,\text{V}$ logic, counter at $1\,\text{Hz}/0.5\,\text{Hz}/0.25\,\text{Hz}$: staircase in steps of $0.625\,\text{V}$, $D = 5$: $3.125\,\text{V}$

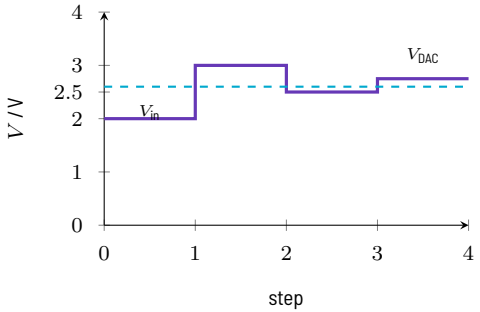
▶ Falstad: `examples/adc/r2r_dac.txt`

Successive approximation (SAR)



- binary search with one comparator and a DAC: one bit per clock cycle, MSB first
- set the bit, compare V_{in} with V_{DAC}: keep the bit if $V_{\text{in}} \geq V_{\text{DAC}}$, otherwise clear it
- N steps for N bits; needs an S&H (built in); 8 ...18 bit, up to some MS/s

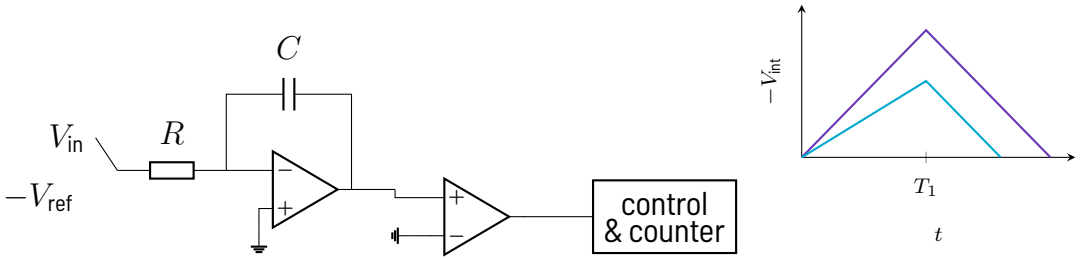
SAR: worked example



step	trial	V_{DAC}	$V_{in} \geq V_{DAC}$?
1	**1**000	$2.00\,V$	yes: keep
2	1**1**00	$3.00\,V$	no: clear
3	10**1**0	$2.50\,V$	yes: keep
4	101**1**	$2.75\,V$	no: clear
	1010	$2.50\,V$	$D = 10$

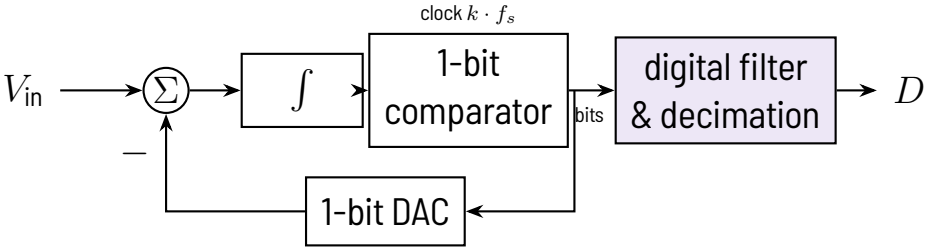
- 4 bit, $V_{FS} = 4\,V$, LSB $= 0.25\,V$, $V_{in} = 2.6\,V$
- result 1010: V_{in} lies between $2.50\,V$ and $2.75\,V$

Dual-slope converter



- integrate V_{in} for a fixed time T_1, then integrate $-V_{ref}$ until the output is zero again (T_2)
- $V_{in}\,T_1 = V_{ref}\,T_2 \quad \Rightarrow \quad D = N_1 \dfrac{V_{in}}{V_{ref}}$: independent of R, C and the clock frequency
- $T_1 = 20$ ms (or 100 ms): 50 Hz interference integrates to zero

Sigma-delta converter (concept)



- oversampling with a 1-bit quantizer in a feedback loop: the density of ones is proportional to V_{in}
- the loop shapes the quantization noise to high frequencies; a digital low-pass removes it
- 16 ...24 bit, audio and precision measurement (weighing scales, thermocouples)
- the digital filter is also the anti-aliasing filter: the analog filter can be simple

Comparison

	flash	SAR	dual-slope	sigma-delta
principle	$2^N - 1$ comparators	binary search	integration	oversampling
steps per sample	1	N	$2 \cdot 2^N$ clocks	many (oversampling)
resolution	4 …8 bit	8 …18 bit	12 …22 bit	16 …24 bit
speed	up to GS/s	kS/s …MS/s	1 …100 S/s	Hz …MHz bandwidth
cost, power	high (exponential)	low	low	medium
typical use	oscilloscopes, radio	microcontrollers	multimeters	audio, scales

- the capstone uses a 3-bit flash converter because every part of it is visible: ladder, comparators, logic

9.7 Exercises

Exercise 9.1: Aliasing

A system samples at $f_s = 5$ Hz.

a) Calculate the alias frequencies of input components at 2 Hz, 4 Hz, 12 Hz, 49 Hz and 50 Hz.

b) The sensor signal is 2 V DC with 0.5 V of 50 Hz hum. Describe the sampled values for $f_s = 5$ Hz and for $f_s = 4.9$ Hz.

c) An RC low-pass with 6.8 kΩ and 10 µF is inserted before the S&H stage. Calculate f_c and the remaining hum amplitude. Compare with the LSB of the 3-bit converter (0.5 V).

d) Check part b) in Falstad.

Solution 9.1

a) $2\,\text{Hz}$: below $f_s/2$, no alias. $4\,\text{Hz}$: $|4-5| = 1\,\text{Hz}$. $12\,\text{Hz}$: $|12 - 2\cdot 5| = 2\,\text{Hz}$. $49\,\text{Hz}$: $|49 - 10\cdot 5| = 1\,\text{Hz}$. $50\,\text{Hz}$: $|50 - 10\cdot 5| = 0\,\text{Hz}$.

b) $f_s = 5\,\text{Hz}$: the hum is sampled at the same phase every time; all samples are equal, $2\,\text{V} + 0.5\,\text{V}\sin\varphi_0$, i.e. a constant offset between $-0.5\,\text{V}$ and $0.5\,\text{V}$. $f_s = 4.9\,\text{Hz}$: the phase advances by 0.204 periods per sample; the samples follow a $1\,\text{Hz}$ sine between $1.5\,\text{V}$ and $2.5\,\text{V}$.

c) $f_c = 1/(2\pi\cdot 6.8\,\text{k}\Omega\cdot 10\,\mu\text{F}) = 2.34\,\text{Hz}$. $|H(50\,\text{Hz})| = 1/\sqrt{1+(50/2.34)^2} = 0.047$ ($-26.6\,\text{dB}$): the hum is reduced to $23\,\text{mV}$, less than $\frac{1}{20}$ LSB. The offset error disappears from the readings.

d) ▶ Falstad: `examples/adc/alias50.txt`

`Vs_5Hz` holds $2.14\,\text{V}$ at every sample (offset $0.14\,\text{V}$ for the phase in the example); `Vs_4.9Hz` shows $2.14\,\text{V}$, $2.50\,\text{V}$, $2.14\,\text{V}$, $1.58\,\text{V}$, $1.62\,\text{V}$, $2.20\,\text{V}$, ...

Exercise 9.2: Quantization

A 3-bit converter has a full scale of 4 V and thresholds at $k \cdot$ LSB.

a) Calculate the LSB and the codes for 0.3 V, 1.8 V, 2.0 V and 3.98 V.
b) Which voltage range does the code 101 represent? What is the largest quantization error?
c) Calculate the SNR of the ideal 3-bit converter. How many bits are needed for an SNR of at least 60 dB?
d) Observe the quantization in Falstad.

Solution 9.2

a) LSB $= 0.5\,\text{V}$. $D = \lfloor V_\text{in}/0.5\,\text{V} \rfloor$: $0.3\,\text{V} \to$ 000, $1.8\,\text{V} \to$ 011, $2.0\,\text{V} \to$ 011 or 100 (exactly on the threshold, the comparator offset decides), $3.98\,\text{V} \to$ 111.

b) 101 = 5: $2.5\,\text{V} \ldots 3.0\,\text{V}$. The error $V_\text{in} - D \cdot \text{LSB}$ is between 0 and $0.5\,\text{V}$ (1 LSB); relative to the bin centre $2.75\,\text{V}$ it is $\pm 0.25\,\text{V}$.

c) SNR $= 6.02 \cdot 3 + 1.76 = 19.8\,\text{dB}$. $6.02\,N + 1.76 \geq 60 \Rightarrow N \geq 9.7$: 10 bits.

d) ▶ Falstad: `examples/adc/quant_error.txt`

At $V_\text{in} = 1.8\,\text{V}$ the DAC output is $V_q = 1.5\,\text{V}$, at $3.98\,\text{V}$ it is $3.5\,\text{V}$; $V_\text{in} - V_q$ stays between $0\,\text{V}$ and $0.5\,\text{V}$.

Exercise 9.3: Designing the 3-bit flash converter

Design the converter of the capstone: full scale $4.00\,\text{V}$, 3 bit, comparators $C_k = (V_\text{in} > k \cdot 0.5\,\text{V})$.

a) Dimension the reference ladder with $1\,\text{k}\Omega$ resistors. Calculate the tap voltages, the ladder current and the power taken from the reference.

b) The comparators have an input bias current of $100\,\text{nA}$. Estimate the worst-case threshold shift. (Hint: Thevenin resistance of a tap.)

c) Write down the truth table from thermometer code to binary and derive the logic equations for B_2, B_1, B_0.

d) How must the ladder be changed for thresholds at $(k - \tfrac{1}{2}) \cdot \text{LSB}$?

e) Check the converter in Falstad at the bin centres.

Solution 9.3

a) 8 resistors of $1\,\mathrm{k\Omega}$ from $4\,\mathrm{V}$ to ground. Taps $V_k = k{\cdot}4\,\mathrm{V}/8 = 0.5\,\mathrm{V}, 1.0\,\mathrm{V}, \ldots, 3.5\,\mathrm{V}$. $I = 4\,\mathrm{V}/8\,\mathrm{k\Omega} = 0.5\,\mathrm{mA}$, $P = 2\,\mathrm{mW}$.

b) The largest Thevenin resistance is at the middle tap: $4\,\mathrm{k\Omega} \parallel 4\,\mathrm{k\Omega} = 2\,\mathrm{k\Omega}$. Shift: $100\,\mathrm{nA} \cdot 2\,\mathrm{k\Omega} = 0.2\,\mathrm{mV}$, negligible compared with $0.5\,\mathrm{V}$.

c) Truth table as on the slide **Thermometer code**. With $H_k = C_k\overline{C_{k+1}}$: $B_2 = H_4 + H_5 + H_6 + H_7 = C_4$; $B_1 = H_2 + H_3 + H_6 + H_7 = C_2\overline{C_4} + C_6$; $B_0 = H_1 + H_3 + H_5 + H_7 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$.

d) Total resistance stays $8R$: bottom resistor $R/2 = 500\,\Omega$, six resistors R, top resistor $3R/2 = 1.5\,\mathrm{k\Omega}$. The taps are then $0.25\,\mathrm{V}$, $0.75\,\mathrm{V}$, ..., $3.25\,\mathrm{V}$.

e) ▶ Falstad: `examples/adc/flash3.txt`

Taps $0.5\,\mathrm{V}$ ...$3.5\,\mathrm{V}$, ladder current $0.50\,\mathrm{mA}$. Slider at $0.25\,\mathrm{V}$, $0.75\,\mathrm{V}$, ..., $3.75\,\mathrm{V}$: the bar graph shows 0 ...7 LEDs, the binary LEDs 000, 001, ..., 111.

Exercise 9.4: Bubble error

Because of noise, the comparators of the 3-bit flash converter deliver the pattern 0010111 (C_7 first), a bubble at C_4.

a) Which input range is plausible? Which code would be correct?
b) Which code does the encoder of the capstone produce?
c) Which code does a priority encoder produce?

Solution 9.4

a) $C_5 = 1$ but $C_4 = 0$ contradict each other. Either C_4 is wrong (input between 2.5 V and 3.0 V, code 5) or C_5 is wrong (input between 1.5 V and 2.0 V, code 3).

b) $B_2 = C_4 = 0$; $B_1 = C_2\overline{C_4} + C_6 = 1$; $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7 = 0 + 1 + 1 + 0 = 1$: code 011 = 3, one of the plausible values. (For the pattern 0000101 of the slide, however, it gives 001 instead of 011.)

c) The priority encoder looks at the highest active input, C_5: code 101 = 5, the other plausible value. For 0000101 it gives 011 = 3, the correct value; it never produces a code below the top of the column.

Exercise 9.5: LED display

The bar graph and the binary LEDs are driven by 5 V logic outputs. Red LEDs with $V_F \approx 2$ V shall carry 5 mA … 10 mA.

a) Choose the series resistor from the E12 series and calculate the LED current.
b) Calculate the total LED current for the code 111.
c) In the Falstad model, C_4 drives its bar LED and also the B_2 LED. Which current does this output deliver?
d) Why must each LED have its own resistor?

Solution 9.5

a) $R = (5\,\text{V} - 2\,\text{V})/I_F$ gives $300\,\Omega$ ...$600\,\Omega$. Choose $330\,\Omega$: $I_F = 3\,\text{V}/330\,\Omega = 9.1\,\text{mA}$.

▶ Falstad: `examples/adc/flash3.txt`

The Falstad LED has $V_F = 1.77\,\text{V}$, so the current is $3.23\,\text{V}/330\,\Omega = 9.8\,\text{mA}$.

b) 10 LEDs: $10 \cdot 9.1\,\text{mA} = 91\,\text{mA}$.

c) $2 \cdot 9.1\,\text{mA} = 18\,\text{mA}$ (plus negligible gate input currents). That is near the limit of a CMOS logic output; a buffer or lower LED currents ($680\,\Omega$: $4.4\,\text{mA}$) are safer.

d) LEDs of different colours and samples have different V_F. With a common resistor the LED with the lowest V_F takes most of the current, and the current per LED changes with the number of LEDs that are on.

Exercise 9.6: Schmitt trigger, R-2R DAC and SAR

a) A non-inverting Schmitt trigger has $V_{\text{ref}} = 2\,\text{V}$, $R_1 = 10\,\text{k}\Omega$ and output levels $0\,\text{V}/5\,\text{V}$. Choose R_2 for a hysteresis of $0.5\,\text{V}$ and calculate both thresholds.

b) A 3-bit R-2R DAC is driven by $5\,\text{V}$ logic levels. Calculate the step size and the output for $D = 5$.

c) A 4-bit SAR converter ($V_{\text{FS}} = 4\,\text{V}$) converts $V_{\text{in}} = 1.3\,\text{V}$. Write down the trial codes, the DAC voltages and the result.

d) Check a) and b) in Falstad.

Solution 9.6

a) Hysteresis $V_{0H} R_1/R_2 = 0.5\,\text{V} \Rightarrow R_2 = 5\,\text{V} \cdot 10\,\text{k}\Omega/0.5\,\text{V} = 100\,\text{k}\Omega$. $V_{T+} = 2\,\text{V} \cdot 1.1 = 2.2\,\text{V}$, $V_{T-} = 1.7\,\text{V}$.

b) LSB $= 5\,\text{V}/8 = 0.625\,\text{V}$; $D = 5$: $3.125\,\text{V}$.

c) LSB $= 0.25\,\text{V}$. 1000: $2.00\,\text{V}$, no $\rightarrow$ 0. 0100: $1.00\,\text{V}$, yes $\rightarrow$ 1. 0110: $1.50\,\text{V}$, no $\rightarrow$ 0. 0101: $1.25\,\text{V}$, yes $\rightarrow$ 1. Result 0101 = 5 ($1.25\,\text{V} \ldots 1.50\,\text{V}$).

d) ▶ Falstad: `examples/adc/comparator_hyst.txt`

The Schmitt trigger switches at $2.2\,\text{V}$ (rising) and $1.7\,\text{V}$ (falling) and gives exactly two edges per period; the plain comparator produces 10 edges.

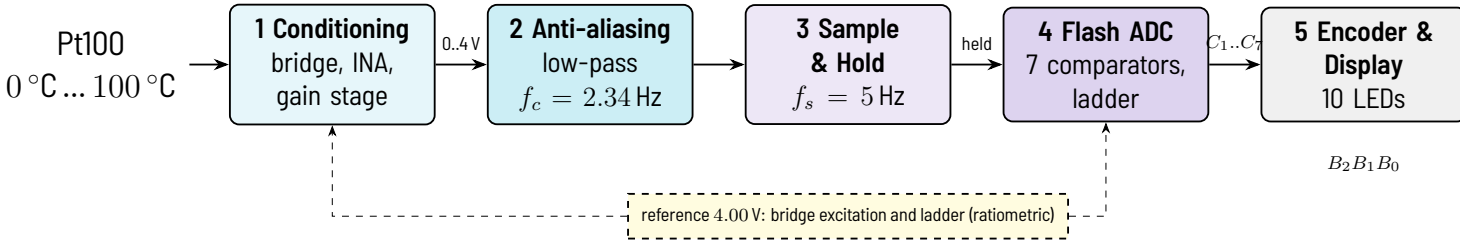
▶ Falstad: `examples/adc/r2r_dac.txt`

`Vout` steps through $0\,\text{V}$, $0.625\,\text{V}$, ..., $4.375\,\text{V}$; at $D = 5$ it shows $3.125\,\text{V}$.

10 From Sensor to Digital Value

10.1 The Measurement Chain

Five stages from sensor to code



- requirement: $0\,°\text{C} \ldots 100\,°\text{C} \rightarrow 0\,\text{V} \ldots 4.00\,\text{V} \rightarrow$ code 000 …111
- $4\,\text{V}$ full scale makes the step size round: $\text{LSB} = 4\,\text{V}/2^3 = 0.5\,\text{V} \mathrel{\widehat{=}} 12.5\,°\text{C}$
- interference: $50\,\text{Hz}$, $5\,\text{mV}$ in series with the sensor (overhead lamps)

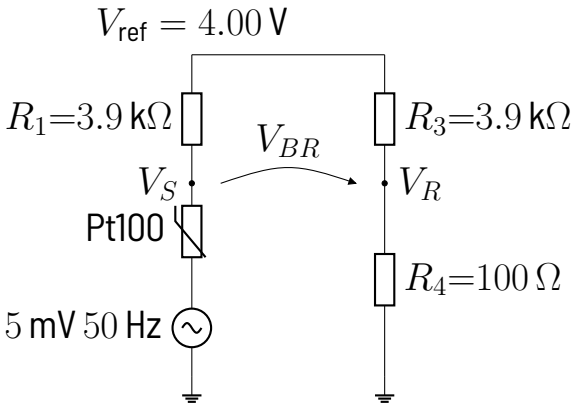
Specification and design values

Quantity	Requirement	Design value
sensor	resistive, slider in Falstad	Pt100, $100\,\Omega$ … $138.5\,\Omega$
sensor current	≤ 1 mA	1.000 mA … 0.990 mA
full scale	0 V …4.00 V	$G = 10 \cdot 10.76 = 107.6$
gain per stage	< 100	1, 10, 10.76
anti-aliasing	$f_c \approx 2.5$ Hz	6.8 kΩ, 10 µF: 2.34 Hz
sampling	$f_s = 5$ Hz	window 20 ms, $C_H = 1$ µF
ADC	3 bit, LSB 0.5 V	8×1 kΩ, 7 comparators
display	LED + resistor	$10\times$ (LED + $330\,\Omega$), 9.8 mA

▶ Falstad: `examples/capstone/capstone.txt`

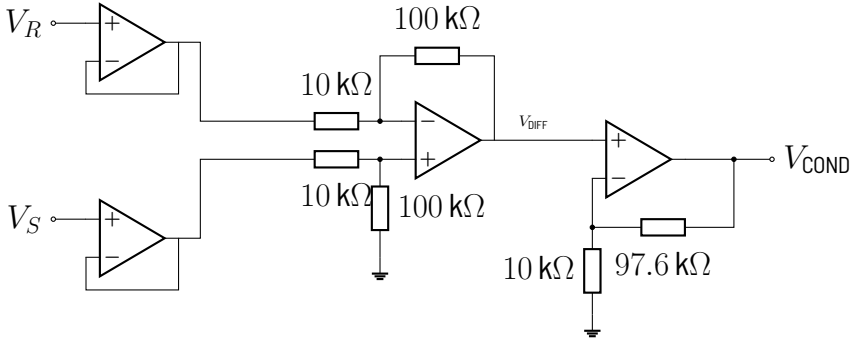
10.2 Stage 1: Signal Conditioning

Bridge design



- sensor current $I_S = V_{\text{ref}}/(R_1 + R) = 4\,\text{V}/4000\,\Omega = 1.000\,\text{mA}$ at $0\,°\text{C}$ ($0.990\,\text{mA}$ at $100\,°\text{C}$)
- balanced at $0\,°\text{C}$: $R_4 = R(0\,°\text{C})$, so $V_{BR} = 0$
- $V_{BR}(T) = V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right)$, $V_{BR}(100\,°\text{C}) = 37.18\,\text{mV}$

Instrumentation amplifier and gain stage



- buffers: input current zero, the bridge is not loaded
- difference amplifier: $V_{\text{DIFF}} = \frac{100\,\text{k}\Omega}{10\,\text{k}\Omega}(V_S - V_R) = 10\,V_{BR}$
- gain stage: $G_2 = 1 + \frac{97.6\,\text{k}\Omega}{10\,\text{k}\Omega} = 10.76$

Gain: how much, and in how many stages?

- required total gain $G = 4.00\,\text{V}/37.18\,\text{mV} = 107.6$
- split: $G_1 = 10$ (difference amplifier), $G_2 = 4.00/(10 \cdot 0.03718) = 10.758$
  - E96: $R_F = 97.6\,\text{k}\Omega \Rightarrow G_2 = 10.76$, full scale $4.0005\,\text{V}$
  - in hardware: $R_F = 91\,\text{k}\Omega + 10\,\text{k}\Omega$ trimmer, adjusted at $100\,°\text{C}$
- finite open-loop gain $A = 10^5$: relative gain error $\approx G/A$
  - one stage $G = 107.6$: $0.11\,\%$; two stages: $2 \times 0.01\,\%$
  - a single stage with $G > 1000$ would lose more than $1\,\%$: keep $G < 100$ per stage

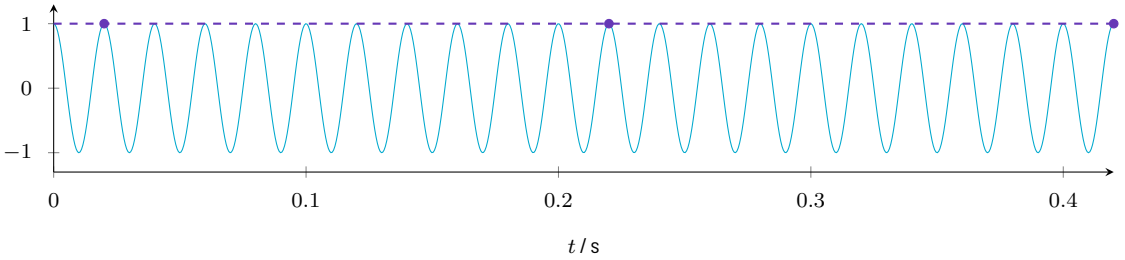
Conditioning

$$V_{\text{COND}} = G_1 G_2\, V_{\text{ref}} \left(\frac{R(T)}{R_1 + R(T)} - \frac{R_4}{R_3 + R_4} \right), \quad G_1 G_2 = 107.6$$

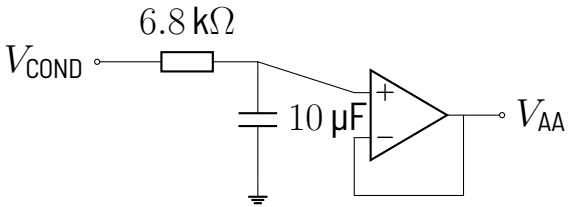
10.3 Stage 2: Anti-Aliasing

Where does the 50 Hz go?

- sampling with f_s folds a frequency f to $f_{\text{alias}} = |f - k f_s|$ (nearest k)
- $f = 50\,\text{Hz}$, $f_s = 5\,\text{Hz}$: $|50 - 10 \cdot 5| = 0\,\text{Hz}$
  - every sample sees the interference at the same phase
  - the result is a **constant offset**: nothing on the scope looks wrong
- amplitude at V_{COND}: $5\,\text{mV} \cdot \frac{R_1}{R_1+R} \cdot 107.6 = 0.52\,\text{V}$ (about 1 LSB)



Anti-aliasing low-pass



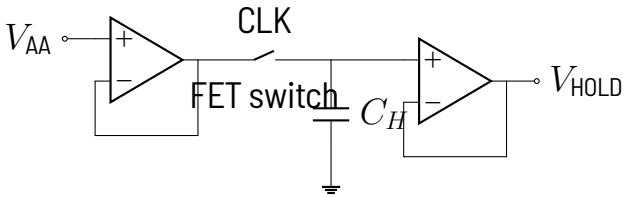
- $f_c = \frac{1}{2\pi RC} = \frac{1}{2\pi \cdot 6.8\,\text{k}\Omega \cdot 10\,\mu\text{F}} = 2.34\,\text{Hz} \leq f_s/2$
- at 50 Hz: $|H| = 1/\sqrt{1 + (50/2.34)^2} = 0.047$ (-26.6 dB)
- residual offset: $0.52\,\text{V} \cdot 0.047 = 25\,\text{mV} < \text{LSB}/2 = 250\,\text{mV}$
- the follower decouples the RC network from the S&H input

Design card: Nyquist & anti-aliasing (capstone values)

$f_s = 5$ Hz $\Rightarrow$ bandwidth ≤ 2.5 Hz; $f_{\text{alias}} = |f - k f_s|$; $f_c = 1/(2\pi RC) \leq f_s/2$; residual $= \hat{v}\,/\sqrt{1 + (f/f_c)^2}$ must stay $<$ LSB$/2$.

10.4 Stage 3: Sample and Hold

Buffered sample and hold



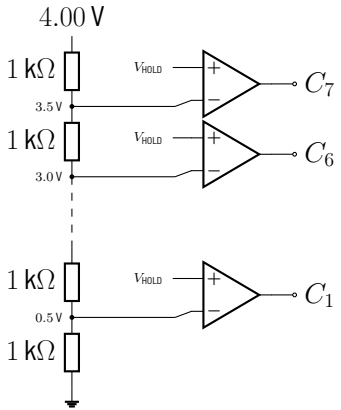
- clock 5 Hz, duty $10\,\%$: switch closed for 20 ms (track), open for 180 ms (hold)
- input buffer: charges C_H quickly; output buffer: no load on C_H

Choosing the hold capacitor

- **charging:** $\tau = R_{\text{on}} C_H = 20\,\Omega \cdot 1\,\mu\text{F} = 20\,\mu\text{s}$
  - settle to LSB$/2$ of 3 bits: $t = \tau \ln 2^4 = 2.8\,\tau = 55\,\mu\text{s} \ll 20\,\text{ms}$
- **droop:** leakage $I_L = 1\,\text{nA}$ (switch + buffer input)
  - $\Delta V = I_L\, t_{\text{hold}} / C_H = 1\,\text{nA} \cdot 0.18\,\text{s} / 1\,\mu\text{F} = 0.18\,\text{mV}$
- both limits leave a wide range: $10\,\text{nF}$ ...$100\,\mu\text{F}$ would work
  - choose a film capacitor with low leakage and dielectric absorption: $C_H = 1\,\mu\text{F}$

10.5 Stages 4 and 5: Flash ADC, Encoder and Display

Reference ladder and comparators



- ladder current $4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; comparator inputs draw no current
- $C_k = 1$ if $V_{\text{HOLD}} > k \cdot 0.5\,\text{V}$; the number of ones is the level 0…7

Quantization and encoder

V_{HOLD} / V	level	C_7	C_6	C_5	C_4	C_3	C_2	C_1	B_2	B_1	B_0
0.0–0.5	0	0	0	0	0	0	0	0	0	0	0
0.5–1.0	1	0	0	0	0	0	0	1	0	0	1
1.0–1.5	2	0	0	0	0	0	1	1	0	1	0
1.5–2.0	3	0	0	0	0	1	1	1	0	1	1
2.0–2.5	4	0	0	0	1	1	1	1	1	0	0
2.5–3.0	5	0	0	1	1	1	1	1	1	0	1
3.0–3.5	6	0	1	1	1	1	1	1	1	1	0
3.5–4.0	7	1	1	1	1	1	1	1	1	1	1

- $B_2 = C_4$ (level ≥ 4)
- $B_1 = C_2\overline{C_4} + C_6$ (levels 2, 3, 6, 7)
- $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$ (odd levels: the edge of the thermometer is at an odd position)

Display

- 7 red LEDs from $C_1 \ldots C_7$: bar graph, a thermometer you can read without decoding
- 3 green LEDs from $B_2 B_1 B_0$: the binary result
- each LED with its own series resistor (chapter 1), driven by a $5\,\mathrm{V}$ logic output: $R = \frac{5\,\mathrm{V} - V_F}{I_F} = \frac{5\,\mathrm{V} - 1.8\,\mathrm{V}}{10\,\mathrm{mA}} = 320\,\Omega \rightarrow$ E12: $330\,\Omega$
  - Falstad: $I_F = 9.8\,\mathrm{mA}$, $V_F = 1.78\,\mathrm{V}$
- never one common resistor for several LEDs: the current would depend on how many are on

10.6 The Complete Chain in Falstad

Using the model

- open ▶ Falstad: `examples/capstone/capstone.txt`
- move the slider from $100\,\Omega$ to $138.5\,\Omega$: the bar graph fills up, the binary LEDs count
- hover over a label to read its voltage; the held value changes every 200 ms
- after a slider step, wait about 0.5 s: the low-pass needs $5\tau = 5 \cdot 68$ ms

Verification table

T / °C	R / Ω	V_{COND} target	V_{HOLD} Falstad	code expected	code Falstad
6.2	102.38	0.25 V	0.251 V	000	000
18.6	107.16	0.75 V	0.751 V	001	001
43.5	116.75	1.75 V	1.749 V	011	011
56.0	121.56	2.25 V	2.250 V	100	100
81.1	131.22	3.25 V	3.249 V	110	110
93.7	136.07	3.75 V	3.750 V	111	111

- the temperatures differ slightly from the linear mapping $25\,°\mathrm{C/V}$ (6, 19, 44, 56, 81, $94\,°$C) because of the bridge nonlinearity

The aliasing experiment

- ▶ Falstad: `examples/capstone/capstone_nofilter.txt`: the S&H takes V_{COND} directly
- V_{COND} carries $0.52\,\mathrm{V}$ of $50\,\mathrm{Hz}$; the S&H always samples at the same phase

	V_{COND} (DC)	V_{HOLD}	code
without filter	$1.75\,\mathrm{V}$	$2.27\,\mathrm{V}$	100 (wrong)
without filter	$0.25\,\mathrm{V}$	$0.77\,\mathrm{V}$	001 (wrong)
with filter	$1.75\,\mathrm{V}$	$1.75\,\mathrm{V}$	011

- without the filter the held value is a clean staircase, just $0.52\,\mathrm{V}$ too high: one full code too high
- the size of the offset depends on the phase between mains and sampling clock (here the worst case)

10.7 Exercises

Exercise 10.1: From sensor to digital value (capstone task 2025/26)

Build a complete data-acquisition chain in Falstad that turns a slowly changing sensor signal ($0\,°\text{C} \ldots 100\,°\text{C}$) into a 3-bit code. Label the five stages Conditioning | Anti-aliasing | Sample & Hold | Flash ADC | Encoder & Display.

a) Conditioning: resistive sensor in a Wheatstone bridge, sensor current ≤ 1 mA, sensor as a resistor with a slider; instrumentation-amplifier front end (two unity-gain buffers and a difference amplifier); gain trimmed to 0 V at the bottom and exactly 4.00 V at the top of the range; less than 100 per stage.

b) Anti-aliasing: sampling clock $f_s = 5$ Hz. Add the 50 Hz, 5 mV noise source in series with the sensor. Observe the held value without a filter, then add an active low-pass with $f_c \approx 2.5$ Hz and justify f_c from the Nyquist criterion.

c) Sample and hold at 5 Hz: follower, FET switch, hold capacitor, follower. Choose the hold capacitor.

d) Flash ADC: ladder 8×1 kΩ from 4 V, seven comparators, thermometer code.

e) Encoder ($B_2 = C_4$, $B_1 = C_2\overline{C_4} + C_6$, $B_0 = C_1\overline{C_2} + C_3\overline{C_4} + C_5\overline{C_6} + C_7$) and display: 7-LED bar graph and three binary LEDs, each LED with its own resistor.

f) Verification table for six inputs near the centres of different code bins, and one line on the aliasing experiment.

Solution 10.1

The design of this chapter is one possible solution; the complete model is ▶ Falstad: `examples/capstone/capstone.txt`.

a) Pt100 with $R_1 = R_3 = 3.9\,\text{k}\Omega$, $R_4 = 100\,\Omega$ at $V_{\text{ref}} = 4.00\,\text{V}$: $I_S = 1.000\,\text{mA}$, bridge output $0\,\text{mV} \ldots 37.18\,\text{mV}$. Buffers, difference amplifier $G_1 = 10$, non-inverting stage $G_2 = 1 + 97.6\,\text{k}\Omega/10\,\text{k}\Omega = 10.76$: $V_{\text{COND}} = 0\,\text{V} \ldots 4.00\,\text{V}$.

b) Without filter the 50 Hz aliases to $|50 - 10 \cdot 5| = 0\,\text{Hz}$; the held value is offset by up to $0.52\,\text{V}$ (Falstad: $1.75\,\text{V}$ becomes $2.27\,\text{V}$, code 100 instead of 011, ▶ Falstad: `examples/capstone/capstone_nofilter.txt`). Filter $R = 6.8\,\text{k}\Omega$, $C = 10\,\mu\text{F}$, $f_c = 2.34\,\text{Hz} \le f_s/2$, 50 Hz attenuated to $4.7\,\%$: $25\,\text{mV}$ residual, below LSB$/2$.

c) $C_H = 1\,\mu\text{F}$: $\tau = 20\,\mu\text{s}$ charges within the $20\,\text{ms}$ window; droop $0.18\,\text{mV}$ per hold period for $1\,\text{nA}$ leakage.

d) Taps at $0.5, 1.0, \ldots, 3.5\,\text{V}$; ladder current $0.5\,\text{mA}$.

e) 3 inverters, 4 AND gates, 2 OR gates (one with 4 inputs); 10 LEDs with $330\,\Omega$, $9.8\,\text{mA}$ each.

f) See the verification table above. Aliasing line: “Without the anti-aliasing filter the held value showed a constant offset of $0.52\,\text{V}$ (one code too high); with the $2.34\,\text{Hz}$ low-pass the offset fell below $30\,\text{mV}$ and all codes were correct.”

Exercise 10.2: Ratiometric measurement

The 4.00 V reference of the capstone is in fact 4.10 V.

a) How large is V_{COND} at $100\,^{\circ}$C?

b) Which code does the converter show at $100\,^{\circ}$C and at $56\,^{\circ}$C? Compare with a design in which the bridge is supplied from a separate, exact 4.00 V source.

Solution 10.2

a) The bridge output scales with the reference: $V_{\mathrm{COND}} = 4.0005\,\mathrm{V} \cdot 4.10/4.00 = 4.10\,\mathrm{V}$.

b) The thresholds scale as well ($k \cdot 0.5125\,\mathrm{V}$). The ratio $V_{\mathrm{COND}}/V_{\mathrm{ref}}$ is unchanged, so both codes are unchanged: 111 at $100\,^{\circ}\mathrm{C}$, 100 at $56\,^{\circ}\mathrm{C}$ ($V_{\mathrm{COND}} = 2.25\,\mathrm{V} \cdot 1.025 = 2.306\,\mathrm{V}$, threshold of level 4: $2.05\,\mathrm{V}$, of level 5: $2.5625\,\mathrm{V}$). With an exact bridge supply and a $4.10\,\mathrm{V}$ ladder the thresholds would be $2.5\,\%$ too high while the signal is correct; a value just above a threshold (e.g. $2.02\,\mathrm{V}$) would then be read one code too low. Ratiometric operation removes the reference from the error budget.

Exercise 10.3: Another sampling rate

The sampling clock of the capstone is changed to $f_s = 4$ Hz and the anti-aliasing filter is removed.

a) To which frequency does the 50 Hz interference alias?
b) Describe what the LEDs show for a constant temperature with $V_{\text{COND}} = 1.75$ V.
c) Is this better or worse than the 0 Hz alias at 5 Hz?

Solution 10.3

a) $|50 - 12 \cdot 4| = 2\,\text{Hz}$, exactly the Nyquist frequency $f_s/2$: consecutive samples see the interference with a phase step of $2\pi \cdot 50/4 = 25\pi$, so its sign alternates from sample to sample.

b) The held value alternates between $1.75 + \hat{v} \sin\varphi$ and $1.75 - \hat{v} \sin\varphi$ ($\hat{v} = 0.52\,\text{V}$): the LEDs blink at $2\,\text{Hz}$ between two codes, e.g. 100 and 010 in the worst case, or stay at 011 if the phase happens to be $\varphi \approx 0$.

c) It is visible, so it is "better" than a silent offset, but the reading is still wrong. Only the low-pass in front of the sampler removes the error for every sampling rate.

Exercise 10.4: Four bits

The capstone shall be extended to 4 bits with the same 4.00 V full scale.

a) What is the new LSB, in volts and in °C? How many comparators and ladder resistors are needed?
b) Is the first-order anti-aliasing filter still sufficient?
c) Is the bridge nonlinearity still acceptable?

Solution 10.4

a) LSB $= 4\,\text{V}/16 = 0.25\,\text{V} \mathrel{\widehat{=}} 6.25\,°\text{C}$; $2^4 - 1 = 15$ comparators, 16 resistors (taps at $k \cdot 0.25\,\text{V}$).

b) Residual $25\,\text{mV} <$ LSB$/2 = 125\,\text{mV}$: yes, still a factor five of margin.

c) The maximum deviation from the straight line is about $0.09\,\text{mV} \cdot 107.6 = 10\,\text{mV} \ll 125\,\text{mV}$: yes.

Exercise 10.5: Hold capacitor at the limits

In the capstone S&H the analog switch has $R_{\text{on}} = 1\,\text{k}\Omega$ and the total leakage current is $10\,\text{nA}$. Determine the smallest and the largest usable hold capacitor if the droop during one hold period and the remaining charging error at the end of the window shall both stay below $\text{LSB}/4 = 125\,\text{mV}$ for a full-scale step.

Solution 10.5

Droop: $I_L\, t_{\text{hold}}/C_H < 0.125\,\text{V} \Rightarrow C_H > 10\,\text{nA} \cdot 0.18\,\text{s}/0.125\,\text{V} = 14.4\,\text{nF}$. Charging: $4\,\text{V} \cdot e^{-t_w/\tau} < 0.125\,\text{V} \Rightarrow t_w > \tau \ln 32 = 3.47\,\tau$, so $\tau < 20\,\text{ms}/3.47 = 5.8\,\text{ms}$ and $C_H < 5.8\,\text{ms}/1\,\text{k}\Omega = 5.8\,\mu\text{F}$. Any value between about $15\,\text{nF}$ and $5.8\,\mu\text{F}$ works; $1\,\mu\text{F}$ is a good choice in the middle (on a logarithmic scale). In Falstad set the switch's on resistance to $1\,\text{k}\Omega$ and C_H to $10\,\mu\text{F}$ in

▶ Falstad: `examples/capstone/capstone.txt`: after a slider step the held staircase needs several samples to reach the new value.